Antroji_uzduotis

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Chapter 1

Aprašymas

1.1 Versija 1.2.

Ši programa skirta studentų duomenų valdymui. Programos Student klasė saugo studento vardą, pavardę, namų darbų pažymius, egzamino pažymius bei apskaičiuoja galutinį įvertinimą pagal medianą ir vidurkį.

1.2 Student klasės struktūra

Kintamasis	Tipas	Aprašymas
name↩ _	std::string	Vardas
surname↩ _	std::string	Pavardė
nd↩ _	std::vector<int>	Namų darbų balai
egz↩ _	int	Egzamino balas
final↩ Avg_	double	Galutinis pagal vidurkį
final↩ Med_	double	Galutinis pagal medianą

1.3 Išvesties ir įvesties operatorių perdengimas

Klasės objektų naudojimosi patogumui buvo perdengti įvedimo ir išvedimo operatoriai.

Įvestis vykdoma per `std::istream`, todėl įvesties operatorius veikia su:

1. rankiniu įvedimu (`cin`)
2. automatiniu įvedimu (`stringstream`)
3. įvedimu iš failo (`ifstream`)

Išvestis vykdoma per std::ostream, todėl išvesties operatorius gali būti naudojamas:

1. išvedimui į ekraną (`cout`)
2. išvedimui į failą (`ofstream`)

Programa taip pat turi rankinius testus, kurie tikrina:

- konstruktorius
- kopijavimą
- move semantiką
- destruktorių
- getter'ius ir setter'ius
- įvesties ir išvesties operatorius
- skaičiavimo tikslumą

Fotografijos:

1.4 Versija 1.5.

Taip pat vėliau buvo pridėta abstrakti klasė Person. Person klasė turi čiuos kintamuosius:

Kintamasis	Tipas	Aprašymas
name↩ _	std::string	Vardas
surname↩ _	std::string	Pavardė

Tada Student klasėje lieka tik šie keturi kintamieji:

Kintamasis	Tipas	Aprašymas
nd↩ _	std::vector<int>	Namų darbų balai
egz↩ _	int	Egzamino balas
final↩ Avg_	double	Galutinis pagal vidurkį
final↩ Med_	double	Galutinis pagal medianą

Kadangi ši klasė yra abstrakti, negalima sukurti šios klasės objektų. Tai pagrindžia šios klasės testas, kuris bando sukurti Person objektą, tačiau tuo atveju testavimų failas nesikompiliuoja.

Fototgrafijos, įrodančios, kad taip tikrai yra:

Programa išliko veiksni po 1.2 versijos logikos įgyvendinimo:

Chapter 2

Directory Documentation

2.1 build_O1/CMakeFiles/4.3.0 Directory Reference

Directories

- directory CompilerIdC
- directory CompilerIdCXX

2.2 build_O2/CMakeFiles/4.3.0 Directory Reference

Directories

- directory CompilerIdC
- directory CompilerIdCXX

2.3 build_O3/CMakeFiles/4.3.0 Directory Reference

Directories

- directory CompilerIdC
- directory CompilerIdCXX

2.4 build_test/CMakeFiles/4.3.0 Directory Reference

Directories

- directory CompilerIdC
- directory CompilerIdCXX

2.5 build_wtests/CMakeFiles/4.3.0 Directory Reference

Directories

- directory CompilerIdC
- directory CompilerIdCXX

2.6 build_O1 Directory Reference

Directories

- directory CMakeFiles

2.7 build_O2 Directory Reference

Directories

- directory CMakeFiles

2.8 build_O3 Directory Reference

Directories

- directory CMakeFiles

2.9 build_test Directory Reference

Directories

- directory CMakeFiles

2.10 build_wtests Directory Reference

Directories

- directory CMakeFiles

2.11 build_O1/CMakeFiles Directory Reference

Directories

- directory 4.3.0
- directory programa.dir

2.12 build_O2/CMakeFiles Directory Reference

Directories

- directory 4.3.0
- directory programa.dir

2.13 build_O3/CMakeFiles Directory Reference

Directories

- directory 4.3.0
- directory programa.dir

2.14 build_test/CMakeFiles Directory Reference

Directories

- directory 4.3.0

2.15 build_wtests/CMakeFiles Directory Reference

Directories

- directory 4.3.0

2.16 build_O1/CMakeFiles/4.3.0/CompilerIdC Directory Reference

Files

- file CMakeCCompilerId.c

2.17 build_O2/CMakeFiles/4.3.0/CompilerIdC Directory Reference

Files

- file CMakeCCompilerId.c

2.18 build_O3/CMakeFiles/4.3.0/CompilerIdC Directory Reference

Files

- file CMakeCCompilerId.c

2.19 build_test/CMakeFiles/4.3.0/CompilerIdC Directory Reference

Files

- file CMakeCCompilerId.c

2.20 build_wtests/CMakeFiles/4.3.0/CompilerIdC Directory Reference

Files

- file CMakeCCompilerId.c

2.21 build_O1/CMakeFiles/4.3.0/CompilerIdCXX Directory Reference

Files

- file CMakeCXXCompilerId.cpp

2.22 build_O2/CMakeFiles/4.3.0/CompilerIdCXX Directory Reference

Files

- file CMakeCXXCompilerId.cpp

2.23 build_O3/CMakeFiles/4.3.0/CompilerIdCXX Directory Reference

Files

- file CMakeCXXCompilerId.cpp

2.24 build_test/CMakeFiles/4.3.0/CompilerIdCXX Directory Reference

Files

- file CMakeCXXCompilerId.cpp

2.25 build_wtests/CMakeFiles/4.3.0/CompilerIdCXX Directory Reference

Files

- file CMakeCXXCompilerId.cpp

2.26 include Directory Reference

Files

- file calculations.h
- file fileHandling.h
- file inputValid.h
- file meniu.h
- file output.h
- file person.h
- file student.h

2.27 obj Directory Reference

Files

- file calculations.d
- file fileHandling.d
- file inputValid.d
- file main.d
- file meniu.d
- file output.d

2.28 build_O1/CMakeFiles/programa.dir Directory Reference

Directories

- directory src

2.29 build_O2/CMakeFiles/programa.dir Directory Reference

Directories

- directory src

2.30 build_O3/CMakeFiles/programa.dir Directory Reference

Directories

- directory src

2.31 build_O1/CMakeFiles/programa.dir/src Directory Reference

Files

- file calculations.cpp.obj.d
- file fileHandling.cpp.obj.d
- file inputValid.cpp.obj.d
- file main.cpp.obj.d
- file meniu.cpp.obj.d
- file output.cpp.obj.d
- file student.cpp.obj.d

2.32 build_O2/CMakeFiles/programa.dir/src Directory Reference

Files

- file calculations.cpp.obj.d
- file fileHandling.cpp.obj.d
- file inputValid.cpp.obj.d
- file main.cpp.obj.d
- file meniu.cpp.obj.d
- file output.cpp.obj.d
- file student.cpp.obj.d

2.33 build_O3/CMakeFiles/programa.dir/src Directory Reference

Files

- file calculations.cpp.obj.d
- file fileHandling.cpp.obj.d
- file inputValid.cpp.obj.d
- file main.cpp.obj.d
- file meniu.cpp.obj.d
- file output.cpp.obj.d
- file student.cpp.obj.d

2.34 src Directory Reference

Files

- file calculations.cpp
- file fileHandling.cpp
- file inputValid.cpp
- file main.cpp
- file meniu.cpp
- file output.cpp
- file person.cpp
- file student.cpp

2.35 tests Directory Reference

Files

- file student_test.cpp

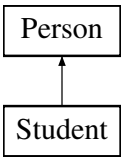
Chapter 3

Class Documentation

3.1 Person Class Reference

```
#include <person.h>
```

Inheritance diagram for Person:



Public Member Functions

- Person (const std::string &name="", const std::string &surname="")
- virtual ~Person ()=default
- Person (const Person &other)
- Person & operator= (const Person &other)
- Person (Person &&other) noexcept
- Person & operator= (Person &&other) noexcept
- virtual void print () const =0
- const std::string & getName () const
- const std::string & getSurname () const

Protected Attributes

- std::string name_
- std::string surname_

3.1.1 Detailed Description

Definition at line 6 of file person.h.

3.1.2 Constructor & Destructor Documentation

3.1.2.1 Person() [1/3]

```
Person::Person (
            const std::string & name = "",
            const std::string & surname = "")  [inline]
```

Definition at line 12 of file person.h.

3.1.2.2 ~Person()

```
virtual Person::~Person ()  [virtual], [default]
```

3.1.2.3 Person() [2/3]

```
Person::Person (
            const Person & other)
```

Definition at line 3 of file person.cpp.

3.1.2.4 Person() [3/3]

```
Person::Person (
            Person && other)  [noexcept]
```

Definition at line 13 of file person.cpp.

3.1.3 Member Function Documentation

3.1.3.1 getName()

```
const std::string & Person::getName () const  [inline]
```

Definition at line 25 of file person.h.

3.1.3.2 getSurname()

```
const std::string & Person::getSurname () const  [inline]
```

Definition at line 26 of file person.h.

3.1.3.3 operator=() [1/2]

```
Person & Person::operator= (
            const Person & other)
```

Definition at line 5 of file person.cpp.

3.1.3.4 operator=() [2/2]

```
Person & Person::operator= (
            Person && other)  [noexcept]
```

Definition at line 19 of file person.cpp.

3.1.3.5 print()

```
virtual void Person::print () const  [pure virtual]
```

Implemented in Student.

3.1.4 Member Data Documentation

3.1.4.1 name_

```
std::string Person::name_  [protected]
```

Definition at line 8 of file person.h.

3.1.4.2 surname_

```
std::string Person::surname_  [protected]
```

Definition at line 9 of file person.h.

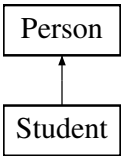
The documentation for this class was generated from the following files:

- include/person.h
- src/person.cpp

3.2 Student Class Reference

```
#include <student.h>
```

Inheritance diagram for Student:



Public Member Functions

- Student (const std::string name="Julius", const std::string surname="Petraitis", const std::vector< int > nd={1, 2}, int egz=5)
- ~Student ()
- Student (const Student &other)
- Student & operator= (const Student &other)
- Student (Student &&other) noexcept
- Student & operator= (Student &&other) noexcept
- int getEgz () const
- const std::vector< int > & getNd () const
- double getFinalAvg () const
- double getFinalMed () const
- void setNd (const std::vector< int > &nd)
- void setEgz (int egz)
- void addNd (int mark)
- void print () const override

Public Member Functions inherited from Person

- Person (const std::string &name="", const std::string &surname="")
- virtual ~Person ()=default
- Person (const Person &other)
- Person & operator= (const Person &other)
- Person (Person &&other) noexcept
- Person & operator= (Person &&other) noexcept
- const std::string & getName () const
- const std::string & getSurname () const

Private Member Functions

- void calculate ()

Private Attributes

- std::vector< int > nd_
- int egz_
- double finalAvg_
- double finalMed_

Friends

- std::ostream & operator<< (std::ostream &os, const Student &s)
- std::istream & operator>> (std::istream &is, Student &s)

Additional Inherited Members

Protected Attributes inherited from Person

- std::string name_
- std::string surname_

3.2.1 Detailed Description

Definition at line 8 of file student.h.

3.2.2 Constructor & Destructor Documentation

3.2.2.1 Student() [1/3]

```
Student::Student (
            const std::string name = "Julius",
            const std::string surname = "Petraitis",
            const std::vector< int > nd = {1, 2},
            int egz = 5)
```

Definition at line 5 of file student.cpp.

3.2.2.2 ~Student()

```
Student::~Student ()
```

Definition at line 9 of file student.cpp.

3.2.2.3 Student() [2/3]

```
Student::Student (
            const Student & other)
```

Definition at line 18 of file student.cpp.

3.2.2.4 Student() [3/3]

```
Student::Student (
            Student && other)  [noexcept]
```

Definition at line 36 of file student.cpp.

3.2.3 Member Function Documentation

3.2.3.1 addNd()

```
void Student::addNd (
            int mark)
```

Definition at line 82 of file student.cpp.

3.2.3.2 calculate()

```
void Student::calculate () [private]
```

Definition at line 87 of file student.cpp.

3.2.3.3 getEgz()

```
int Student::getEgz () const
```

Definition at line 67 of file student.cpp.

3.2.3.4 getFinalAvg()

```
double Student::getFinalAvg () const
```

Definition at line 69 of file student.cpp.

3.2.3.5 getFinalMed()

```
double Student::getFinalMed () const
```

Definition at line 70 of file student.cpp.

3.2.3.6 getNd()

```
const std::vector< int > & Student::getNd () const
```

Definition at line 66 of file student.cpp.

3.2.3.7 operator=() [1/2]

```
Student & Student::operator= (
            const Student & other)
```

Definition at line 25 of file student.cpp.

3.2.3.8 operator=() [2/2]

```
Student & Student::operator= (
            Student && other) [noexcept]
```

Definition at line 49 of file student.cpp.

3.2.3.9 print()

```
void Student::print () const  [override], [virtual]
```

Implements Person.

Definition at line 136 of file student.cpp.

3.2.3.10 setEgz()

```
void Student::setEgz (
            int egz)
```

Definition at line 77 of file student.cpp.

3.2.3.11 setNd()

```
void Student::setNd (
            const std::vector< int > & nd)
```

Definition at line 72 of file student.cpp.

3.2.4 Friends And Related Symbol Documentation

3.2.4.1 operator<<

```
std::ostream & operator<< (
            std::ostream & os,
            const Student & s)  [friend]
```

Definition at line 109 of file student.cpp.

3.2.4.2 operator>>

```
std::istream & operator>> (
            std::istream & is,
            Student & s)  [friend]
```

Definition at line 118 of file student.cpp.

3.2.5 Member Data Documentation

3.2.5.1 egz_

```
int Student::egz_  [private]
```

Definition at line 11 of file student.h.

3.2.5.2 finalAvg_

```
double Student::finalAvg_ [private]
```

Definition at line 13 of file student.h.

3.2.5.3 finalMed_

```
double Student::finalMed_ [private]
```

Definition at line 14 of file student.h.

3.2.5.4 nd_

```
std::vector<int> Student::nd_ [private]
```

Definition at line 10 of file student.h.

The documentation for this class was generated from the following files:

- include/student.h
- src/student.cpp

Chapter 4

File Documentation

4.1 build_O1/CMakeFiles/4.3.0/CompilerIdC/CMakeCCompilerId.c File Reference

Macros

- #define __has_include(x)
- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define C_STD_99 199901L
- #define C_STD_11 201112L
- #define C_STD_17 201710L
- #define C_STD_23 202311L
- #define C_VERSION

Functions

- int main (int argc, char *argv[ ])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.1.1 Macro Definition Documentation

4.1.1.1 __has_include

```
#define __has_include(
              x)
```

Value:

```
0
```

Definition at line 17 of file CMakeCCompilerId.c.

4.1.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 775 of file CMakeCCompilerId.c.

4.1.1.3 C_STD_11

```
#define C_STD_11 201112L
```

Definition at line 861 of file CMakeCCompilerId.c.

4.1.1.4 C_STD_17

```
#define C_STD_17 201710L
```

Definition at line 862 of file CMakeCCompilerId.c.

4.1.1.5 C_STD_23

```
#define C_STD_23 202311L
```

Definition at line 863 of file CMakeCCompilerId.c.

4.1.1.6 C_STD_99

```
#define C_STD_99 199901L
```

Definition at line 860 of file CMakeCCompilerId.c.

4.1.1.7 C_VERSION

```
#define C_VERSION
```

Definition at line 873 of file CMakeCCompilerId.c.

4.1.1.8 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 464 of file CMakeCCompilerId.c.

4.1.1.9 DEC

```
#define DEC(
                n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 779 of file CMakeCCompilerId.c.

4.1.1.10 HEX

```
#define HEX(
                n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 790 of file CMakeCCompilerId.c.

4.1.1.11 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 595 of file CMakeCCompilerId.c.

4.1.1.12 STRINGIFY

```
#define STRINGIFY(
                X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 485 of file CMakeCCompilerId.c.

4.1.1.13 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
              X)
```

Value:

```
#X
```

Definition at line 484 of file CMakeCCompilerId.c.

4.1.2 Function Documentation

4.1.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 907 of file CMakeCCompilerId.c.

4.1.3 Variable Documentation

4.1.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 856 of file CMakeCCompilerId.c.

4.1.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 471 of file CMakeCCompilerId.c.

4.1.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 889 of file CMakeCCompilerId.c.

4.1.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
=
"INFO" ":" "standard_default[" C_VERSION "]"
```

Definition at line 886 of file CMakeCCompilerId.c.

4.1.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 855 of file CMakeCCompilerId.c.

4.2 CMakeCCompilerId.c

Go to the documentation of this file.

```
00001 #ifdef __cplusplus
00002 # error "A C++ compiler has been selected for C."
00003 #endif
00004
00005 #if defined(__18CXX)
00006 # define ID_VOID_MAIN
00007 #endif
00008 #if defined(__CLASSIC_C__)
00009 /* cv-qualifiers did not exist in K&R C */
00010 # define const
00011 # define volatile
00012 #endif
00013
00014 #if !defined(__has_include)
00015 /* If the compiler does not have __has_include, pretend the answer is
00016    always no.  */
00017 #  define __has_include(x) 0
00018 #endif
00019
00020
00021 /* Version number components: V=Version, R=Revision, P=Patch
00022    Version date components:   YYYY=Year, MM=Month,   DD=Day  */
00023
00024 #if defined(__INTEL_COMPILER) || defined(__ICC)
00025 # define COMPILER_ID "Intel"
00026 # if defined(_MSC_VER)
00027 #  define SIMULATE_ID "MSVC"
00028 # endif
00029 # if defined(__GNUC__)
00030 #  define SIMULATE_ID "GNU"
00031 # endif
00032   /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
00033      except that a few beta releases use the old format with V=2021.  */
00034 # if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
00035 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
00036 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
00037 #  if defined(__INTEL_COMPILER_UPDATE)
00038 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
00039 #  else
00040 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
00041 #  endif
00042 # else
00043 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
00044 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
00045   /* The third version component from --version is an update index,
00046      but no macro is provided for it.  */
00047 #  define COMPILER_VERSION_PATCH DEC(0)
00048 # endif
00049 # if defined(__INTEL_COMPILER_BUILD_DATE)
00050   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
00051 #  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
00052 # endif
```

```
# if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# if defined(__GNUC__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
# elif defined(__GNUG__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
  /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
  /* __WATCOMC__ = VVRP + 1100 */
```

```
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_C)
# define COMPILER_ID "SunPro"
# if __SUNPRO_C >= 0x5100
   /* __SUNPRO_C = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# endif

#elif defined(__HP_cc)
# define COMPILER_ID "HP"
  /* __HP_cc = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_cc/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_cc/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_cc     % 100)

#elif defined(__DECC)
# define COMPILER_ID "Compaq"
  /* __DECC_VER = VVRRTPPPP */
# define COMPILER_VERSION_MAJOR DEC(__DECC_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECC_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECC_VER         % 10000)

#elif defined(__IBMC__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ >= 800
# define COMPILER_ID "XL"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif
```

```

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__TINYC__)
# define COMPILER_ID "TinyCC"

#elif defined(__BCC__)
# define COMPILER_ID "Bruce"

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
```

```
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__)
# define COMPILER_ID "GNU"
# define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
```

```
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
    defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
    defined(__ICC8051__) || defined(__ICCSTM8__))
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)


#elif defined(__SDCC_VERSION_MAJOR) || defined(SDCC)
# define COMPILER_ID "SDCC"
# if defined(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MAJOR DEC(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MINOR DEC(__SDCC_VERSION_MINOR)
#  define COMPILER_VERSION_PATCH DEC(__SDCC_VERSION_PATCH)
# else
  /* SDCC = VRP */
#  define COMPILER_VERSION_MAJOR DEC(SDCC/100)
#  define COMPILER_VERSION_MINOR DEC(SDCC/10 % 10)
#  define COMPILER_VERSION_PATCH DEC(SDCC    % 10)
# endif


/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)
```

```

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

```

```
# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

```

```
# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

```

```
# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif

/* Convert integer to decimal digit literals.  */
#define DEC(n)                   \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)             \
  ('0' + ((n)»28 & 0xF)), \
  ('0' + ((n)»24 & 0xF)), \
  ('0' + ((n)»20 & 0xF)), \
  ('0' + ((n)»16 & 0xF)), \
  ('0' + ((n)»12 & 0xF)), \
  ('0' + ((n)»8  & 0xF)), \
  ('0' + ((n)»4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif
```

```c
/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define C_STD_99 199901L
#define C_STD_11 201112L
#define C_STD_17 201710L
#define C_STD_23 202311L

#ifdef __STDC_VERSION__
#  define C_STD __STDC_VERSION__
#endif

#if !defined(__STDC__) && !defined(__clang__) && !defined(__RENESAS__)
# if defined(_MSC_VER) || defined(__ibmxl__) || defined(__IBMC__)
#  define C_VERSION "90"
# else
#  define C_VERSION
# endif
#elif C_STD > C_STD_17
# define C_VERSION "23"
#elif C_STD > C_STD_11
# define C_VERSION "17"
#elif C_STD > C_STD_99
# define C_VERSION "11"
#elif C_STD >= C_STD_99
# define C_VERSION "99"
#else
# define C_VERSION "90"
#endif
const char* info_language_standard_default =
  "INFO" ":" "standard_default[" C_VERSION "]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||            \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&              \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

#ifdef ID_VOID_MAIN
void main() {}
#else
# if defined(__CLASSIC_C__)
int main(argc, argv) int argc; char *argv[];
# else
int main(int argc, char* argv[])
# endif
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
```

```
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
#endif
```

4.3 build_O2/CMakeFiles/4.3.0/CompilerIdC/CMakeCCompilerId.c File Reference

Macros

- #define __has_include(x)
- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define C_STD_99 199901L
- #define C_STD_11 201112L
- #define C_STD_17 201710L
- #define C_STD_23 202311L
- #define C_VERSION

Functions

- int main (int argc, char *argv[])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.3.1 Macro Definition Documentation

4.3.1.1 __has_include

```
#define __has_include(
              x)
```

Value:

0

Definition at line 17 of file CMakeCCompilerId.c.

4.3.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 775 of file CMakeCCompilerId.c.

4.3.1.3 C_STD_11

```
#define C_STD_11 201112L
```

Definition at line 861 of file CMakeCCompilerId.c.

4.3.1.4 C_STD_17

```
#define C_STD_17 201710L
```

Definition at line 862 of file CMakeCCompilerId.c.

4.3.1.5 C_STD_23

```
#define C_STD_23 202311L
```

Definition at line 863 of file CMakeCCompilerId.c.

4.3.1.6 C_STD_99

```
#define C_STD_99 199901L
```

Definition at line 860 of file CMakeCCompilerId.c.

4.3.1.7 C_VERSION

```
#define C_VERSION
```

Definition at line 873 of file CMakeCCompilerId.c.

4.3.1.8 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 464 of file CMakeCCompilerId.c.

4.3.1.9 DEC

```
#define DEC(
            n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 779 of file CMakeCCompilerId.c.

4.3.1.10 HEX

```
#define HEX(
            n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 790 of file CMakeCCompilerId.c.

4.3.1.11 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 595 of file CMakeCCompilerId.c.

4.3.1.12 STRINGIFY

```
#define STRINGIFY(
            X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 485 of file CMakeCCompilerId.c.

4.3.1.13 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
            X)
```

Value:

```
#X
```

Definition at line 484 of file CMakeCCompilerId.c.

4.3.2 Function Documentation

4.3.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 907 of file CMakeCCompilerId.c.

4.3.3 Variable Documentation

4.3.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 856 of file CMakeCCompilerId.c.

4.3.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 471 of file CMakeCCompilerId.c.

4.3.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 889 of file CMakeCCompilerId.c.

4.3.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
                                        =
"INFO" ":" "standard_default[" C_VERSION "]"
```

Definition at line 886 of file CMakeCCompilerId.c.

4.3.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 855 of file CMakeCCompilerId.c.

4.4 CMakeCCompilerId.c

Go to the documentation of this file.

```
00001 #ifdef __cplusplus
00002 # error "A C++ compiler has been selected for C."
00003 #endif
00004
00005 #if defined(__18CXX)
00006 # define ID_VOID_MAIN
00007 #endif
00008 #if defined(__CLASSIC_C__)
00009 /* cv-qualifiers did not exist in K&R C */
00010 # define const
00011 # define volatile
00012 #endif
00013
00014 #if !defined(__has_include)
00015 /* If the compiler does not have __has_include, pretend the answer is
00016    always no.  */
00017 #  define __has_include(x) 0
00018 #endif
00019
00020
00021 /* Version number components: V=Version, R=Revision, P=Patch
00022    Version date components:   YYYY=Year, MM=Month,   DD=Day  */
00023
00024 #if defined(__INTEL_COMPILER) || defined(__ICC)
00025 # define COMPILER_ID "Intel"
00026 # if defined(_MSC_VER)
00027 #  define SIMULATE_ID "MSVC"
00028 # endif
00029 # if defined(__GNUC__)
00030 #  define SIMULATE_ID "GNU"
00031 # endif
00032   /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
00033      except that a few beta releases use the old format with V=2021.  */
00034 # if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
00035 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
00036 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
00037 #  if defined(__INTEL_COMPILER_UPDATE)
00038 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
00039 #  else
00040 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
00041 #  endif
00042 # else
00043 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
00044 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
00045   /* The third version component from --version is an update index,
00046      but no macro is provided for it.  */
00047 #  define COMPILER_VERSION_PATCH DEC(0)
00048 # endif
00049 # if defined(__INTEL_COMPILER_BUILD_DATE)
00050   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
00051 #  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
00052 # endif
00053 # if defined(_MSC_VER)
00054   /* _MSC_VER = VVRR */
00055 #  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
00056 #  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
00057 # endif
00058 # if defined(__GNUC__)
00059 #  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
00060 # elif defined(__GNUG__)
00061 #  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
00062 # endif
00063 # if defined(__GNUC_MINOR__)
00064 #  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
00065 # endif
00066 # if defined(__GNUC_PATCHLEVEL__)
00067 #  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
00068 # endif
00069
```

```
#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_C)
# define COMPILER_ID "SunPro"
# if __SUNPRO_C >= 0x5100
   /* __SUNPRO_C = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xF)
```

```
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# endif

#elif defined(__HP_cc)
# define COMPILER_ID "HP"
  /* __HP_cc = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_cc/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_cc/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_cc     % 100)

#elif defined(__DECC)
# define COMPILER_ID "Compaq"
  /* __DECC_VER = VVRRTPPPP */
# define COMPILER_VERSION_MAJOR DEC(__DECC_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECC_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECC_VER         % 10000)

#elif defined(__IBMC__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ >= 800
# define COMPILER_ID "XL"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
```

```
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__TINYC__)
# define COMPILER_ID "TinyCC"

#elif defined(__BCC__)
# define COMPILER_ID "Bruce"

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
```

```
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__)
# define COMPILER_ID "GNU"
# define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
```

```
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
    defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
    defined(__ICC8051__) || defined(__ICCSTM8__))
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)


#elif defined(__SDCC_VERSION_MAJOR) || defined(SDCC)
# define COMPILER_ID "SDCC"
# if defined(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MAJOR DEC(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MINOR DEC(__SDCC_VERSION_MINOR)
#  define COMPILER_VERSION_PATCH DEC(__SDCC_VERSION_PATCH)
# else
  /* SDCC = VRP */
#  define COMPILER_VERSION_MAJOR DEC(SDCC/100)
#  define COMPILER_VERSION_MINOR DEC(SDCC/10 % 10)
#  define COMPILER_VERSION_PATCH DEC(SDCC    % 10)
# endif


/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

```

```
#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif
```

```

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"
```

```

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

```

```
# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif

/* Convert integer to decimal digit literals.  */
#define DEC(n)                   \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)              \
  ('0' + ((n)>>28 & 0xF)), \
  ('0' + ((n)>>24 & 0xF)), \
  ('0' + ((n)>>20 & 0xF)), \
  ('0' + ((n)>>16 & 0xF)), \
  ('0' + ((n)>>12 & 0xF)), \
  ('0' + ((n)>>8  & 0xF)), \
  ('0' + ((n)>>4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif
```

```c

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define C_STD_99 199901L
#define C_STD_11 201112L
#define C_STD_17 201710L
#define C_STD_23 202311L

#ifdef __STDC_VERSION__
#  define C_STD __STDC_VERSION__
#endif

#if !defined(__STDC__) && !defined(__clang__) && !defined(__RENESAS__)
# if defined(_MSC_VER) || defined(__ibmxl__) || defined(__IBMC__)
#  define C_VERSION "90"
# else
#  define C_VERSION
# endif
#elif C_STD > C_STD_17
# define C_VERSION "23"
#elif C_STD > C_STD_11
# define C_VERSION "17"
#elif C_STD > C_STD_99
# define C_VERSION "11"
#elif C_STD >= C_STD_99
# define C_VERSION "99"
#else
# define C_VERSION "90"
#endif
const char* info_language_standard_default =
  "INFO" ":" "standard_default[" C_VERSION "]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||            \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&              \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

#ifdef ID_VOID_MAIN
void main() {}
#else
# if defined(__CLASSIC_C__)
int main(argc, argv) int argc; char *argv[];
# else
int main(int argc, char* argv[])
# endif
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
#endif
```

4.5 build_O3/CMakeFiles/4.3.0/CompilerIdC/CMakeCCompilerId.c File Reference

Macros

- #define __has_include(x)
- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define C_STD_99 199901L
- #define C_STD_11 201112L
- #define C_STD_17 201710L
- #define C_STD_23 202311L
- #define C_VERSION

Functions

- int main (int argc, char *argv[ ])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.5.1 Macro Definition Documentation

4.5.1.1 __has_include

```
#define __has_include(
            x)
```

Value:

```
0
```

Definition at line 17 of file CMakeCCompilerId.c.

4.5.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 775 of file CMakeCCompilerId.c.

4.5.1.3 C_STD_11

```
#define C_STD_11 201112L
```

Definition at line 861 of file CMakeCCompilerId.c.

4.5.1.4 C_STD_17

```
#define C_STD_17 201710L
```

Definition at line 862 of file CMakeCCompilerId.c.

4.5.1.5 C_STD_23

```
#define C_STD_23 202311L
```

Definition at line 863 of file CMakeCCompilerId.c.

4.5.1.6 C_STD_99

```
#define C_STD_99 199901L
```

Definition at line 860 of file CMakeCCompilerId.c.

4.5.1.7 C_VERSION

```
#define C_VERSION
```

Definition at line 873 of file CMakeCCompilerId.c.

4.5.1.8 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 464 of file CMakeCCompilerId.c.

4.5.1.9 DEC

```
#define DEC(
              n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 779 of file CMakeCCompilerId.c.

4.5.1.10 HEX

```
#define HEX(
            n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 790 of file CMakeCCompilerId.c.

4.5.1.11 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 595 of file CMakeCCompilerId.c.

4.5.1.12 STRINGIFY

```
#define STRINGIFY(
            X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 485 of file CMakeCCompilerId.c.

4.5.1.13 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
            X)
```

Value:

```
#X
```

Definition at line 484 of file CMakeCCompilerId.c.

4.5.2 Function Documentation

4.5.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 907 of file CMakeCCompilerId.c.

4.5.3 Variable Documentation

4.5.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 856 of file CMakeCCompilerId.c.

4.5.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 471 of file CMakeCCompilerId.c.

4.5.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 889 of file CMakeCCompilerId.c.

4.5.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
                                        =
"INFO" ":" "standard_default[" C_VERSION "]"
```

Definition at line 886 of file CMakeCCompilerId.c.

4.5.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 855 of file CMakeCCompilerId.c.

4.6 CMakeCCompilerId.c

Go to the documentation of this file.

```
#ifdef __cplusplus
# error "A C++ compiler has been selected for C."
#endif

#if defined(__18CXX)
# define ID_VOID_MAIN
#endif
#if defined(__CLASSIC_C__)
/* cv-qualifiers did not exist in K&R C */
# define const
# define volatile
#endif

#if !defined(__has_include)
/* If the compiler does not have __has_include, pretend the answer is
   always no.  */
#  define __has_include(x) 0
#endif


/* Version number components: V=Version, R=Revision, P=Patch
   Version date components:   YYYY=Year, MM=Month,   DD=Day  */

#if defined(__INTEL_COMPILER) || defined(__ICC)
# define COMPILER_ID "Intel"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# if defined(__GNUC__)
#  define SIMULATE_ID "GNU"
# endif
  /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
     except that a few beta releases use the old format with V=2021.  */
# if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
#  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
#  if defined(__INTEL_COMPILER_UPDATE)
#   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
#  else
#   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
#  endif
# else
#  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
   /* The third version component from --version is an update index,
      but no macro is provided for it.  */
#  define COMPILER_VERSION_PATCH DEC(0)
# endif
# if defined(__INTEL_COMPILER_BUILD_DATE)
   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
#  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
# endif
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# if defined(__GNUC__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
# elif defined(__GNUG__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
```

```
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_C)
# define COMPILER_ID "SunPro"
# if __SUNPRO_C >= 0x5100
   /* __SUNPRO_C = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# endif

#elif defined(__HP_cc)
# define COMPILER_ID "HP"
  /* __HP_cc = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_cc/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_cc/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_cc     % 100)

#elif defined(__DECC)
# define COMPILER_ID "Compaq"
  /* __DECC_VER = VVRRTPPPP */
```

```
# define COMPILER_VERSION_MAJOR DEC(__DECC_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECC_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECC_VER         % 10000)

#elif defined(__IBMC__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ >= 800
# define COMPILER_ID "XL"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
```

```
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__TINYC__)
# define COMPILER_ID "TinyCC"

#elif defined(__BCC__)
# define COMPILER_ID "Bruce"

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
```

```
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__)
# define COMPILER_ID "GNU"
# define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
  defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
  defined(__ICC8051__) || defined(__ICCSTM8__))
```

```
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)


#elif defined(__SDCC_VERSION_MAJOR) || defined(SDCC)
# define COMPILER_ID "SDCC"
# if defined(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MAJOR DEC(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MINOR DEC(__SDCC_VERSION_MINOR)
#  define COMPILER_VERSION_PATCH DEC(__SDCC_VERSION_PATCH)
# else
  /* SDCC = VRP */
#  define COMPILER_VERSION_MAJOR DEC(SDCC/100)
#  define COMPILER_VERSION_MINOR DEC(SDCC/10 % 10)
#  define COMPILER_VERSION_PATCH DEC(SDCC    % 10)
# endif


/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
```

```
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
```

```
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
```

```
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif
```

```
/* Convert integer to decimal digit literals.  */
#define DEC(n)                          \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)             \
  ('0' + ((n)»28 & 0xF)), \
  ('0' + ((n)»24 & 0xF)), \
  ('0' + ((n)»20 & 0xF)), \
  ('0' + ((n)»16 & 0xF)), \
  ('0' + ((n)»12 & 0xF)), \
  ('0' + ((n)»8  & 0xF)), \
  ('0' + ((n)»4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define C_STD_99 199901L
#define C_STD_11 201112L
#define C_STD_17 201710L
```

```
#define C_STD_23 202311L

#ifdef __STDC_VERSION__
#  define C_STD __STDC_VERSION__
#endif

#if !defined(__STDC__) && !defined(__clang__) && !defined(__RENESAS__)
# if defined(_MSC_VER) || defined(__ibmxl__) || defined(__IBMC__)
#  define C_VERSION "90"
# else
#  define C_VERSION
# endif
#elif C_STD > C_STD_17
# define C_VERSION "23"
#elif C_STD > C_STD_11
# define C_VERSION "17"
#elif C_STD > C_STD_99
# define C_VERSION "11"
#elif C_STD >= C_STD_99
# define C_VERSION "99"
#else
# define C_VERSION "90"
#endif
const char* info_language_standard_default =
  "INFO" ":" "standard_default[" C_VERSION "]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||           \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&             \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*------------------------------------------------------------------------*/

#ifdef ID_VOID_MAIN
void main() {}
#else
# if defined(__CLASSIC_C__)
int main(argc, argv) int argc; char *argv[];
# else
int main(int argc, char* argv[])
# endif
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
#endif
```

4.7 build_test/CMakeFiles/4.3.0/CompilerIdC/CMakeCCompilerId.c File Reference

Macros

- #define __has_include(x)

- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define C_STD_99 199901L
- #define C_STD_11 201112L
- #define C_STD_17 201710L
- #define C_STD_23 202311L
- #define C_VERSION

Functions

- int main (int argc, char *argv[ ])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.7.1 Macro Definition Documentation

4.7.1.1 __has_include

```
#define __has_include(
            x)
```

Value:

```
0
```

Definition at line 17 of file CMakeCCompilerId.c.

4.7.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 775 of file CMakeCCompilerId.c.

4.7.1.3 C_STD_11

```
#define C_STD_11 201112L
```

Definition at line 861 of file CMakeCCompilerId.c.

4.7.1.4 C_STD_17

```
#define C_STD_17 201710L
```

Definition at line 862 of file CMakeCCompilerId.c.

4.7.1.5 C_STD_23

```
#define C_STD_23 202311L
```

Definition at line 863 of file CMakeCCompilerId.c.

4.7.1.6 C_STD_99

```
#define C_STD_99 199901L
```

Definition at line 860 of file CMakeCCompilerId.c.

4.7.1.7 C_VERSION

```
#define C_VERSION
```

Definition at line 873 of file CMakeCCompilerId.c.

4.7.1.8 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 464 of file CMakeCCompilerId.c.

4.7.1.9 DEC

```
#define DEC(
                n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 779 of file CMakeCCompilerId.c.

4.7.1.10 HEX

```
#define HEX(
            n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 790 of file CMakeCCompilerId.c.

4.7.1.11 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 595 of file CMakeCCompilerId.c.

4.7.1.12 STRINGIFY

```
#define STRINGIFY(
            X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 485 of file CMakeCCompilerId.c.

4.7.1.13 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
            X)
```

Value:

```
#X
```

Definition at line 484 of file CMakeCCompilerId.c.

4.7.2 Function Documentation

4.7.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 907 of file CMakeCCompilerId.c.

4.7.3 Variable Documentation

4.7.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 856 of file CMakeCCompilerId.c.

4.7.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 471 of file CMakeCCompilerId.c.

4.7.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["


  "OFF"
"]"
```

Definition at line 889 of file CMakeCCompilerId.c.

4.7.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
                                        =
"INFO" ":" "standard_default[" C_VERSION "]"
```

Definition at line 886 of file CMakeCCompilerId.c.

4.7.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 855 of file CMakeCCompilerId.c.

4.8 CMakeCCompilerId.c

Go to the documentation of this file.

```
#ifdef __cplusplus
# error "A C++ compiler has been selected for C."
#endif

#if defined(__18CXX)
# define ID_VOID_MAIN
#endif
#if defined(__CLASSIC_C__)
/* cv-qualifiers did not exist in K&R C */
# define const
# define volatile
#endif

#if !defined(__has_include)
/* If the compiler does not have __has_include, pretend the answer is
   always no.  */
#  define __has_include(x) 0
#endif


/* Version number components: V=Version, R=Revision, P=Patch
   Version date components:   YYYY=Year, MM=Month,   DD=Day  */

#if defined(__INTEL_COMPILER) || defined(__ICC)
# define COMPILER_ID "Intel"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# if defined(__GNUC__)
#  define SIMULATE_ID "GNU"
# endif
  /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
     except that a few beta releases use the old format with V=2021.  */
# if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
#  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
#  if defined(__INTEL_COMPILER_UPDATE)
#   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
#  else
#   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
#  endif
# else
#  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
   /* The third version component from --version is an update index,
      but no macro is provided for it.  */
#  define COMPILER_VERSION_PATCH DEC(0)
# endif
# if defined(__INTEL_COMPILER_BUILD_DATE)
   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
#  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
# endif
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# if defined(__GNUC__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
# elif defined(__GNUG__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
```

```
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_C)
# define COMPILER_ID "SunPro"
# if __SUNPRO_C >= 0x5100
   /* __SUNPRO_C = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# endif

#elif defined(__HP_cc)
# define COMPILER_ID "HP"
  /* __HP_cc = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_cc/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_cc/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_cc     % 100)

#elif defined(__DECC)
# define COMPILER_ID "Compaq"
  /* __DECC_VER = VVRRTPPPP */
```

```
# define COMPILER_VERSION_MAJOR DEC(__DECC_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECC_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECC_VER         % 10000)

#elif defined(__IBMC__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ >= 800
# define COMPILER_ID "XL"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
```

```
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__TINYC__)
# define COMPILER_ID "TinyCC"

#elif defined(__BCC__)
# define COMPILER_ID "Bruce"

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
```

```
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__)
# define COMPILER_ID "GNU"
# define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
  defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
  defined(__ICC8051__) || defined(__ICCSTM8__))
```

```
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)


#elif defined(__SDCC_VERSION_MAJOR) || defined(SDCC)
# define COMPILER_ID "SDCC"
# if defined(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MAJOR DEC(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MINOR DEC(__SDCC_VERSION_MINOR)
#  define COMPILER_VERSION_PATCH DEC(__SDCC_VERSION_PATCH)
# else
  /* SDCC = VRP */
#  define COMPILER_VERSION_MAJOR DEC(SDCC/100)
#  define COMPILER_VERSION_MINOR DEC(SDCC/10 % 10)
#  define COMPILER_VERSION_PATCH DEC(SDCC    % 10)
# endif


/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
```

```
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
```

```
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
```

```
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif
```

```c

/* Convert integer to decimal digit literals.  */
#define DEC(n)                       \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)             \
  ('0' + ((n)»28 & 0xF)), \
  ('0' + ((n)»24 & 0xF)), \
  ('0' + ((n)»20 & 0xF)), \
  ('0' + ((n)»16 & 0xF)), \
  ('0' + ((n)»12 & 0xF)), \
  ('0' + ((n)»8  & 0xF)), \
  ('0' + ((n)»4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define C_STD_99 199901L
#define C_STD_11 201112L
#define C_STD_17 201710L
```

```
#define C_STD_23 202311L

#ifdef __STDC_VERSION__
#  define C_STD __STDC_VERSION__
#endif

#if !defined(__STDC__) && !defined(__clang__) && !defined(__RENESAS__)
# if defined(_MSC_VER) || defined(__ibmxl__) || defined(__IBMC__)
#  define C_VERSION "90"
# else
#  define C_VERSION
# endif
#elif C_STD > C_STD_17
# define C_VERSION "23"
#elif C_STD > C_STD_11
# define C_VERSION "17"
#elif C_STD > C_STD_99
# define C_VERSION "11"
#elif C_STD >= C_STD_99
# define C_VERSION "99"
#else
# define C_VERSION "90"
#endif
const char* info_language_standard_default =
  "INFO" ":" "standard_default[" C_VERSION "]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||           \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&             \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

#ifdef ID_VOID_MAIN
void main() {}
#else
# if defined(__CLASSIC_C__)
int main(argc, argv) int argc; char *argv[];
# else
int main(int argc, char* argv[])
# endif
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
#endif
```

4.9 build_wtests/CMakeFiles/4.3.0/CompilerIdC/CMakeCCompilerId.c File Reference

Macros

- #define __has_include(x)

- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define C_STD_99 199901L
- #define C_STD_11 201112L
- #define C_STD_17 201710L
- #define C_STD_23 202311L
- #define C_VERSION

Functions

- int main (int argc, char *argv[ ])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.9.1 Macro Definition Documentation

4.9.1.1 __has_include

```
#define __has_include(
            x)
```

Value:

```
0
```

Definition at line 17 of file CMakeCCompilerId.c.

4.9.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 775 of file CMakeCCompilerId.c.

4.9.1.3 C_STD_11

```
#define C_STD_11 201112L
```

Definition at line 861 of file CMakeCCompilerId.c.

4.9.1.4 C_STD_17

```
#define C_STD_17 201710L
```

Definition at line 862 of file CMakeCCompilerId.c.

4.9.1.5 C_STD_23

```
#define C_STD_23 202311L
```

Definition at line 863 of file CMakeCCompilerId.c.

4.9.1.6 C_STD_99

```
#define C_STD_99 199901L
```

Definition at line 860 of file CMakeCCompilerId.c.

4.9.1.7 C_VERSION

```
#define C_VERSION
```

Definition at line 873 of file CMakeCCompilerId.c.

4.9.1.8 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 464 of file CMakeCCompilerId.c.

4.9.1.9 DEC

```
#define DEC(
                n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 779 of file CMakeCCompilerId.c.

4.9.1.10 HEX

```
#define HEX(
            n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 790 of file CMakeCCompilerId.c.

4.9.1.11 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 595 of file CMakeCCompilerId.c.

4.9.1.12 STRINGIFY

```
#define STRINGIFY(
            X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 485 of file CMakeCCompilerId.c.

4.9.1.13 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
            X)
```

Value:

```
#X
```

Definition at line 484 of file CMakeCCompilerId.c.

4.9.2 Function Documentation

4.9.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 907 of file CMakeCCompilerId.c.

4.9.3 Variable Documentation

4.9.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 856 of file CMakeCCompilerId.c.

4.9.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 471 of file CMakeCCompilerId.c.

4.9.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 889 of file CMakeCCompilerId.c.

4.9.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
                                        =
"INFO" ":" "standard_default[" C_VERSION "]"
```

Definition at line 886 of file CMakeCCompilerId.c.

4.9.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 855 of file CMakeCCompilerId.c.

4.10 CMakeCCompilerId.c

Go to the documentation of this file.

```
#ifdef __cplusplus
# error "A C++ compiler has been selected for C."
#endif

#if defined(__18CXX)
# define ID_VOID_MAIN
#endif
#if defined(__CLASSIC_C__)
/* cv-qualifiers did not exist in K&R C */
# define const
# define volatile
#endif

#if !defined(__has_include)
/* If the compiler does not have __has_include, pretend the answer is
   always no.  */
#  define __has_include(x) 0
#endif


/* Version number components: V=Version, R=Revision, P=Patch
   Version date components:   YYYY=Year, MM=Month,   DD=Day  */

#if defined(__INTEL_COMPILER) || defined(__ICC)
# define COMPILER_ID "Intel"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# if defined(__GNUC__)
#  define SIMULATE_ID "GNU"
# endif
  /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
     except that a few beta releases use the old format with V=2021.  */
# if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
#  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
#  if defined(__INTEL_COMPILER_UPDATE)
#   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
#  else
#   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
#  endif
# else
#  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
   /* The third version component from --version is an update index,
      but no macro is provided for it.  */
#  define COMPILER_VERSION_PATCH DEC(0)
# endif
# if defined(__INTEL_COMPILER_BUILD_DATE)
   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
#  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
# endif
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# if defined(__GNUC__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
# elif defined(__GNUG__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
```

```
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_C)
# define COMPILER_ID "SunPro"
# if __SUNPRO_C >= 0x5100
   /* __SUNPRO_C = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_C»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_C»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_C    & 0xF)
# endif

#elif defined(__HP_cc)
# define COMPILER_ID "HP"
  /* __HP_cc = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_cc/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_cc/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_cc     % 100)

#elif defined(__DECC)
# define COMPILER_ID "Compaq"
  /* __DECC_VER = VVRRTPPPP */
```

```
# define COMPILER_VERSION_MAJOR DEC(__DECC_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECC_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECC_VER         % 10000)

#elif defined(__IBMC__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ >= 800
# define COMPILER_ID "XL"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__IBMC__) && !defined(__COMPILER_VER__) && __IBMC__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMC__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMC__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMC__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMC__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
```

```
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__TINYC__)
# define COMPILER_ID "TinyCC"

#elif defined(__BCC__)
# define COMPILER_ID "Bruce"

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
```

```
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__)
# define COMPILER_ID "GNU"
# define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
  defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
  defined(__ICC8051__) || defined(__ICCSTM8__))
```

```
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)


#elif defined(__SDCC_VERSION_MAJOR) || defined(SDCC)
# define COMPILER_ID "SDCC"
# if defined(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MAJOR DEC(__SDCC_VERSION_MAJOR)
#  define COMPILER_VERSION_MINOR DEC(__SDCC_VERSION_MINOR)
#  define COMPILER_VERSION_PATCH DEC(__SDCC_VERSION_PATCH)
# else
  /* SDCC = VRP */
#  define COMPILER_VERSION_MAJOR DEC(SDCC/100)
#  define COMPILER_VERSION_MINOR DEC(SDCC/10 % 10)
#  define COMPILER_VERSION_PATCH DEC(SDCC    % 10)
# endif


/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
```

```
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
```

```
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
```

```
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif
```

```
/* Convert integer to decimal digit literals.  */
#define DEC(n)                          \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)             \
  ('0' + ((n)»28 & 0xF)), \
  ('0' + ((n)»24 & 0xF)), \
  ('0' + ((n)»20 & 0xF)), \
  ('0' + ((n)»16 & 0xF)), \
  ('0' + ((n)»12 & 0xF)), \
  ('0' + ((n)»8  & 0xF)), \
  ('0' + ((n)»4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define C_STD_99 199901L
#define C_STD_11 201112L
#define C_STD_17 201710L
```

```
#define C_STD_23 202311L

#ifdef __STDC_VERSION__
#  define C_STD __STDC_VERSION__
#endif

#if !defined(__STDC__) && !defined(__clang__) && !defined(__RENESAS__)
# if defined(_MSC_VER) || defined(__ibmxl__) || defined(__IBMC__)
#  define C_VERSION "90"
# else
#  define C_VERSION
# endif
#elif C_STD > C_STD_17
# define C_VERSION "23"
#elif C_STD > C_STD_11
# define C_VERSION "17"
#elif C_STD > C_STD_99
# define C_VERSION "11"
#elif C_STD >= C_STD_99
# define C_VERSION "99"
#else
# define C_VERSION "90"
#endif
const char* info_language_standard_default =
  "INFO" ":" "standard_default[" C_VERSION "]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||           \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&             \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

#ifdef ID_VOID_MAIN
void main() {}
#else
# if defined(__CLASSIC_C__)
int main(argc, argv) int argc; char *argv[];
# else
int main(int argc, char* argv[])
# endif
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
#endif
```

4.11 build_O1/CMakeFiles/4.3.0/CompilerIdCXX/CMakeCXXCompilerId.cpp File Reference

Macros

- #define __has_include(x)

- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define CXX_STD_98 199711L
- #define CXX_STD_11 201103L
- #define CXX_STD_14 201402L
- #define CXX_STD_17 201703L
- #define CXX_STD_20 202002L
- #define CXX_STD_23 202302L
- #define CXX_STD __cplusplus

Functions

- int main (int argc, char *argv[])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.11.1 Macro Definition Documentation

4.11.1.1 __has_include

```
#define __has_include(
            x)
```

Value:

```
0
```

Definition at line 11 of file CMakeCXXCompilerId.cpp.

4.11.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 754 of file CMakeCXXCompilerId.cpp.

4.11.1.3 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 443 of file CMakeCXXCompilerId.cpp.

4.11.1.4 CXX_STD

```
#define CXX_STD __cplusplus
```

Definition at line 891 of file CMakeCXXCompilerId.cpp.

4.11.1.5 CXX_STD_11

```
#define CXX_STD_11 201103L
```

Definition at line 840 of file CMakeCXXCompilerId.cpp.

4.11.1.6 CXX_STD_14

```
#define CXX_STD_14 201402L
```

Definition at line 841 of file CMakeCXXCompilerId.cpp.

4.11.1.7 CXX_STD_17

```
#define CXX_STD_17 201703L
```

Definition at line 842 of file CMakeCXXCompilerId.cpp.

4.11.1.8 CXX_STD_20

```
#define CXX_STD_20 202002L
```

Definition at line 843 of file CMakeCXXCompilerId.cpp.

4.11.1.9 CXX_STD_23

```
#define CXX_STD_23 202302L
```

Definition at line 844 of file CMakeCXXCompilerId.cpp.

4.11.1.10 CXX_STD_98

```
#define CXX_STD_98 199711L
```

Definition at line 839 of file CMakeCXXCompilerId.cpp.

4.11.1.11 DEC

```
#define DEC(
            n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 758 of file CMakeCXXCompilerId.cpp.

4.11.1.12 HEX

```
#define HEX(
            n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 769 of file CMakeCXXCompilerId.cpp.

4.11.1.13 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 574 of file CMakeCXXCompilerId.cpp.

4.11.1.14 STRINGIFY

```
#define STRINGIFY(
            X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 464 of file CMakeCXXCompilerId.cpp.

4.11.1.15 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
            X)
```

Value:

```
#X
```

Definition at line 463 of file CMakeCXXCompilerId.cpp.

4.11.2 Function Documentation

4.11.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 924 of file CMakeCXXCompilerId.cpp.

4.11.3 Variable Documentation

4.11.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 835 of file CMakeCXXCompilerId.cpp.

4.11.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 450 of file CMakeCXXCompilerId.cpp.

4.11.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 912 of file CMakeCXXCompilerId.cpp.

4.11.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
                                      = "INFO" ":" "standard_default["









  "98"
"]"
```

Definition at line 894 of file CMakeCXXCompilerId.cpp.

4.11.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 834 of file CMakeCXXCompilerId.cpp.

4.12 CMakeCXXCompilerId.cpp

Go to the documentation of this file.

```
00001 /* This source file must have a .cpp extension so that all C++ compilers
00002    recognize the extension without flags.  Borland does not know .cxx for
00003    example.  */
00004 #ifndef __cplusplus
00005 # error "A C compiler has been selected for C++."
00006 #endif
00007
00008 #if !defined(__has_include)
00009 /* If the compiler does not have __has_include, pretend the answer is
00010    always no.  */
00011 #  define __has_include(x) 0
00012 #endif
00013
00014
00015 /* Version number components: V=Version, R=Revision, P=Patch
00016    Version date components:   YYYY=Year, MM=Month,   DD=Day  */
00017
00018 #if defined(__INTEL_COMPILER) || defined(__ICC)
00019 # define COMPILER_ID "Intel"
00020 # if defined(_MSC_VER)
00021 #  define SIMULATE_ID "MSVC"
00022 # endif
00023 # if defined(__GNUC__)
00024 #  define SIMULATE_ID "GNU"
00025 # endif
00026   /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
00027      except that a few beta releases use the old format with V=2021.  */
00028 # if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
00029 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
00030 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
00031 #  if defined(__INTEL_COMPILER_UPDATE)
00032 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
00033 #  else
00034 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
00035 #  endif
00036 # else
00037 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
00038 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
00039   /* The third version component from --version is an update index,
00040      but no macro is provided for it.  */
00041 #  define COMPILER_VERSION_PATCH DEC(0)
00042 # endif
00043 # if defined(__INTEL_COMPILER_BUILD_DATE)
00044   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
00045 #  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
00046 # endif
00047 # if defined(_MSC_VER)
00048   /* _MSC_VER = VVRR */
00049 #  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
00050 #  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
00051 # endif
00052 # if defined(__GNUC__)
00053 #  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
00054 # elif defined(__GNUG__)
00055 #  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
00056 # endif
00057 # if defined(__GNUC_MINOR__)
00058 #  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
00059 # endif
00060 # if defined(__GNUC_PATCHLEVEL__)
00061 #  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
00062 # endif
00063
00064 #elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
00065 # define COMPILER_ID "IntelLLVM"
00066 #if defined(_MSC_VER)
00067 # define SIMULATE_ID "MSVC"
00068 #endif
00069 #if defined(__GNUC__)
```

```
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_CC)
# define COMPILER_ID "SunPro"
# if __SUNPRO_CC >= 0x5100
   /* __SUNPRO_CC = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# endif

#elif defined(__HP_aCC)
# define COMPILER_ID "HP"
  /* __HP_aCC = VVRRPP */
```

```
# define COMPILER_VERSION_MAJOR DEC(__HP_aCC/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_aCC/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_aCC     % 100)

#elif defined(__DECCXX)
# define COMPILER_ID "Compaq"
  /* __DECCXX_VER = VVRRTPPPP */
# define COMPILER_VERSION_MAJOR DEC(__DECCXX_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECCXX_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECCXX_VER         % 10000)

#elif defined(__IBMCPP__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ >= 800
# define COMPILER_ID "XL"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
```

```
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
```

```
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__) || defined(__GNUG__)
# define COMPILER_ID "GNU"
# if defined(__GNUC__)
#  define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# else
#  define COMPILER_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
```

```
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
defined(__ICC8051__) || defined(__ICCSTM8__))
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)



/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

```

```
#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

```

```
# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

```

```
# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif

/* Convert integer to decimal digit literals.  */
#define DEC(n)                   \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
```

```
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)              \
  ('0' + ((n)»28 & 0xF)), \
  ('0' + ((n)»24 & 0xF)), \
  ('0' + ((n)»20 & 0xF)), \
  ('0' + ((n)»16 & 0xF)), \
  ('0' + ((n)»12 & 0xF)), \
  ('0' + ((n)»8  & 0xF)), \
  ('0' + ((n)»4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define CXX_STD_98 199711L
#define CXX_STD_11 201103L
#define CXX_STD_14 201402L
#define CXX_STD_17 201703L
#define CXX_STD_20 202002L
#define CXX_STD_23 202302L

#if defined(__INTEL_COMPILER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > CXX_STD_17
#    define CXX_STD _MSVC_LANG
#  elif _MSVC_LANG == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
```

```
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14 && __cplusplus > CXX_STD_17
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14
#    define CXX_STD CXX_STD_17
#  elif defined(__INTEL_CXX11_MODE__) && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  elif defined(__INTEL_CXX11_MODE__)
#    define CXX_STD CXX_STD_11
#  else
#    define CXX_STD CXX_STD_98
#  endif
#elif defined(_MSC_VER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > __cplusplus
#    define CXX_STD _MSVC_LANG
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__NVCOMPILER)
#  if __cplusplus == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__INTEL_COMPILER) || defined(__PGI)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_namespace_attributes)
#    define CXX_STD CXX_STD_17
#  elif __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif (defined(__IBMCPP__) || defined(__ibmxl__)) && defined(__linux__)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif __cplusplus == 1 && defined(__GXX_EXPERIMENTAL_CXX0X__)
#  define CXX_STD CXX_STD_11
#else
#  define CXX_STD __cplusplus
#endif

const char* info_language_standard_default = "INFO" ":" "standard_default["
#if CXX_STD > CXX_STD_23
  "26"
#elif CXX_STD > CXX_STD_20
  "23"
#elif CXX_STD > CXX_STD_17
  "20"
#elif CXX_STD > CXX_STD_14
  "17"
#elif CXX_STD > CXX_STD_11
  "14"
#elif CXX_STD >= CXX_STD_11
  "11"
#else
  "98"
#endif
"]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||            \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&              \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

int main(int argc, char* argv[])
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
```

```
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
```

4.13 build_O2/CMakeFiles/4.3.0/CompilerIdCXX/CMakeCXXCompilerId.cpp File Reference

Macros

- #define __has_include(x)
- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define CXX_STD_98 199711L
- #define CXX_STD_11 201103L
- #define CXX_STD_14 201402L
- #define CXX_STD_17 201703L
- #define CXX_STD_20 202002L
- #define CXX_STD_23 202302L
- #define CXX_STD __cplusplus

Functions

- int main (int argc, char *argv[])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.13.1 Macro Definition Documentation

4.13.1.1 __has_include

```
#define __has_include(
            x)
```

Value:

```
0
```

Definition at line 11 of file CMakeCXXCompilerId.cpp.

4.13.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 754 of file CMakeCXXCompilerId.cpp.

4.13.1.3 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 443 of file CMakeCXXCompilerId.cpp.

4.13.1.4 CXX_STD

```
#define CXX_STD __cplusplus
```

Definition at line 891 of file CMakeCXXCompilerId.cpp.

4.13.1.5 CXX_STD_11

```
#define CXX_STD_11 201103L
```

Definition at line 840 of file CMakeCXXCompilerId.cpp.

4.13.1.6 CXX_STD_14

```
#define CXX_STD_14 201402L
```

Definition at line 841 of file CMakeCXXCompilerId.cpp.

4.13.1.7 CXX_STD_17

```
#define CXX_STD_17 201703L
```

Definition at line 842 of file CMakeCXXCompilerId.cpp.

4.13.1.8 CXX_STD_20

```
#define CXX_STD_20 202002L
```

Definition at line 843 of file CMakeCXXCompilerId.cpp.

4.13.1.9 CXX_STD_23

```
#define CXX_STD_23 202302L
```

Definition at line 844 of file CMakeCXXCompilerId.cpp.

4.13.1.10 CXX_STD_98

```
#define CXX_STD_98 199711L
```

Definition at line 839 of file CMakeCXXCompilerId.cpp.

4.13.1.11 DEC

```
#define DEC(
                n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 758 of file CMakeCXXCompilerId.cpp.

4.13.1.12 HEX

```
#define HEX(
                n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 769 of file CMakeCXXCompilerId.cpp.

4.13.1.13 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 574 of file CMakeCXXCompilerId.cpp.

4.13.1.14 STRINGIFY

```
#define STRINGIFY(
                X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 464 of file CMakeCXXCompilerId.cpp.

4.13.1.15 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
              X)
```

Value:

```
#X
```

Definition at line 463 of file CMakeCXXCompilerId.cpp.

4.13.2 Function Documentation

4.13.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 924 of file CMakeCXXCompilerId.cpp.

4.13.3 Variable Documentation

4.13.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 835 of file CMakeCXXCompilerId.cpp.

4.13.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 450 of file CMakeCXXCompilerId.cpp.

4.13.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 912 of file CMakeCXXCompilerId.cpp.

4.13.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
= "INFO" ":" "standard_default["
```

```
  "98"
"]"
```

Definition at line 894 of file CMakeCXXCompilerId.cpp.

4.13.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 834 of file CMakeCXXCompilerId.cpp.

4.14 CMakeCXXCompilerId.cpp

Go to the documentation of this file.

```
00001 /* This source file must have a .cpp extension so that all C++ compilers
00002    recognize the extension without flags.  Borland does not know .cxx for
00003    example.  */
00004 #ifndef __cplusplus
00005 # error "A C compiler has been selected for C++."
00006 #endif
00007
00008 #if !defined(__has_include)
00009 /* If the compiler does not have __has_include, pretend the answer is
00010    always no.  */
00011 #  define __has_include(x) 0
00012 #endif
00013
00014
00015 /* Version number components: V=Version, R=Revision, P=Patch
00016    Version date components:   YYYY=Year, MM=Month,   DD=Day  */
00017
00018 #if defined(__INTEL_COMPILER) || defined(__ICC)
00019 # define COMPILER_ID "Intel"
00020 # if defined(_MSC_VER)
00021 #  define SIMULATE_ID "MSVC"
00022 # endif
00023 # if defined(__GNUC__)
00024 #  define SIMULATE_ID "GNU"
00025 # endif
00026   /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
00027      except that a few beta releases use the old format with V=2021.  */
00028 # if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
00029 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
00030 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
00031 #  if defined(__INTEL_COMPILER_UPDATE)
00032 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
00033 #  else
00034 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
00035 #  endif
00036 # else
00037 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
```

```
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
   /* The third version component from --version is an update index,
      but no macro is provided for it.  */
#  define COMPILER_VERSION_PATCH DEC(0)
# endif
# if defined(__INTEL_COMPILER_BUILD_DATE)
   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
#  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
# endif
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# if defined(__GNUC__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
# elif defined(__GNUG__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
```

```
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_CC)
# define COMPILER_ID "SunPro"
# if __SUNPRO_CC >= 0x5100
   /* __SUNPRO_CC = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# endif

#elif defined(__HP_aCC)
# define COMPILER_ID "HP"
  /* __HP_aCC = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_aCC/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_aCC/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_aCC     % 100)

#elif defined(__DECCXX)
# define COMPILER_ID "Compaq"
  /* __DECCXX_VER = VVRRTPPPP */
# define COMPILER_VERSION_MAJOR DEC(__DECCXX_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECCXX_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECCXX_VER         % 10000)

#elif defined(__IBMCPP__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ >= 800
# define COMPILER_ID "XL"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
```

```
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
```

```
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__) || defined(__GNUG__)
# define COMPILER_ID "GNU"
# if defined(__GNUC__)
#  define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# else
#  define COMPILER_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
```

```
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
  defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
  defined(__ICC8051__) || defined(__ICCSTM8__))
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)



/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
```

```
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

```

```
# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

```

```
# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

```

```
# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif

/* Convert integer to decimal digit literals.  */
#define DEC(n)                    \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)             \
  ('0' + ((n)>>28 & 0xF)), \
  ('0' + ((n)>>24 & 0xF)), \
  ('0' + ((n)>>20 & 0xF)), \
  ('0' + ((n)>>16 & 0xF)), \
  ('0' + ((n)>>12 & 0xF)), \
  ('0' + ((n)>>8  & 0xF)), \
  ('0' + ((n)>>4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
```

```
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define CXX_STD_98 199711L
#define CXX_STD_11 201103L
#define CXX_STD_14 201402L
#define CXX_STD_17 201703L
#define CXX_STD_20 202002L
#define CXX_STD_23 202302L

#if defined(__INTEL_COMPILER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > CXX_STD_17
#    define CXX_STD _MSVC_LANG
#  elif _MSVC_LANG == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14 && __cplusplus > CXX_STD_17
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14
#    define CXX_STD CXX_STD_17
#  elif defined(__INTEL_CXX11_MODE__) && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  elif defined(__INTEL_CXX11_MODE__)
#    define CXX_STD CXX_STD_11
#  else
#    define CXX_STD CXX_STD_98
#  endif
#elif defined(_MSC_VER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > __cplusplus
#    define CXX_STD _MSVC_LANG
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__NVCOMPILER)
#  if __cplusplus == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__INTEL_COMPILER) || defined(__PGI)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_namespace_attributes)
#    define CXX_STD CXX_STD_17
#  elif __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif (defined(__IBMCPP__) || defined(__ibmxl__)) && defined(__linux__)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif __cplusplus == 1 && defined(__GXX_EXPERIMENTAL_CXX0X__)
#  define CXX_STD CXX_STD_11
#else
#  define CXX_STD __cplusplus
#endif

const char* info_language_standard_default = "INFO" ":" "standard_default["
#if CXX_STD > CXX_STD_23
  "26"
#elif CXX_STD > CXX_STD_20
  "23"
#elif CXX_STD > CXX_STD_17
  "20"
#elif CXX_STD > CXX_STD_14
  "17"
#elif CXX_STD > CXX_STD_11
  "14"
```

```
#elif CXX_STD >= CXX_STD_11
  "11"
#else
  "98"
#endif
"]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||            \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&              \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

int main(int argc, char* argv[])
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
```

4.15 build_O3/CMakeFiles/4.3.0/CompilerIdCXX/CMakeCXXCompilerId.cpp File Reference

Macros

- #define __has_include(x)
- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define CXX_STD_98 199711L
- #define CXX_STD_11 201103L
- #define CXX_STD_14 201402L
- #define CXX_STD_17 201703L
- #define CXX_STD_20 202002L
- #define CXX_STD_23 202302L
- #define CXX_STD __cplusplus

Functions

- int main (int argc, char *argv[])

Variables

- char const ∗ info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const ∗ info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const ∗ info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char ∗ info_language_standard_default
- const char ∗ info_language_extensions_default

4.15.1 Macro Definition Documentation

4.15.1.1 __has_include

```
#define __has_include(
            x)
```

Value:

```
0
```

Definition at line 11 of file CMakeCXXCompilerId.cpp.

4.15.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 754 of file CMakeCXXCompilerId.cpp.

4.15.1.3 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 443 of file CMakeCXXCompilerId.cpp.

4.15.1.4 CXX_STD

```
#define CXX_STD __cplusplus
```

Definition at line 891 of file CMakeCXXCompilerId.cpp.

4.15.1.5 CXX_STD_11

```
#define CXX_STD_11 201103L
```

Definition at line 840 of file CMakeCXXCompilerId.cpp.

4.15.1.6 CXX_STD_14

```
#define CXX_STD_14 201402L
```

Definition at line 841 of file CMakeCXXCompilerId.cpp.

4.15.1.7 CXX_STD_17

```
#define CXX_STD_17 201703L
```

Definition at line 842 of file CMakeCXXCompilerId.cpp.

4.15.1.8 CXX_STD_20

```
#define CXX_STD_20 202002L
```

Definition at line 843 of file CMakeCXXCompilerId.cpp.

4.15.1.9 CXX_STD_23

```
#define CXX_STD_23 202302L
```

Definition at line 844 of file CMakeCXXCompilerId.cpp.

4.15.1.10 CXX_STD_98

```
#define CXX_STD_98 199711L
```

Definition at line 839 of file CMakeCXXCompilerId.cpp.

4.15.1.11 DEC

```
#define DEC(
                n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 758 of file CMakeCXXCompilerId.cpp.

4.15.1.12 HEX

```
#define HEX(
            n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 769 of file CMakeCXXCompilerId.cpp.

4.15.1.13 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 574 of file CMakeCXXCompilerId.cpp.

4.15.1.14 STRINGIFY

```
#define STRINGIFY(
            X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 464 of file CMakeCXXCompilerId.cpp.

4.15.1.15 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
            X)
```

Value:

```
#X
```

Definition at line 463 of file CMakeCXXCompilerId.cpp.

4.15.2 Function Documentation

4.15.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 924 of file CMakeCXXCompilerId.cpp.

4.15.3 Variable Documentation

4.15.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 835 of file CMakeCXXCompilerId.cpp.

4.15.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 450 of file CMakeCXXCompilerId.cpp.

4.15.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 912 of file CMakeCXXCompilerId.cpp.

4.15.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
                                      = "INFO" ":" "standard_default["









  "98"
"]"
```

Definition at line 894 of file CMakeCXXCompilerId.cpp.

4.15.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 834 of file CMakeCXXCompilerId.cpp.

4.16 CMakeCXXCompilerId.cpp

Go to the documentation of this file.

```
/* This source file must have a .cpp extension so that all C++ compilers
   recognize the extension without flags.  Borland does not know .cxx for
   example.  */
#ifndef __cplusplus
# error "A C compiler has been selected for C++."
#endif

#if !defined(__has_include)
/* If the compiler does not have __has_include, pretend the answer is
   always no.  */
#  define __has_include(x) 0
#endif


/* Version number components: V=Version, R=Revision, P=Patch
   Version date components:   YYYY=Year, MM=Month,   DD=Day  */

#if defined(__INTEL_COMPILER) || defined(__ICC)
# define COMPILER_ID "Intel"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# if defined(__GNUC__)
#  define SIMULATE_ID "GNU"
# endif
  /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
     except that a few beta releases use the old format with V=2021.  */
# if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
#  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
#  if defined(__INTEL_COMPILER_UPDATE)
#   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
#  else
#   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
#  endif
# else
#  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
   /* The third version component from --version is an update index,
      but no macro is provided for it.  */
#  define COMPILER_VERSION_PATCH DEC(0)
# endif
# if defined(__INTEL_COMPILER_BUILD_DATE)
   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
#  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
# endif
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# if defined(__GNUC__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
# elif defined(__GNUG__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
```

```
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_CC)
# define COMPILER_ID "SunPro"
# if __SUNPRO_CC >= 0x5100
   /* __SUNPRO_CC = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# endif

#elif defined(__HP_aCC)
# define COMPILER_ID "HP"
  /* __HP_aCC = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_aCC/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_aCC/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_aCC     % 100)

#elif defined(__DECCXX)
# define COMPILER_ID "Compaq"
  /* __DECCXX_VER = VVRRTPPPP */
# define COMPILER_VERSION_MAJOR DEC(__DECCXX_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECCXX_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECCXX_VER         % 10000)

#elif defined(__IBMCPP__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
```

```
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ >= 800
# define COMPILER_ID "XL"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
```

```
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
```

```
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__) || defined(__GNUG__)
# define COMPILER_ID "GNU"
# if defined(__GNUC__)
#  define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# else
#  define COMPILER_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
    defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
    defined(__ICC8051__) || defined(__ICCSTM8__))
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
```

```
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)



/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
```

```
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
```

```
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
```

```
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif

/* Convert integer to decimal digit literals.  */
#define DEC(n)                   \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)                   \
  ('0' + ((n)>>28 & 0xF)), \
  ('0' + ((n)>>24 & 0xF)), \
  ('0' + ((n)>>20 & 0xF)), \
  ('0' + ((n)>>16 & 0xF)), \
  ('0' + ((n)>>12 & 0xF)), \
  ('0' + ((n)>>8  & 0xF)), \
  ('0' + ((n)>>4  & 0xF)), \
```

```
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define CXX_STD_98 199711L
#define CXX_STD_11 201103L
#define CXX_STD_14 201402L
#define CXX_STD_17 201703L
#define CXX_STD_20 202002L
#define CXX_STD_23 202302L

#if defined(__INTEL_COMPILER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > CXX_STD_17
#    define CXX_STD _MSVC_LANG
#  elif _MSVC_LANG == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14 && __cplusplus > CXX_STD_17
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14
#    define CXX_STD CXX_STD_17
#  elif defined(__INTEL_CXX11_MODE__) && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  elif defined(__INTEL_CXX11_MODE__)
#    define CXX_STD CXX_STD_11
#  else
#    define CXX_STD CXX_STD_98
#  endif
#elif defined(_MSC_VER) && defined(_MSVC_LANG)
```

```
#  if _MSVC_LANG > __cplusplus
#    define CXX_STD _MSVC_LANG
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__NVCOMPILER)
#  if __cplusplus == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__INTEL_COMPILER) || defined(__PGI)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_namespace_attributes)
#    define CXX_STD CXX_STD_17
#  elif __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif (defined(__IBMCPP__) || defined(__ibmxl__)) && defined(__linux__)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif __cplusplus == 1 && defined(__GXX_EXPERIMENTAL_CXX0X__)
#  define CXX_STD CXX_STD_11
#else
#  define CXX_STD __cplusplus
#endif

const char* info_language_standard_default = "INFO" ":" "standard_default["
#if CXX_STD > CXX_STD_23
  "26"
#elif CXX_STD > CXX_STD_20
  "23"
#elif CXX_STD > CXX_STD_17
  "20"
#elif CXX_STD > CXX_STD_14
  "17"
#elif CXX_STD > CXX_STD_11
  "14"
#elif CXX_STD >= CXX_STD_11
  "11"
#else
  "98"
#endif
"]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||           \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&             \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

int main(int argc, char* argv[])
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
```

4.17 build_test/CMakeFiles/4.3.0/CompilerIdCXX/CMakeCXXCompilerId.cpp File Reference

Macros

- #define __has_include(x)
- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define CXX_STD_98 199711L
- #define CXX_STD_11 201103L
- #define CXX_STD_14 201402L
- #define CXX_STD_17 201703L
- #define CXX_STD_20 202002L
- #define CXX_STD_23 202302L
- #define CXX_STD __cplusplus

Functions

- int main (int argc, char *argv[ ])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.17.1 Macro Definition Documentation

4.17.1.1 __has_include

```
#define __has_include(
            x)
```

Value:

```
0
```

Definition at line 11 of file CMakeCXXCompilerId.cpp.

4.17.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 754 of file CMakeCXXCompilerId.cpp.

4.17.1.3 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 443 of file CMakeCXXCompilerId.cpp.

4.17.1.4 CXX_STD

```
#define CXX_STD __cplusplus
```

Definition at line 891 of file CMakeCXXCompilerId.cpp.

4.17.1.5 CXX_STD_11

```
#define CXX_STD_11 201103L
```

Definition at line 840 of file CMakeCXXCompilerId.cpp.

4.17.1.6 CXX_STD_14

```
#define CXX_STD_14 201402L
```

Definition at line 841 of file CMakeCXXCompilerId.cpp.

4.17.1.7 CXX_STD_17

```
#define CXX_STD_17 201703L
```

Definition at line 842 of file CMakeCXXCompilerId.cpp.

4.17.1.8 CXX_STD_20

```
#define CXX_STD_20 202002L
```

Definition at line 843 of file CMakeCXXCompilerId.cpp.

4.17.1.9 CXX_STD_23

```
#define CXX_STD_23 202302L
```

Definition at line 844 of file CMakeCXXCompilerId.cpp.

4.17.1.10 CXX_STD_98

```
#define CXX_STD_98 199711L
```

Definition at line 839 of file CMakeCXXCompilerId.cpp.

4.17.1.11 DEC

```
#define DEC(
            n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 758 of file CMakeCXXCompilerId.cpp.

4.17.1.12 HEX

```
#define HEX(
            n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 769 of file CMakeCXXCompilerId.cpp.

4.17.1.13 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 574 of file CMakeCXXCompilerId.cpp.

4.17.1.14 STRINGIFY

```
#define STRINGIFY(
            X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 464 of file CMakeCXXCompilerId.cpp.

4.17.1.15 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
            X)
```

Value:

```
#X
```

Definition at line 463 of file CMakeCXXCompilerId.cpp.

4.17.2 Function Documentation

4.17.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 924 of file CMakeCXXCompilerId.cpp.

4.17.3 Variable Documentation

4.17.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 835 of file CMakeCXXCompilerId.cpp.

4.17.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 450 of file CMakeCXXCompilerId.cpp.

4.17.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 912 of file CMakeCXXCompilerId.cpp.

4.17.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
                                      = "INFO" ":" "standard_default["









  "98"
"]"
```

Definition at line 894 of file CMakeCXXCompilerId.cpp.

4.17.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 834 of file CMakeCXXCompilerId.cpp.

4.18 CMakeCXXCompilerId.cpp

Go to the documentation of this file.

```
00001 /* This source file must have a .cpp extension so that all C++ compilers
00002    recognize the extension without flags.  Borland does not know .cxx for
00003    example.  */
00004 #ifndef __cplusplus
00005 # error "A C compiler has been selected for C++."
00006 #endif
00007
00008 #if !defined(__has_include)
00009 /* If the compiler does not have __has_include, pretend the answer is
00010    always no.  */
00011 #  define __has_include(x) 0
00012 #endif
00013
00014
00015 /* Version number components: V=Version, R=Revision, P=Patch
00016    Version date components:   YYYY=Year, MM=Month,   DD=Day  */
00017
00018 #if defined(__INTEL_COMPILER) || defined(__ICC)
00019 # define COMPILER_ID "Intel"
00020 # if defined(_MSC_VER)
00021 #  define SIMULATE_ID "MSVC"
00022 # endif
00023 # if defined(__GNUC__)
00024 #  define SIMULATE_ID "GNU"
00025 # endif
00026   /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
00027      except that a few beta releases use the old format with V=2021.  */
00028 # if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
00029 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
00030 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
00031 #  if defined(__INTEL_COMPILER_UPDATE)
00032 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
00033 #  else
00034 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
00035 #  endif
00036 # else
00037 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
00038 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
00039   /* The third version component from --version is an update index,
00040      but no macro is provided for it.  */
00041 #  define COMPILER_VERSION_PATCH DEC(0)
00042 # endif
00043 # if defined(__INTEL_COMPILER_BUILD_DATE)
00044   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
00045 #  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
00046 # endif
00047 # if defined(_MSC_VER)
00048   /* _MSC_VER = VVRR */
00049 #  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
00050 #  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
00051 # endif
00052 # if defined(__GNUC__)
00053 #  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
00054 # elif defined(__GNUG__)
00055 #  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
00056 # endif
00057 # if defined(__GNUC_MINOR__)
00058 #  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
00059 # endif
00060 # if defined(__GNUC_PATCHLEVEL__)
00061 #  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
00062 # endif
00063
00064 #elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
00065 # define COMPILER_ID "IntelLLVM"
00066 #if defined(_MSC_VER)
00067 # define SIMULATE_ID "MSVC"
00068 #endif
00069 #if defined(__GNUC__)
```

```
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_CC)
# define COMPILER_ID "SunPro"
# if __SUNPRO_CC >= 0x5100
   /* __SUNPRO_CC = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# endif

#elif defined(__HP_aCC)
# define COMPILER_ID "HP"
  /* __HP_aCC = VVRRPP */
```

```
# define COMPILER_VERSION_MAJOR DEC(__HP_aCC/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_aCC/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_aCC     % 100)

#elif defined(__DECCXX)
# define COMPILER_ID "Compaq"
  /* __DECCXX_VER = VVRRTPPPP */
# define COMPILER_VERSION_MAJOR DEC(__DECCXX_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECCXX_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECCXX_VER         % 10000)

#elif defined(__IBMCPP__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ >= 800
# define COMPILER_ID "XL"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
```

```
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
```

```
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__) || defined(__GNUG__)
# define COMPILER_ID "GNU"
# if defined(__GNUC__)
#  define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# else
#  define COMPILER_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
```

```
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
defined(__ICC8051__) || defined(__ICCSTM8__))
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)



/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

```

```
#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

```

```
# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

```

```
# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif

/* Convert integer to decimal digit literals.  */
#define DEC(n)                   \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
```

```
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)              \
  ('0' + ((n)»28 & 0xF)), \
  ('0' + ((n)»24 & 0xF)), \
  ('0' + ((n)»20 & 0xF)), \
  ('0' + ((n)»16 & 0xF)), \
  ('0' + ((n)»12 & 0xF)), \
  ('0' + ((n)»8  & 0xF)), \
  ('0' + ((n)»4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define CXX_STD_98 199711L
#define CXX_STD_11 201103L
#define CXX_STD_14 201402L
#define CXX_STD_17 201703L
#define CXX_STD_20 202002L
#define CXX_STD_23 202302L

#if defined(__INTEL_COMPILER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > CXX_STD_17
#    define CXX_STD _MSVC_LANG
#  elif _MSVC_LANG == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
```

```
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14 && __cplusplus > CXX_STD_17
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14
#    define CXX_STD CXX_STD_17
#  elif defined(__INTEL_CXX11_MODE__) && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  elif defined(__INTEL_CXX11_MODE__)
#    define CXX_STD CXX_STD_11
#  else
#    define CXX_STD CXX_STD_98
#  endif
#elif defined(_MSC_VER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > __cplusplus
#    define CXX_STD _MSVC_LANG
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__NVCOMPILER)
#  if __cplusplus == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__INTEL_COMPILER) || defined(__PGI)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_namespace_attributes)
#    define CXX_STD CXX_STD_17
#  elif __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif (defined(__IBMCPP__) || defined(__ibmxl__)) && defined(__linux__)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif __cplusplus == 1 && defined(__GXX_EXPERIMENTAL_CXX0X__)
#  define CXX_STD CXX_STD_11
#else
#  define CXX_STD __cplusplus
#endif

const char* info_language_standard_default = "INFO" ":" "standard_default["
#if CXX_STD > CXX_STD_23
  "26"
#elif CXX_STD > CXX_STD_20
  "23"
#elif CXX_STD > CXX_STD_17
  "20"
#elif CXX_STD > CXX_STD_14
  "17"
#elif CXX_STD > CXX_STD_11
  "14"
#elif CXX_STD >= CXX_STD_11
  "11"
#else
  "98"
#endif
"]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||            \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&              \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*--------------------------------------------------------------------------*/

int main(int argc, char* argv[])
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
```

```
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
```

4.19 build_wtests/CMakeFiles/4.3.0/CompilerIdCXX/CMake↩ CXXCompilerId.cpp File Reference

Macros

- #define __has_include(x)
- #define COMPILER_ID ""
- #define STRINGIFY_HELPER(X)
- #define STRINGIFY(X)
- #define PLATFORM_ID
- #define ARCHITECTURE_ID
- #define DEC(n)
- #define HEX(n)
- #define CXX_STD_98 199711L
- #define CXX_STD_11 201103L
- #define CXX_STD_14 201402L
- #define CXX_STD_17 201703L
- #define CXX_STD_20 202002L
- #define CXX_STD_23 202302L
- #define CXX_STD __cplusplus

Functions

- int main (int argc, char *argv[])

Variables

- char const * info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
- char const * info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
- char const * info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
- const char * info_language_standard_default
- const char * info_language_extensions_default

4.19.1 Macro Definition Documentation

4.19.1.1 __has_include

```
#define __has_include(
            x)
```

Value:

```
0
```

Definition at line 11 of file CMakeCXXCompilerId.cpp.

4.19.1.2 ARCHITECTURE_ID

```
#define ARCHITECTURE_ID
```

Definition at line 754 of file CMakeCXXCompilerId.cpp.

4.19.1.3 COMPILER_ID

```
#define COMPILER_ID ""
```

Definition at line 443 of file CMakeCXXCompilerId.cpp.

4.19.1.4 CXX_STD

```
#define CXX_STD __cplusplus
```

Definition at line 891 of file CMakeCXXCompilerId.cpp.

4.19.1.5 CXX_STD_11

```
#define CXX_STD_11 201103L
```

Definition at line 840 of file CMakeCXXCompilerId.cpp.

4.19.1.6 CXX_STD_14

```
#define CXX_STD_14 201402L
```

Definition at line 841 of file CMakeCXXCompilerId.cpp.

4.19.1.7 CXX_STD_17

```
#define CXX_STD_17 201703L
```

Definition at line 842 of file CMakeCXXCompilerId.cpp.

4.19.1.8 CXX_STD_20

```
#define CXX_STD_20 202002L
```

Definition at line 843 of file CMakeCXXCompilerId.cpp.

4.19.1.9 CXX_STD_23

```
#define CXX_STD_23 202302L
```

Definition at line 844 of file CMakeCXXCompilerId.cpp.

4.19.1.10 CXX_STD_98

```
#define CXX_STD_98 199711L
```

Definition at line 839 of file CMakeCXXCompilerId.cpp.

4.19.1.11 DEC

```
#define DEC(
                n)
```

Value:

```
('0' + (((n) / 10000000)%10)), \
('0' + (((n) / 1000000)%10)),  \
('0' + (((n) / 100000)%10)),   \
('0' + (((n) / 10000)%10)),    \
('0' + (((n) / 1000)%10)),     \
('0' + (((n) / 100)%10)),      \
('0' + (((n) / 10)%10)),       \
('0' +  ((n) % 10))
```

Definition at line 758 of file CMakeCXXCompilerId.cpp.

4.19.1.12 HEX

```
#define HEX(
                n)
```

Value:

```
('0' + ((n)»28 & 0xF)), \
('0' + ((n)»24 & 0xF)), \
('0' + ((n)»20 & 0xF)), \
('0' + ((n)»16 & 0xF)), \
('0' + ((n)»12 & 0xF)), \
('0' + ((n)»8  & 0xF)), \
('0' + ((n)»4  & 0xF)), \
('0' + ((n)     & 0xF))
```

Definition at line 769 of file CMakeCXXCompilerId.cpp.

4.19.1.13 PLATFORM_ID

```
#define PLATFORM_ID
```

Definition at line 574 of file CMakeCXXCompilerId.cpp.

4.19.1.14 STRINGIFY

```
#define STRINGIFY(
                X)
```

Value:

```
STRINGIFY_HELPER(X)
```

Definition at line 464 of file CMakeCXXCompilerId.cpp.

4.19.1.15 STRINGIFY_HELPER

```
#define STRINGIFY_HELPER(
              X)
```

Value:

```
#X
```

Definition at line 463 of file CMakeCXXCompilerId.cpp.

4.19.2 Function Documentation

4.19.2.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 924 of file CMakeCXXCompilerId.cpp.

4.19.3 Variable Documentation

4.19.3.1 info_arch

```
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]"
```

Definition at line 835 of file CMakeCXXCompilerId.cpp.

4.19.3.2 info_compiler

```
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]"
```

Definition at line 450 of file CMakeCXXCompilerId.cpp.

4.19.3.3 info_language_extensions_default

```
const char* info_language_extensions_default
```

Initial value:

```
                                        = "INFO" ":" "extensions_default["



  "OFF"
"]"
```

Definition at line 912 of file CMakeCXXCompilerId.cpp.

4.19.3.4 info_language_standard_default

```
const char* info_language_standard_default
```

Initial value:

```
= "INFO" ":" "standard_default["
```

```
  "98"
"]"
```

Definition at line 894 of file CMakeCXXCompilerId.cpp.

4.19.3.5 info_platform

```
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]"
```

Definition at line 834 of file CMakeCXXCompilerId.cpp.

4.20 CMakeCXXCompilerId.cpp

Go to the documentation of this file.

```
00001 /* This source file must have a .cpp extension so that all C++ compilers
00002    recognize the extension without flags.  Borland does not know .cxx for
00003    example.  */
00004 #ifndef __cplusplus
00005 # error "A C compiler has been selected for C++."
00006 #endif
00007
00008 #if !defined(__has_include)
00009 /* If the compiler does not have __has_include, pretend the answer is
00010    always no.  */
00011 #  define __has_include(x) 0
00012 #endif
00013
00014
00015 /* Version number components: V=Version, R=Revision, P=Patch
00016    Version date components:   YYYY=Year, MM=Month,   DD=Day  */
00017
00018 #if defined(__INTEL_COMPILER) || defined(__ICC)
00019 # define COMPILER_ID "Intel"
00020 # if defined(_MSC_VER)
00021 #  define SIMULATE_ID "MSVC"
00022 # endif
00023 # if defined(__GNUC__)
00024 #  define SIMULATE_ID "GNU"
00025 # endif
00026   /* __INTEL_COMPILER = VRP prior to 2021, and then VVVV for 2021 and later,
00027      except that a few beta releases use the old format with V=2021.  */
00028 # if __INTEL_COMPILER < 2021 || __INTEL_COMPILER == 202110 || __INTEL_COMPILER == 202111
00029 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER/100)
00030 #  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER/10 % 10)
00031 #  if defined(__INTEL_COMPILER_UPDATE)
00032 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER_UPDATE)
00033 #  else
00034 #   define COMPILER_VERSION_PATCH DEC(__INTEL_COMPILER   % 10)
00035 #  endif
00036 # else
00037 #  define COMPILER_VERSION_MAJOR DEC(__INTEL_COMPILER)
```

```
#  define COMPILER_VERSION_MINOR DEC(__INTEL_COMPILER_UPDATE)
   /* The third version component from --version is an update index,
      but no macro is provided for it.  */
#  define COMPILER_VERSION_PATCH DEC(0)
# endif
# if defined(__INTEL_COMPILER_BUILD_DATE)
   /* __INTEL_COMPILER_BUILD_DATE = YYYYMMDD */
#  define COMPILER_VERSION_TWEAK DEC(__INTEL_COMPILER_BUILD_DATE)
# endif
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# if defined(__GNUC__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
# elif defined(__GNUG__)
#  define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif (defined(__clang__) && defined(__INTEL_CLANG_COMPILER)) || defined(__INTEL_LLVM_COMPILER)
# define COMPILER_ID "IntelLLVM"
#if defined(_MSC_VER)
# define SIMULATE_ID "MSVC"
#endif
#if defined(__GNUC__)
# define SIMULATE_ID "GNU"
#endif
/* __INTEL_LLVM_COMPILER = VVVVRP prior to 2021.2.0, VVVVRRPP for 2021.2.0 and
 * later.  Look for 6 digit vs. 8 digit version number to decide encoding.
 * VVVV is no smaller than the current year when a version is released.
 */
#if __INTEL_LLVM_COMPILER < 1000000L
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/100)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER    % 10)
#else
# define COMPILER_VERSION_MAJOR DEC(__INTEL_LLVM_COMPILER/10000)
# define COMPILER_VERSION_MINOR DEC(__INTEL_LLVM_COMPILER/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__INTEL_LLVM_COMPILER     % 100)
#endif
#if defined(_MSC_VER)
  /* _MSC_VER = VVRR */
# define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
# define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
#endif
#if defined(__GNUC__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#elif defined(__GNUG__)
# define SIMULATE_VERSION_MAJOR DEC(__GNUG__)
#endif
#if defined(__GNUC_MINOR__)
# define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#endif
#if defined(__GNUC_PATCHLEVEL__)
# define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#endif

#elif defined(__PATHCC__)
# define COMPILER_ID "PathScale"
# define COMPILER_VERSION_MAJOR DEC(__PATHCC__)
# define COMPILER_VERSION_MINOR DEC(__PATHCC_MINOR__)
# if defined(__PATHCC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PATHCC_PATCHLEVEL__)
# endif

#elif defined(__BORLANDC__) && defined(__CODEGEARC_VERSION__)
# define COMPILER_ID "Embarcadero"
# define COMPILER_VERSION_MAJOR HEX(__CODEGEARC_VERSION__»24 & 0x00FF)
# define COMPILER_VERSION_MINOR HEX(__CODEGEARC_VERSION__»16 & 0x00FF)
# define COMPILER_VERSION_PATCH DEC(__CODEGEARC_VERSION__     & 0xFFFF)

#elif defined(__BORLANDC__)
# define COMPILER_ID "Borland"
  /* __BORLANDC__ = 0xVRR */
# define COMPILER_VERSION_MAJOR HEX(__BORLANDC__»8)
# define COMPILER_VERSION_MINOR HEX(__BORLANDC__ & 0xFF)

#elif defined(__WATCOMC__) && __WATCOMC__ < 1200
# define COMPILER_ID "Watcom"
   /* __WATCOMC__ = VVRR */
```

```
# define COMPILER_VERSION_MAJOR DEC(__WATCOMC__ / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__WATCOMC__)
# define COMPILER_ID "OpenWatcom"
   /* __WATCOMC__ = VVRP + 1100 */
# define COMPILER_VERSION_MAJOR DEC((__WATCOMC__ - 1100) / 100)
# define COMPILER_VERSION_MINOR DEC((__WATCOMC__ / 10) % 10)
# if (__WATCOMC__ % 10) > 0
#  define COMPILER_VERSION_PATCH DEC(__WATCOMC__ % 10)
# endif

#elif defined(__SUNPRO_CC)
# define COMPILER_ID "SunPro"
# if __SUNPRO_CC >= 0x5100
   /* __SUNPRO_CC = 0xVRRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»12)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xFF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# else
   /* __SUNPRO_CC = 0xVRP */
#  define COMPILER_VERSION_MAJOR HEX(__SUNPRO_CC»8)
#  define COMPILER_VERSION_MINOR HEX(__SUNPRO_CC»4 & 0xF)
#  define COMPILER_VERSION_PATCH HEX(__SUNPRO_CC    & 0xF)
# endif

#elif defined(__HP_aCC)
# define COMPILER_ID "HP"
  /* __HP_aCC = VVRRPP */
# define COMPILER_VERSION_MAJOR DEC(__HP_aCC/10000)
# define COMPILER_VERSION_MINOR DEC(__HP_aCC/100 % 100)
# define COMPILER_VERSION_PATCH DEC(__HP_aCC     % 100)

#elif defined(__DECCXX)
# define COMPILER_ID "Compaq"
  /* __DECCXX_VER = VVRRTPPPP */
# define COMPILER_VERSION_MAJOR DEC(__DECCXX_VER/10000000)
# define COMPILER_VERSION_MINOR DEC(__DECCXX_VER/100000  % 100)
# define COMPILER_VERSION_PATCH DEC(__DECCXX_VER         % 10000)

#elif defined(__IBMCPP__) && defined(__COMPILER_VER__)
# define COMPILER_ID "zOS"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__open_xl__) && defined(__clang__)
# define COMPILER_ID "IBMClang"
# define COMPILER_VERSION_MAJOR DEC(__open_xl_version__)
# define COMPILER_VERSION_MINOR DEC(__open_xl_release__)
# define COMPILER_VERSION_PATCH DEC(__open_xl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__open_xl_ptf_fix_level__)
# define COMPILER_VERSION_INTERNAL_STR  __clang_version__


#elif defined(__ibmxl__) && defined(__clang__)
# define COMPILER_ID "XLClang"
# define COMPILER_VERSION_MAJOR DEC(__ibmxl_version__)
# define COMPILER_VERSION_MINOR DEC(__ibmxl_release__)
# define COMPILER_VERSION_PATCH DEC(__ibmxl_modification__)
# define COMPILER_VERSION_TWEAK DEC(__ibmxl_ptf_fix_level__)


#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ >= 800
# define COMPILER_ID "XL"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__IBMCPP__) && !defined(__COMPILER_VER__) && __IBMCPP__ < 800
# define COMPILER_ID "VisualAge"
  /* __IBMCPP__ = VRP */
# define COMPILER_VERSION_MAJOR DEC(__IBMCPP__/100)
# define COMPILER_VERSION_MINOR DEC(__IBMCPP__/10 % 10)
# define COMPILER_VERSION_PATCH DEC(__IBMCPP__    % 10)

#elif defined(__NVCOMPILER)
# define COMPILER_ID "NVHPC"
# define COMPILER_VERSION_MAJOR DEC(__NVCOMPILER_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__NVCOMPILER_MINOR__)
# if defined(__NVCOMPILER_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__NVCOMPILER_PATCHLEVEL__)
```

```
# endif

#elif defined(__PGI)
# define COMPILER_ID "PGI"
# define COMPILER_VERSION_MAJOR DEC(__PGIC__)
# define COMPILER_VERSION_MINOR DEC(__PGIC_MINOR__)
# if defined(__PGIC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__PGIC_PATCHLEVEL__)
# endif

#elif defined(__clang__) && defined(__cray__)
# define COMPILER_ID "CrayClang"
# define COMPILER_VERSION_MAJOR DEC(__cray_major__)
# define COMPILER_VERSION_MINOR DEC(__cray_minor__)
# define COMPILER_VERSION_PATCH DEC(__cray_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(_CRAYC)
# define COMPILER_ID "Cray"
# define COMPILER_VERSION_MAJOR DEC(_RELEASE_MAJOR)
# define COMPILER_VERSION_MINOR DEC(_RELEASE_MINOR)

#elif defined(__TI_COMPILER_VERSION__)
# define COMPILER_ID "TI"
  /* __TI_COMPILER_VERSION__ = VVVRRRPPP */
# define COMPILER_VERSION_MAJOR DEC(__TI_COMPILER_VERSION__/1000000)
# define COMPILER_VERSION_MINOR DEC(__TI_COMPILER_VERSION__/1000   % 1000)
# define COMPILER_VERSION_PATCH DEC(__TI_COMPILER_VERSION__        % 1000)

#elif defined(__CLANG_FUJITSU)
# define COMPILER_ID "FujitsuClang"
# define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
# define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
# define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# define COMPILER_VERSION_INTERNAL_STR __clang_version__


#elif defined(__FUJITSU)
# define COMPILER_ID "Fujitsu"
# if defined(__FCC_version__)
#   define COMPILER_VERSION __FCC_version__
# elif defined(__FCC_major__)
#   define COMPILER_VERSION_MAJOR DEC(__FCC_major__)
#   define COMPILER_VERSION_MINOR DEC(__FCC_minor__)
#   define COMPILER_VERSION_PATCH DEC(__FCC_patchlevel__)
# endif
# if defined(__fcc_version)
#   define COMPILER_VERSION_INTERNAL DEC(__fcc_version)
# elif defined(__FCC_VERSION)
#   define COMPILER_VERSION_INTERNAL DEC(__FCC_VERSION)
# endif


#elif defined(__ghs__)
# define COMPILER_ID "GHS"
/* __GHS_VERSION_NUMBER = VVVVRP */
# ifdef __GHS_VERSION_NUMBER
# define COMPILER_VERSION_MAJOR DEC(__GHS_VERSION_NUMBER / 100)
# define COMPILER_VERSION_MINOR DEC(__GHS_VERSION_NUMBER / 10 % 10)
# define COMPILER_VERSION_PATCH DEC(__GHS_VERSION_NUMBER      % 10)
# endif

#elif defined(__TASKING__)
# define COMPILER_ID "Tasking"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION__/1000)
  # define COMPILER_VERSION_MINOR DEC(__VERSION__ % 100)
# define COMPILER_VERSION_INTERNAL DEC(__VERSION__)

#elif defined(__ORANGEC__)
# define COMPILER_ID "OrangeC"
# define COMPILER_VERSION_MAJOR DEC(__ORANGEC_MAJOR__)
# define COMPILER_VERSION_MINOR DEC(__ORANGEC_MINOR__)
# define COMPILER_VERSION_PATCH DEC(__ORANGEC_PATCHLEVEL__)

#elif defined(__RENESAS__)
# define COMPILER_ID "Renesas"
/* __RENESAS_VERSION__ = 0xVVRRPP00 */
# define COMPILER_VERSION_MAJOR HEX(__RENESAS_VERSION__ » 24 & 0xFF)
# define COMPILER_VERSION_MINOR HEX(__RENESAS_VERSION__ » 16 & 0xFF)
# define COMPILER_VERSION_PATCH HEX(__RENESAS_VERSION__ » 8  & 0xFF)

#elif defined(__SCO_VERSION__)
# define COMPILER_ID "SCO"

#elif defined(__ARMCC_VERSION) && !defined(__clang__)
# define COMPILER_ID "ARMCC"
```

```
#if __ARMCC_VERSION >= 1000000
  /* __ARMCC_VERSION = VRRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION     % 10000)
#else
  /* __ARMCC_VERSION = VRPPPP */
  # define COMPILER_VERSION_MAJOR DEC(__ARMCC_VERSION/100000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCC_VERSION/10000 % 10)
  # define COMPILER_VERSION_PATCH DEC(__ARMCC_VERSION    % 10000)
#endif


#elif defined(__clang__) && defined(__apple_build_version__)
# define COMPILER_ID "AppleClang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif
# define COMPILER_VERSION_TWEAK DEC(__apple_build_version__)

#elif defined(__clang__) && defined(__ARMCOMPILER_VERSION)
# define COMPILER_ID "ARMClang"
  # define COMPILER_VERSION_MAJOR DEC(__ARMCOMPILER_VERSION/1000000)
  # define COMPILER_VERSION_MINOR DEC(__ARMCOMPILER_VERSION/10000 % 100)
  # define COMPILER_VERSION_PATCH DEC(__ARMCOMPILER_VERSION/100   % 100)
# define COMPILER_VERSION_INTERNAL DEC(__ARMCOMPILER_VERSION)

#elif defined(__clang__) && defined(__ti__)
# define COMPILER_ID "TIClang"
  # define COMPILER_VERSION_MAJOR DEC(__ti_major__)
  # define COMPILER_VERSION_MINOR DEC(__ti_minor__)
  # define COMPILER_VERSION_PATCH DEC(__ti_patchlevel__)
# define COMPILER_VERSION_INTERNAL DEC(__ti_version__)

#elif defined(__clang__)
# define COMPILER_ID "Clang"
# if defined(_MSC_VER)
#  define SIMULATE_ID "MSVC"
# endif
# define COMPILER_VERSION_MAJOR DEC(__clang_major__)
# define COMPILER_VERSION_MINOR DEC(__clang_minor__)
# define COMPILER_VERSION_PATCH DEC(__clang_patchlevel__)
# if defined(_MSC_VER)
   /* _MSC_VER = VVRR */
#  define SIMULATE_VERSION_MAJOR DEC(_MSC_VER / 100)
#  define SIMULATE_VERSION_MINOR DEC(_MSC_VER % 100)
# endif

#elif defined(__LCC__) && (defined(__GNUC__) || defined(__GNUG__) || defined(__MCST__))
# define COMPILER_ID "LCC"
# define COMPILER_VERSION_MAJOR DEC(__LCC__ / 100)
# define COMPILER_VERSION_MINOR DEC(__LCC__ % 100)
# if defined(__LCC_MINOR__)
#  define COMPILER_VERSION_PATCH DEC(__LCC_MINOR__)
# endif
# if defined(__GNUC__) && defined(__GNUC_MINOR__)
#  define SIMULATE_ID "GNU"
#  define SIMULATE_VERSION_MAJOR DEC(__GNUC__)
#  define SIMULATE_VERSION_MINOR DEC(__GNUC_MINOR__)
#  if defined(__GNUC_PATCHLEVEL__)
#   define SIMULATE_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
#  endif
# endif

#elif defined(__GNUC__) || defined(__GNUG__)
# define COMPILER_ID "GNU"
# if defined(__GNUC__)
#  define COMPILER_VERSION_MAJOR DEC(__GNUC__)
# else
#  define COMPILER_VERSION_MAJOR DEC(__GNUG__)
# endif
# if defined(__GNUC_MINOR__)
#  define COMPILER_VERSION_MINOR DEC(__GNUC_MINOR__)
# endif
# if defined(__GNUC_PATCHLEVEL__)
#  define COMPILER_VERSION_PATCH DEC(__GNUC_PATCHLEVEL__)
# endif

#elif defined(_MSC_VER)
```

```
# define COMPILER_ID "MSVC"
  /* _MSC_VER = VVRR */
# define COMPILER_VERSION_MAJOR DEC(_MSC_VER / 100)
# define COMPILER_VERSION_MINOR DEC(_MSC_VER % 100)
# if defined(_MSC_FULL_VER)
#  if _MSC_VER >= 1400
    /* _MSC_FULL_VER = VVRRPPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 100000)
#  else
    /* _MSC_FULL_VER = VVRRPPPP */
#   define COMPILER_VERSION_PATCH DEC(_MSC_FULL_VER % 10000)
#  endif
# endif
# if defined(_MSC_BUILD)
#  define COMPILER_VERSION_TWEAK DEC(_MSC_BUILD)
# endif

#elif defined(_ADI_COMPILER)
# define COMPILER_ID "ADSP"
#if defined(__VERSIONNUM__)
  /* __VERSIONNUM__ = 0xVVRRPTT */
#  define COMPILER_VERSION_MAJOR DEC(__VERSIONNUM__ » 24 & 0xFF)
#  define COMPILER_VERSION_MINOR DEC(__VERSIONNUM__ » 16 & 0xFF)
#  define COMPILER_VERSION_PATCH DEC(__VERSIONNUM__ » 8 & 0xFF)
#  define COMPILER_VERSION_TWEAK DEC(__VERSIONNUM__ & 0xFF)
#endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# define COMPILER_ID "IAR"
# if defined(__VER__) && defined(__ICCARM__)
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 1000000)
#  define COMPILER_VERSION_MINOR DEC(((__VER__) / 1000) % 1000)
#  define COMPILER_VERSION_PATCH DEC((__VER__) % 1000)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# elif defined(__VER__) && (defined(__ICCAVR__) || defined(__ICCRX__) || defined(__ICCRH850__) ||
  defined(__ICCRL78__) || defined(__ICC430__) || defined(__ICCRISCV__) || defined(__ICCV850__) ||
  defined(__ICC8051__) || defined(__ICCSTM8__))
#  define COMPILER_VERSION_MAJOR DEC((__VER__) / 100)
#  define COMPILER_VERSION_MINOR DEC((__VER__) - (((__VER__) / 100)*100))
#  define COMPILER_VERSION_PATCH DEC(__SUBVERSION__)
#  define COMPILER_VERSION_INTERNAL DEC(__IAR_SYSTEMS_ICC__)
# endif

#elif defined(__DCC__) && defined(_DIAB_TOOL)
# define COMPILER_ID "Diab"
  # define COMPILER_VERSION_MAJOR DEC(__VERSION_MAJOR_NUMBER__)
  # define COMPILER_VERSION_MINOR DEC(__VERSION_MINOR_NUMBER__)
  # define COMPILER_VERSION_PATCH DEC(__VERSION_ARCH_FEATURE_NUMBER__)
  # define COMPILER_VERSION_TWEAK DEC(__VERSION_BUG_FIX_NUMBER__)



/* These compilers are either not known or too old to define an
  identification macro.  Try to identify the platform and guess that
  it is the native compiler.  */
#elif defined(__hpux) || defined(__hpua)
# define COMPILER_ID "HP"

#else /* unknown compiler */
# define COMPILER_ID ""
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_compiler = "INFO" ":" "compiler[" COMPILER_ID "]";
#ifdef SIMULATE_ID
char const* info_simulate = "INFO" ":" "simulate[" SIMULATE_ID "]";
#endif

#ifdef __QNXNTO__
char const* qnxnto = "INFO" ":" "qnxnto[]";
#endif

#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
char const *info_cray = "INFO" ":" "compiler_wrapper[CrayPrgEnv]";
#endif

#define STRINGIFY_HELPER(X) #X
#define STRINGIFY(X) STRINGIFY_HELPER(X)

/* Identify known platforms by name.  */
#if defined(__linux) || defined(__linux__) || defined(linux)
# define PLATFORM_ID "Linux"

#elif defined(__MSYS__)
```

```
# define PLATFORM_ID "MSYS"

#elif defined(__CYGWIN__)
# define PLATFORM_ID "Cygwin"

#elif defined(__MINGW32__)
# define PLATFORM_ID "MinGW"

#elif defined(__APPLE__)
# define PLATFORM_ID "Darwin"

#elif defined(_WIN32) || defined(__WIN32__) || defined(WIN32)
# define PLATFORM_ID "Windows"

#elif defined(__FreeBSD__) || defined(__FreeBSD)
# define PLATFORM_ID "FreeBSD"

#elif defined(__NetBSD__) || defined(__NetBSD)
# define PLATFORM_ID "NetBSD"

#elif defined(__OpenBSD__) || defined(__OPENBSD)
# define PLATFORM_ID "OpenBSD"

#elif defined(__sun) || defined(sun)
# define PLATFORM_ID "SunOS"

#elif defined(_AIX) || defined(__AIX) || defined(__AIX__) || defined(__aix) || defined(__aix__)
# define PLATFORM_ID "AIX"

#elif defined(__hpux) || defined(__hpux__)
# define PLATFORM_ID "HP-UX"

#elif defined(__HAIKU__)
# define PLATFORM_ID "Haiku"

#elif defined(__BeOS) || defined(__BEOS__) || defined(_BEOS)
# define PLATFORM_ID "BeOS"

#elif defined(__QNX__) || defined(__QNXNTO__)
# define PLATFORM_ID "QNX"

#elif defined(__tru64) || defined(_tru64) || defined(__TRU64__)
# define PLATFORM_ID "Tru64"

#elif defined(__riscos) || defined(__riscos__)
# define PLATFORM_ID "RISCos"

#elif defined(__sinix) || defined(__sinix__) || defined(__SINIX__)
# define PLATFORM_ID "SINIX"

#elif defined(__UNIX_SV__)
# define PLATFORM_ID "UNIX_SV"

#elif defined(__bsdos__)
# define PLATFORM_ID "BSDOS"

#elif defined(_MPRAS) || defined(MPRAS)
# define PLATFORM_ID "MP-RAS"

#elif defined(__osf) || defined(__osf__)
# define PLATFORM_ID "OSF1"

#elif defined(_SCO_SV) || defined(SCO_SV) || defined(sco_sv)
# define PLATFORM_ID "SCO_SV"

#elif defined(__ultrix) || defined(__ultrix__) || defined(_ULTRIX)
# define PLATFORM_ID "ULTRIX"

#elif defined(__XENIX__) || defined(_XENIX) || defined(XENIX)
# define PLATFORM_ID "Xenix"

#elif defined(__WATCOMC__)
# if defined(__LINUX__)
#  define PLATFORM_ID "Linux"

# elif defined(__DOS__)
#  define PLATFORM_ID "DOS"

# elif defined(__OS2__)
#  define PLATFORM_ID "OS2"

# elif defined(__WINDOWS__)
#  define PLATFORM_ID "Windows3x"

# elif defined(__VXWORKS__)
#  define PLATFORM_ID "VxWorks"

```

```
# else /* unknown platform */
#  define PLATFORM_ID
# endif

#elif defined(__INTEGRITY)
# if defined(INT_178B)
#  define PLATFORM_ID "Integrity178"

# else /* regular Integrity */
#  define PLATFORM_ID "Integrity"
# endif

# elif defined(_ADI_COMPILER)
#  define PLATFORM_ID "ADSP"

#else /* unknown platform */
# define PLATFORM_ID

#endif

/* For windows compilers MSVC and Intel we can determine
   the architecture of the compiler being used.  This is because
   the compilers do not have flags that can change the architecture,
   but rather depend on which compiler is being used
*/
#if defined(_WIN32) && defined(_MSC_VER)
# if defined(_M_IA64)
#  define ARCHITECTURE_ID "IA64"

# elif defined(_M_ARM64EC)
#  define ARCHITECTURE_ID "ARM64EC"

# elif defined(_M_X64) || defined(_M_AMD64)
#  define ARCHITECTURE_ID "x64"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# elif defined(_M_ARM64)
#  define ARCHITECTURE_ID "ARM64"

# elif defined(_M_ARM)
#  if _M_ARM == 4
#   define ARCHITECTURE_ID "ARMV4I"
#  elif _M_ARM == 5
#   define ARCHITECTURE_ID "ARMV5I"
#  else
#   define ARCHITECTURE_ID "ARMV" STRINGIFY(_M_ARM)
#  endif

# elif defined(_M_MIPS)
#  define ARCHITECTURE_ID "MIPS"

# elif defined(_M_SH)
#  define ARCHITECTURE_ID "SHx"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__WATCOMC__)
# if defined(_M_I86)
#  define ARCHITECTURE_ID "I86"

# elif defined(_M_IX86)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__IAR_SYSTEMS_ICC__) || defined(__IAR_SYSTEMS_ICC)
# if defined(__ICCARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__ICCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__ICCRH850__)
#  define ARCHITECTURE_ID "RH850"

# elif defined(__ICCRL78__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__ICCRISCV__)
#  define ARCHITECTURE_ID "RISCV"

```

```
# elif defined(__ICCAVR__)
#  define ARCHITECTURE_ID "AVR"

# elif defined(__ICC430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__ICCV850__)
#  define ARCHITECTURE_ID "V850"

# elif defined(__ICC8051__)
#  define ARCHITECTURE_ID "8051"

# elif defined(__ICCSTM8__)
#  define ARCHITECTURE_ID "STM8"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__ghs__)
# if defined(__PPC64__)
#  define ARCHITECTURE_ID "PPC64"

# elif defined(__ppc__)
#  define ARCHITECTURE_ID "PPC"

# elif defined(__ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__x86_64__)
#  define ARCHITECTURE_ID "x64"

# elif defined(__i386__)
#  define ARCHITECTURE_ID "X86"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__clang__) && defined(__ti__)
# if defined(__ARM_ARCH)
#  define ARCHITECTURE_ID "ARM"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__TI_COMPILER_VERSION__)
# if defined(__TI_ARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__MSP430__)
#  define ARCHITECTURE_ID "MSP430"

# elif defined(__TMS320C28XX__)
#  define ARCHITECTURE_ID "TMS320C28x"

# elif defined(__TMS320C6X__) || defined(_TMS320C6X)
#  define ARCHITECTURE_ID "TMS320C6x"

# else /* unknown architecture */
#  define ARCHITECTURE_ID ""
# endif

# elif defined(__ADSPSHARC__)
#  define ARCHITECTURE_ID "SHARC"

# elif defined(__ADSPBLACKFIN__)
#  define ARCHITECTURE_ID "Blackfin"

#elif defined(__TASKING__)

# if defined(__CTC__) || defined(__CPTC__)
#  define ARCHITECTURE_ID "TriCore"

# elif defined(__CMCS__)
#  define ARCHITECTURE_ID "MCS"

# elif defined(__CARM__) || defined(__CPARM__)
#  define ARCHITECTURE_ID "ARM"

# elif defined(__CARC__)
#  define ARCHITECTURE_ID "ARC"

# elif defined(__C51__)
#  define ARCHITECTURE_ID "8051"

```

```
# elif defined(__CPCP__)
#  define ARCHITECTURE_ID "PCP"

# else
#  define ARCHITECTURE_ID ""
# endif

#elif defined(__RENESAS__)
# if defined(__CCRX__)
#  define ARCHITECTURE_ID "RX"

# elif defined(__CCRL__)
#  define ARCHITECTURE_ID "RL78"

# elif defined(__CCRH__)
#  define ARCHITECTURE_ID "RH850"

# else
#  define ARCHITECTURE_ID ""
# endif

#else
#  define ARCHITECTURE_ID
#endif

/* Convert integer to decimal digit literals.  */
#define DEC(n)                    \
  ('0' + (((n) / 10000000)%10)), \
  ('0' + (((n) / 1000000)%10)),  \
  ('0' + (((n) / 100000)%10)),   \
  ('0' + (((n) / 10000)%10)),    \
  ('0' + (((n) / 1000)%10)),     \
  ('0' + (((n) / 100)%10)),      \
  ('0' + (((n) / 10)%10)),       \
  ('0' +  ((n) % 10))

/* Convert integer to hex digit literals.  */
#define HEX(n)             \
  ('0' + ((n)>>28 & 0xF)), \
  ('0' + ((n)>>24 & 0xF)), \
  ('0' + ((n)>>20 & 0xF)), \
  ('0' + ((n)>>16 & 0xF)), \
  ('0' + ((n)>>12 & 0xF)), \
  ('0' + ((n)>>8  & 0xF)), \
  ('0' + ((n)>>4  & 0xF)), \
  ('0' + ((n)     & 0xF))

/* Construct a string literal encoding the version number. */
#ifdef COMPILER_VERSION
char const* info_version = "INFO" ":" "compiler_version[" COMPILER_VERSION "]";

/* Construct a string literal encoding the version number components. */
#elif defined(COMPILER_VERSION_MAJOR)
char const info_version[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','[',
  COMPILER_VERSION_MAJOR,
# ifdef COMPILER_VERSION_MINOR
  '.', COMPILER_VERSION_MINOR,
#  ifdef COMPILER_VERSION_PATCH
   '.', COMPILER_VERSION_PATCH,
#   ifdef COMPILER_VERSION_TWEAK
    '.', COMPILER_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct a string literal encoding the internal version number. */
#ifdef COMPILER_VERSION_INTERNAL
char const info_version_internal[] = {
  'I', 'N', 'F', 'O', ':',
  'c','o','m','p','i','l','e','r','_','v','e','r','s','i','o','n','_',
  'i','n','t','e','r','n','a','l','[',
  COMPILER_VERSION_INTERNAL,']','\0'};
#elif defined(COMPILER_VERSION_INTERNAL_STR)
char const* info_version_internal = "INFO" ":" "compiler_version_internal["
COMPILER_VERSION_INTERNAL_STR "]";
#endif

/* Construct a string literal encoding the version number components. */
#ifdef SIMULATE_VERSION_MAJOR
char const info_simulate_version[] = {
  'I', 'N', 'F', 'O', ':',
  's','i','m','u','l','a','t','e','_','v','e','r','s','i','o','n','[',
  SIMULATE_VERSION_MAJOR,
```

```
# ifdef SIMULATE_VERSION_MINOR
  '.', SIMULATE_VERSION_MINOR,
#  ifdef SIMULATE_VERSION_PATCH
   '.', SIMULATE_VERSION_PATCH,
#   ifdef SIMULATE_VERSION_TWEAK
    '.', SIMULATE_VERSION_TWEAK,
#   endif
#  endif
# endif
  ']','\0'};
#endif

/* Construct the string literal in pieces to prevent the source from
   getting matched.  Store it in a pointer rather than an array
   because some compilers will just produce instructions to fill the
   array rather than assigning a pointer to a static array.  */
char const* info_platform = "INFO" ":" "platform[" PLATFORM_ID "]";
char const* info_arch = "INFO" ":" "arch[" ARCHITECTURE_ID "]";



#define CXX_STD_98 199711L
#define CXX_STD_11 201103L
#define CXX_STD_14 201402L
#define CXX_STD_17 201703L
#define CXX_STD_20 202002L
#define CXX_STD_23 202302L

#if defined(__INTEL_COMPILER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > CXX_STD_17
#    define CXX_STD _MSVC_LANG
#  elif _MSVC_LANG == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14 && __cplusplus > CXX_STD_17
#    define CXX_STD CXX_STD_20
#  elif _MSVC_LANG > CXX_STD_14
#    define CXX_STD CXX_STD_17
#  elif defined(__INTEL_CXX11_MODE__) && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  elif defined(__INTEL_CXX11_MODE__)
#    define CXX_STD CXX_STD_11
#  else
#    define CXX_STD CXX_STD_98
#  endif
#elif defined(_MSC_VER) && defined(_MSVC_LANG)
#  if _MSVC_LANG > __cplusplus
#    define CXX_STD _MSVC_LANG
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__NVCOMPILER)
#  if __cplusplus == CXX_STD_17 && defined(__cpp_aggregate_paren_init)
#    define CXX_STD CXX_STD_20
#  else
#    define CXX_STD __cplusplus
#  endif
#elif defined(__INTEL_COMPILER) || defined(__PGI)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_namespace_attributes)
#    define CXX_STD CXX_STD_17
#  elif __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif (defined(__IBMCPP__) || defined(__ibmxl__)) && defined(__linux__)
#  if __cplusplus == CXX_STD_11 && defined(__cpp_aggregate_nsdmi)
#    define CXX_STD CXX_STD_14
#  else
#    define CXX_STD __cplusplus
#  endif
#elif __cplusplus == 1 && defined(__GXX_EXPERIMENTAL_CXX0X__)
#  define CXX_STD CXX_STD_11
#else
#  define CXX_STD __cplusplus
#endif

const char* info_language_standard_default = "INFO" ":" "standard_default["
#if CXX_STD > CXX_STD_23
  "26"
#elif CXX_STD > CXX_STD_20
  "23"
#elif CXX_STD > CXX_STD_17
  "20"
#elif CXX_STD > CXX_STD_14
  "17"
#elif CXX_STD > CXX_STD_11
  "14"
```

```
#elif CXX_STD >= CXX_STD_11
  "11"
#else
  "98"
#endif
"]";

const char* info_language_extensions_default = "INFO" ":" "extensions_default["
#if (defined(__clang__) || defined(__GNUC__) || defined(__xlC__) ||            \
     defined(__TI_COMPILER_VERSION__) || defined(__RENESAS__)) &&              \
  !defined(__STRICT_ANSI__)
  "ON"
#else
  "OFF"
#endif
"]";

/*------------------------------------------------------------------------*/

int main(int argc, char* argv[])
{
  int require = 0;
  require += info_compiler[argc];
  require += info_platform[argc];
  require += info_arch[argc];
#ifdef COMPILER_VERSION_MAJOR
  require += info_version[argc];
#endif
#if defined(COMPILER_VERSION_INTERNAL) || defined(COMPILER_VERSION_INTERNAL_STR)
  require += info_version_internal[argc];
#endif
#ifdef SIMULATE_ID
  require += info_simulate[argc];
#endif
#ifdef SIMULATE_VERSION_MAJOR
  require += info_simulate_version[argc];
#endif
#if defined(__CRAYXT_COMPUTE_LINUX_TARGET)
  require += info_cray[argc];
#endif
  require += info_language_standard_default[argc];
  require += info_language_extensions_default[argc];
  (void)argv;
  return require;
}
```

4.21 build_O1/CMakeFiles/programa.dir/src/calculations.cpp.obj.d File Reference

4.22 calculations.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/calculations.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\calculations.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
C:/msys64/mingw64/include/c++/15.2.0/bit \
C:/msys64/mingw64/include/c++/15.2.0/concepts \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/new \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/random \
C:/msys64/mingw64/include/c++/15.2.0/cmath \
C:/msys64/mingw64/include/math.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/specfun.h \
C:/msys64/mingw64/include/c++/15.2.0/limits \
C:/msys64/mingw64/include/c++/15.2.0/tr1/gamma.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/special_function_util.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/bessel_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/beta_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/ell_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/exp_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/hypergeometric.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/legendre_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/modified_bessel_func.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_hermite.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_laguerre.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/riemann_zeta.tcc \
C:/msys64/mingw64/include/c++/15.2.0/cstdint \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stdint.h \
C:/msys64/mingw64/include/stdint.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/opt_random.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/pmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/emmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/xmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mwaitintrin.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.tcc \
C:/msys64/mingw64/include/c++/15.2.0/numeric \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_numeric.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_numeric_defs.h
```

4.23 build_O2/CMakeFiles/programa.dir/src/calculations.cpp.obj.d File Reference

4.24 calculations.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/calculations.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\calculations.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/random \
C:/msys64/mingw64/include/c++/15.2.0/cmath \
C:/msys64/mingw64/include/math.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/specfun.h \
C:/msys64/mingw64/include/c++/15.2.0/limits \
C:/msys64/mingw64/include/c++/15.2.0/tr1/gamma.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/special_function_util.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/bessel_function.tcc \
```

```
C:/msys64/mingw64/include/c++/15.2.0/tr1/beta_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/ell_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/exp_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/hypergeometric.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/legendre_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/modified_bessel_func.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_hermite.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_laguerre.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/riemann_zeta.tcc \
C:/msys64/mingw64/include/c++/15.2.0/cstdint \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stdint.h \
C:/msys64/mingw64/include/stdint.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/opt_random.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/pmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/emmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/xmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mwaitintrin.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.tcc \
C:/msys64/mingw64/include/c++/15.2.0/numeric \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_numeric.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_numeric_defs.h
```

4.25 build_O3/CMakeFiles/programa.dir/src/calculations.cpp.obj.d File Reference

4.26 calculations.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/calculations.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\calculations.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
 C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/random \
C:/msys64/mingw64/include/c++/15.2.0/cmath \
C:/msys64/mingw64/include/math.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/specfun.h \
C:/msys64/mingw64/include/c++/15.2.0/limits \
C:/msys64/mingw64/include/c++/15.2.0/tr1/gamma.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/special_function_util.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/bessel_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/beta_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/ell_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/exp_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/hypergeometric.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/legendre_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/modified_bessel_func.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_hermite.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_laguerre.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/riemann_zeta.tcc \
C:/msys64/mingw64/include/c++/15.2.0/cstdint \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stdint.h \
```

```
C:/msys64/mingw64/include/stdint.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/opt_random.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/pmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/emmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/xmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mwaitintrin.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.tcc \
C:/msys64/mingw64/include/c++/15.2.0/numeric \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_numeric.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_numeric_defs.h
```

4.27 build_O1/CMakeFiles/programa.dir/src/fileHandling.cpp.obj.d File Reference

4.28 fileHandling.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/fileHandling.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\fileHandling.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/fileHandling.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
 C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
 C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
 C:/msys64/mingw64/include/c++/15.2.0/cstddef \
 C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
 C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
 C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
 C:/msys64/mingw64/include/_mingw_mac.h \
```

```
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
```

```
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h
```

4.29 build_O2/CMakeFiles/programa.dir/src/fileHandling.cpp.obj.d File Reference

4.30 fileHandling.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/fileHandling.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\fileHandling.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/fileHandling.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h
```

4.31 build_O3/CMakeFiles/programa.dir/src/fileHandling.cpp.obj.d File Reference

4.32 fileHandling.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/fileHandling.cpp.obj: \
  C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\fileHandling.cpp \
  C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/fileHandling.h \
  C:/msys64/mingw64/include/c++/15.2.0/vector \
  C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
  C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
  C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
  C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
  C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
  C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
  C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
  C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
  C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
C:/msys64/mingw64/include/c++/15.2.0/type_traits \
C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
C:/msys64/mingw64/include/c++/15.2.0/bit \
C:/msys64/mingw64/include/c++/15.2.0/concepts \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/new \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
```

```
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h
```

4.33 build_O1/CMakeFiles/programa.dir/src/inputValid.cpp.obj.d File Reference

4.34 inputValid.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/inputValid.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\inputValid.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
 C:/msys64/mingw64/include/c++/15.2.0/string \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
 C:/msys64/mingw64/include/c++/15.2.0/cwchar \
 C:/msys64/mingw64/include/wchar.h C:/msys64/mingw64/include/corecrt.h \
 C:/msys64/mingw64/include/_mingw.h \
 C:/msys64/mingw64/include/_mingw_mac.h \
 C:/msys64/mingw64/include/_mingw_secapi.h \
 C:/msys64/mingw64/include/vadefs.h \
 C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
 C:/msys64/mingw64/include/corecrt_stdio_config.h \
 C:/msys64/mingw64/include/corecrt_wstdlib.h \
 C:/msys64/mingw64/include/corecrt_wctype.h \
 C:/msys64/mingw64/include/_mingw_off_t.h \
 C:/msys64/mingw64/include/_mingw_stat64.h \
 C:/msys64/mingw64/include/swprintf.inl \
 C:/msys64/mingw64/include/sec_api/wchar_s.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
 C:/msys64/mingw64/include/c++/15.2.0/clocale \
 C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/crtdefs.h \
 C:/msys64/mingw64/include/stdio.h \
 C:/msys64/mingw64/include/sec_api/stdio_s.h \
 C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
 C:/msys64/mingw64/include/c++/15.2.0/cctype \
 C:/msys64/mingw64/include/ctype.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
 C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
 C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc
```

4.35 build_O2/CMakeFiles/programa.dir/src/inputValid.cpp.obj.d File Reference

4.36 inputValid.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/inputValid.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\inputValid.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
 C:/msys64/mingw64/include/c++/15.2.0/string \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
 C:/msys64/mingw64/include/c++/15.2.0/cwchar \
 C:/msys64/mingw64/include/wchar.h C:/msys64/mingw64/include/corecrt.h \
 C:/msys64/mingw64/include/_mingw.h \
 C:/msys64/mingw64/include/_mingw_mac.h \
 C:/msys64/mingw64/include/_mingw_secapi.h \
 C:/msys64/mingw64/include/vadefs.h \
 C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
 C:/msys64/mingw64/include/corecrt_stdio_config.h \
 C:/msys64/mingw64/include/corecrt_wstdlib.h \
 C:/msys64/mingw64/include/corecrt_wctype.h \
 C:/msys64/mingw64/include/_mingw_off_t.h \
 C:/msys64/mingw64/include/_mingw_stat64.h \
 C:/msys64/mingw64/include/swprintf.inl \
 C:/msys64/mingw64/include/sec_api/wchar_s.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
 C:/msys64/mingw64/include/c++/15.2.0/clocale \
 C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/crtdefs.h \
 C:/msys64/mingw64/include/stdio.h \
 C:/msys64/mingw64/include/sec_api/stdio_s.h \
 C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
 C:/msys64/mingw64/include/c++/15.2.0/cctype \
 C:/msys64/mingw64/include/ctype.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
 C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
 C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
 C:/msys64/mingw64/include/c++/15.2.0/string_view \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
 C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
 C:/msys64/mingw64/include/stdlib.h \
 C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
 C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
 C:/msys64/mingw64/include/limits.h \
 C:/msys64/mingw64/include/sec_api/stdlib_s.h \
 C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
 C:/msys64/mingw64/include/malloc.h \
```

```
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc
```

4.37 build_O3/CMakeFiles/programa.dir/src/inputValid.cpp.obj.d File Reference

4.38 inputValid.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/inputValid.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\inputValid.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
 C:/msys64/mingw64/include/c++/15.2.0/string \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h C:/msys64/mingw64/include/corecrt.h \
C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/type_traits \
C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/new \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
C:/msys64/mingw64/include/c++/15.2.0/bit \
C:/msys64/mingw64/include/c++/15.2.0/concepts \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
```

```
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc
```

4.39 build_O1/CMakeFiles/programa.dir/src/main.cpp.obj.d File Reference

4.40 main.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/main.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\main.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/meniu.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
C:/msys64/mingw64/include/c++/15.2.0/bit \
C:/msys64/mingw64/include/c++/15.2.0/concepts \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/new \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h
```

4.41 build_O2/CMakeFiles/programa.dir/src/main.cpp.obj.d File Reference

4.42 main.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/main.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\main.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/meniu.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
 C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
 C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h
```

4.43 build_O3/CMakeFiles/programa.dir/src/main.cpp.obj.d File Reference

4.44 main.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/main.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\main.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/meniu.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
 C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
 C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
 C:/msys64/mingw64/include/c++/15.2.0/cstddef \
 C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
 C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
 C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
 C:/msys64/mingw64/include/_mingw_mac.h \
 C:/msys64/mingw64/include/_mingw_secapi.h \
 C:/msys64/mingw64/include/vadefs.h \
 C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
 C:/msys64/mingw64/include/c++/15.2.0/tuple \
 C:/msys64/mingw64/include/c++/15.2.0/list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/deque \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
 C:/msys64/mingw64/include/c++/15.2.0/string \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
 C:/msys64/mingw64/include/c++/15.2.0/cwchar \
 C:/msys64/mingw64/include/wchar.h \
 C:/msys64/mingw64/include/corecrt_stdio_config.h \
 C:/msys64/mingw64/include/corecrt_wstdlib.h \
 C:/msys64/mingw64/include/corecrt_wctype.h \
 C:/msys64/mingw64/include/_mingw_off_t.h \
 C:/msys64/mingw64/include/_mingw_stat64.h \
 C:/msys64/mingw64/include/swprintf.inl \
 C:/msys64/mingw64/include/sec_api/wchar_s.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h
```

4.45 build_O1/CMakeFiles/programa.dir/src/meniu.cpp.obj.d File Reference

4.46 meniu.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/meniu.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\meniu.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/meniu.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/fileHandling.h \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/chrono \
C:/msys64/mingw64/include/c++/15.2.0/bits/chrono.h \
C:/msys64/mingw64/include/c++/15.2.0/ratio \
C:/msys64/mingw64/include/c++/15.2.0/cstdint \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stdint.h \
C:/msys64/mingw64/include/stdint.h \
C:/msys64/mingw64/include/c++/15.2.0/limits \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/bits/parse_numbers.h \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/random \
C:/msys64/mingw64/include/c++/15.2.0/cmath \
C:/msys64/mingw64/include/math.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/specfun.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/gamma.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/special_function_util.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/bessel_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/beta_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/ell_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/exp_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/hypergeometric.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/legendre_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/modified_bessel_func.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_hermite.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_laguerre.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/riemann_zeta.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/opt_random.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/pmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/emmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/xmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mwaitintrin.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/random.tcc \
C:/msys64/mingw64/include/c++/15.2.0/numeric \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_numeric.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_numeric_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h
```

4.47 build_O2/CMakeFiles/programa.dir/src/meniu.cpp.obj.d File Reference

4.48 meniu.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/meniu.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\meniu.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/meniu.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
 C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
 C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
 C:/msys64/mingw64/include/c++/15.2.0/cstddef \
 C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
```

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C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/fileHandling.h \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
```

```
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/chrono \
C:/msys64/mingw64/include/c++/15.2.0/bits/chrono.h \
C:/msys64/mingw64/include/c++/15.2.0/ratio \
C:/msys64/mingw64/include/c++/15.2.0/cstdint \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stdint.h \
C:/msys64/mingw64/include/stdint.h \
C:/msys64/mingw64/include/c++/15.2.0/limits \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/bits/parse_numbers.h \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/random \
C:/msys64/mingw64/include/c++/15.2.0/cmath \
C:/msys64/mingw64/include/math.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/specfun.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/gamma.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/special_function_util.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/bessel_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/beta_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/ell_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/exp_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/hypergeometric.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/legendre_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/modified_bessel_func.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_hermite.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_laguerre.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/riemann_zeta.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/opt_random.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/pmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/emmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/xmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mwaitintrin.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.tcc \
C:/msys64/mingw64/include/c++/15.2.0/numeric \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_numeric.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_numeric_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h
```

4.49 build_O3/CMakeFiles/programa.dir/src/meniu.cpp.obj.d File Reference

4.50 meniu.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/meniu.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\meniu.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/meniu.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
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 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
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 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
 C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
 C:/msys64/mingw64/include/c++/15.2.0/cstddef \
 C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
 C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
 C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
 C:/msys64/mingw64/include/_mingw_mac.h \
 C:/msys64/mingw64/include/_mingw_secapi.h \
 C:/msys64/mingw64/include/vadefs.h \
 C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
 C:/msys64/mingw64/include/c++/15.2.0/tuple \
 C:/msys64/mingw64/include/c++/15.2.0/list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/deque \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
 C:/msys64/mingw64/include/c++/15.2.0/string \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
```

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C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/fileHandling.h \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/chrono \
C:/msys64/mingw64/include/c++/15.2.0/bits/chrono.h \
C:/msys64/mingw64/include/c++/15.2.0/ratio \
C:/msys64/mingw64/include/c++/15.2.0/cstdint \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stdint.h \
C:/msys64/mingw64/include/stdint.h \
C:/msys64/mingw64/include/c++/15.2.0/limits \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/bits/parse_numbers.h \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/random \
C:/msys64/mingw64/include/c++/15.2.0/cmath \
C:/msys64/mingw64/include/math.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/specfun.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/gamma.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/special_function_util.h \
C:/msys64/mingw64/include/c++/15.2.0/tr1/bessel_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/beta_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/ell_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/exp_integral.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/hypergeometric.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/legendre_function.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/modified_bessel_func.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_hermite.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/poly_laguerre.tcc \
C:/msys64/mingw64/include/c++/15.2.0/tr1/riemann_zeta.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/opt_random.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/pmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/emmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/xmmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mmintrin.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mwaitintrin.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/random.tcc \
C:/msys64/mingw64/include/c++/15.2.0/numeric \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_numeric.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_numeric_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h
```

4.51 build_O1/CMakeFiles/programa.dir/src/output.cpp.obj.d File Reference

4.52 output.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/output.cpp.obj: \
  C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\output.cpp \
  C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
  C:/msys64/mingw64/include/c++/15.2.0/vector \
  C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
  C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
  C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
C:/msys64/mingw64/include/c++/15.2.0/type_traits \
C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
C:/msys64/mingw64/include/c++/15.2.0/bit \
C:/msys64/mingw64/include/c++/15.2.0/concepts \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/new \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc
```

4.53 build_O2/CMakeFiles/programa.dir/src/output.cpp.obj.d File Reference

4.54 output.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/output.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\output.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
 C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
 C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
 C:/msys64/mingw64/include/c++/15.2.0/cstddef \
 C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
 C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
 C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
 C:/msys64/mingw64/include/_mingw_mac.h \
 C:/msys64/mingw64/include/_mingw_secapi.h \
 C:/msys64/mingw64/include/vadefs.h \
 C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
 C:/msys64/mingw64/include/c++/15.2.0/tuple \
 C:/msys64/mingw64/include/c++/15.2.0/list \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc
```

4.55 build_O3/CMakeFiles/programa.dir/src/output.cpp.obj.d File Reference

4.56 output.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/output.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\output.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/output.h \
 C:/msys64/mingw64/include/c++/15.2.0/vector \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
 C:/msys64/mingw64/include/c++/15.2.0/bit \
 C:/msys64/mingw64/include/c++/15.2.0/concepts \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/corecrt.h C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/list \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_list.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocated_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/aligned_buffer.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/list.tcc \
C:/msys64/mingw64/include/c++/15.2.0/deque \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_deque.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/deque.tcc \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/inputValid.h \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/calculations.h \
C:/msys64/mingw64/include/c++/15.2.0/iostream \
C:/msys64/mingw64/include/c++/15.2.0/ostream \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.h \
C:/msys64/mingw64/include/c++/15.2.0/ios \
C:/msys64/mingw64/include/c++/15.2.0/exception \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_ptr.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_init_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/typeinfo \
C:/msys64/mingw64/include/c++/15.2.0/bits/nested_exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ios_base.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/atomicity.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/gthr-default.h \
C:/msys64/mingw64/include/pthread.h \
C:/msys64/mingw64/include/sys/types.h \
C:/msys64/mingw64/include/process.h \
C:/msys64/mingw64/include/corecrt_startup.h \
C:/msys64/mingw64/include/signal.h \
C:/msys64/mingw64/include/pthread_signal.h \
C:/msys64/mingw64/include/time.h C:/msys64/mingw64/include/sys/timeb.h \
C:/msys64/mingw64/include/sec_api/sys/timeb_s.h \
C:/msys64/mingw64/include/_timeval.h \
C:/msys64/mingw64/include/pthread_time.h \
C:/msys64/mingw64/include/pthread_compat.h \
C:/msys64/mingw64/include/sched.h \
C:/msys64/mingw64/include/pthread_unistd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/atomic_word.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_classes.tcc \
C:/msys64/mingw64/include/c++/15.2.0/system_error \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/error_constants.h \
C:/msys64/mingw64/include/c++/15.2.0/stdexcept \
C:/msys64/mingw64/include/c++/15.2.0/streambuf \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.h \
C:/msys64/mingw64/include/c++/15.2.0/cwctype \
C:/msys64/mingw64/include/wctype.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_base.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/streambuf_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/ctype_inline.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_ios.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/istream \
C:/msys64/mingw64/include/c++/15.2.0/bits/istream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/iomanip \
C:/msys64/mingw64/include/c++/15.2.0/locale \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.h \
C:/msys64/mingw64/include/c++/15.2.0/ctime \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/time_members.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/messages_members.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/codecvt.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_facets_nonio.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/locale_conv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/quoted_string.h \
C:/msys64/mingw64/include/c++/15.2.0/sstream \
C:/msys64/mingw64/include/c++/15.2.0/bits/sstream.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/fstream \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/basic_file.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++io.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/fstream.tcc
```

4.57 build_O1/CMakeFiles/programa.dir/src/student.cpp.obj.d File Reference

4.58 student.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/student.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\student.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
 C:/msys64/mingw64/include/c++/15.2.0/string \
```

```
C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h C:/msys64/mingw64/include/corecrt.h \
C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/type_traits \
C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/new \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
C:/msys64/mingw64/include/c++/15.2.0/bit \
C:/msys64/mingw64/include/c++/15.2.0/concepts \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
```

```
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/vector \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h
```

4.59 build_O2/CMakeFiles/programa.dir/src/student.cpp.obj.d File Reference

4.60 student.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/student.cpp.obj: \
 C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\student.cpp \
 C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
 C:/msys64/mingw64/include/c++/15.2.0/string \
 C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
 C:/msys64/mingw64/include/c++/15.2.0/cwchar \
 C:/msys64/mingw64/include/wchar.h C:/msys64/mingw64/include/corecrt.h \
 C:/msys64/mingw64/include/_mingw.h \
 C:/msys64/mingw64/include/_mingw_mac.h \
 C:/msys64/mingw64/include/_mingw_secapi.h \
 C:/msys64/mingw64/include/vadefs.h \
 C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
 C:/msys64/mingw64/include/corecrt_stdio_config.h \
 C:/msys64/mingw64/include/corecrt_wstdlib.h \
 C:/msys64/mingw64/include/corecrt_wctype.h \
 C:/msys64/mingw64/include/_mingw_off_t.h \
 C:/msys64/mingw64/include/_mingw_stat64.h \
 C:/msys64/mingw64/include/swprintf.inl \
 C:/msys64/mingw64/include/sec_api/wchar_s.h \
 C:/msys64/mingw64/include/c++/15.2.0/type_traits \
 C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
 C:/msys64/mingw64/include/c++/15.2.0/new \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
 C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
 C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
 C:/msys64/mingw64/include/c++/15.2.0/clocale \
 C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/crtdefs.h \
 C:/msys64/mingw64/include/stdio.h \
 C:/msys64/mingw64/include/sec_api/stdio_s.h \
 C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
 C:/msys64/mingw64/include/c++/15.2.0/cctype \
```

```
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
C:/msys64/mingw64/include/c++/15.2.0/bit \
C:/msys64/mingw64/include/c++/15.2.0/concepts \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/vector \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h
```

4.61 build_O3/CMakeFiles/programa.dir/src/student.cpp.obj.d File Reference

4.62 student.cpp.obj.d

Go to the documentation of this file.

```
CMakeFiles/programa.dir/src/student.cpp.obj: \
```

```
C:\Users\rzvil\OneDrive\Desktop\antroji_uzduotis\src\student.cpp \
C:/Users/rzvil/OneDrive/Desktop/antroji_uzduotis/include/student.h \
C:/msys64/mingw64/include/c++/15.2.0/string \
C:/msys64/mingw64/include/c++/15.2.0/bits/requires_hosted.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++config.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/os_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/cpu_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/pstl_config.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stringfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/memoryfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/char_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/postypes.h \
C:/msys64/mingw64/include/c++/15.2.0/cwchar \
C:/msys64/mingw64/include/wchar.h C:/msys64/mingw64/include/corecrt.h \
C:/msys64/mingw64/include/_mingw.h \
C:/msys64/mingw64/include/_mingw_mac.h \
C:/msys64/mingw64/include/_mingw_secapi.h \
C:/msys64/mingw64/include/vadefs.h \
C:/msys64/mingw64/include/sdks/_mingw_ddk.h \
C:/msys64/mingw64/include/corecrt_stdio_config.h \
C:/msys64/mingw64/include/corecrt_wstdlib.h \
C:/msys64/mingw64/include/corecrt_wctype.h \
C:/msys64/mingw64/include/_mingw_off_t.h \
C:/msys64/mingw64/include/_mingw_stat64.h \
C:/msys64/mingw64/include/swprintf.inl \
C:/msys64/mingw64/include/sec_api/wchar_s.h \
C:/msys64/mingw64/include/c++/15.2.0/type_traits \
C:/msys64/mingw64/include/c++/15.2.0/bits/version.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/new_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/new \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/functexcept.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/exception_defines.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/move.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cpp_type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/localefwd.h \
C:/msys64/mingw64/include/c++/15.2.0/x86_64-w64-mingw32/bits/c++locale.h \
C:/msys64/mingw64/include/c++/15.2.0/clocale \
C:/msys64/mingw64/include/locale.h C:/msys64/mingw64/include/crtdefs.h \
C:/msys64/mingw64/include/stdio.h \
C:/msys64/mingw64/include/sec_api/stdio_s.h \
C:/msys64/mingw64/include/c++/15.2.0/iosfwd \
C:/msys64/mingw64/include/c++/15.2.0/cctype \
C:/msys64/mingw64/include/ctype.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ostream_insert.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/cxxabi_forced.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_funcs.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/concept_check.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/assertions.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator_base_types.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_iterator.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/type_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/ptr_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_function.h \
C:/msys64/mingw64/include/c++/15.2.0/backward/binders.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/numeric_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algobase.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_pair.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/utility.h \
C:/msys64/mingw64/include/c++/15.2.0/debug/debug.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/predefined_ops.h \
C:/msys64/mingw64/include/c++/15.2.0/bit \
C:/msys64/mingw64/include/c++/15.2.0/concepts \
C:/msys64/mingw64/include/c++/15.2.0/bits/refwrap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/invoke.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/range_access.h \
C:/msys64/mingw64/include/c++/15.2.0/initializer_list \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.h \
C:/msys64/mingw64/include/c++/15.2.0/ext/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/alloc_traits.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_construct.h \
C:/msys64/mingw64/include/c++/15.2.0/string_view \
C:/msys64/mingw64/include/c++/15.2.0/bits/functional_hash.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/hash_bytes.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/string_view.tcc \
C:/msys64/mingw64/include/c++/15.2.0/ext/string_conversions.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdlib \
C:/msys64/mingw64/include/stdlib.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/limits.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/syslimits.h \
C:/msys64/mingw64/include/limits.h \
C:/msys64/mingw64/include/sec_api/stdlib_s.h \
C:/msys64/mingw64/include/c++/15.2.0/stdlib.h \
C:/msys64/mingw64/include/malloc.h \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/mm_malloc.h \
```

```
C:/msys64/mingw64/include/errno.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/std_abs.h \
C:/msys64/mingw64/include/c++/15.2.0/cstdio \
C:/msys64/mingw64/include/c++/15.2.0/cerrno \
C:/msys64/mingw64/include/c++/15.2.0/bits/charconv.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/basic_string.tcc \
C:/msys64/mingw64/include/c++/15.2.0/bits/memory_resource.h \
C:/msys64/mingw64/include/c++/15.2.0/cstddef \
C:/msys64/mingw64/lib/gcc/x86_64-w64-mingw32/15.2.0/include/stddef.h \
C:/msys64/mingw64/include/stddef.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uses_allocator_args.h \
C:/msys64/mingw64/include/c++/15.2.0/tuple \
C:/msys64/mingw64/include/c++/15.2.0/vector \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_uninitialized.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_vector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_bvector.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/vector.tcc \
C:/msys64/mingw64/include/c++/15.2.0/algorithm \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_algo.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/algorithmfwd.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_heap.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/uniform_int_dist.h \
C:/msys64/mingw64/include/c++/15.2.0/bits/stl_tempbuf.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/glue_algorithm_defs.h \
C:/msys64/mingw64/include/c++/15.2.0/pstl/execution_defs.h
```

4.63 include/calculations.h File Reference

```
#include <vector>
#include "student.h"
```

Functions

- double med (const std::vector< int > &nd, int egz)
- double avg (const std::vector< int > &nd, int egz)
- std::string GenerateName (const std::string &vardas, const std::vector< std::string > &saknys)
- bool compareByName (const Student &a, const Student &b)
- bool compareBySurname (const Student &a, const Student &b)
- bool compareByMed (const Student &a, const Student &b)
- bool compareByAvg (const Student &a, const Student &b)

4.63.1 Function Documentation

4.63.1.1 avg()

```
double avg (
            const std::vector< int > & nd,
            int egz)
```

4.63.1.2 compareByAvg()

```
bool compareByAvg (
            const Student & a,
            const Student & b)
```

Definition at line 40 of file calculations.cpp.

4.63.1.3 compareByMed()

```
bool compareByMed (
            const Student & a,
            const Student & b)
```

Definition at line 36 of file calculations.cpp.

4.63.1.4 compareByName()

```
bool compareByName (
            const Student & a,
            const Student & b)
```

Definition at line 22 of file calculations.cpp.

4.63.1.5 compareBySurname()

```
bool compareBySurname (
            const Student & a,
            const Student & b)
```

Definition at line 29 of file calculations.cpp.

4.63.1.6 GenerateName()

```
std::string GenerateName (
            const std::string & vardas,
            const std::vector< std::string > & saknys)
```

Definition at line 6 of file calculations.cpp.

4.63.1.7 med()

```
double med (
            const std::vector< int > & nd,
            int egz)
```

4.64 calculations.h

Go to the documentation of this file.

```
#ifndef CALCULATIONS_H
#define CALCULATIONS_H

#include <vector>
#include "student.h"

double med(const std::vector<int>& nd, int egz);
double avg(const std::vector<int>& nd, int egz);
std::string GenerateName(const std::string& vardas, const std::vector<std::string>& saknys);

bool compareByName(const Student &a, const Student &b);
bool compareBySurname(const Student &a, const Student &b);
bool compareByMed(const Student &a, const Student &b);
bool compareByAvg(const Student &a, const Student &b);

#endif
```

4.65 include/fileHandling.h File Reference

```
#include <vector>
#include <list>
#include <deque>
#include <string>
#include "student.h"
```

Functions

- template<typename Container>
  bool createFile (const Container &Students, int n)
- template<typename Container>
  bool readFile (const std::string &filename, Container &Students)

4.65.1 Function Documentation

4.65.1.1 createFile()

```
template<typename Container>
bool createFile (
            const Container & Students,
            int n)
```

Definition at line 13 of file fileHandling.cpp.

4.65.1.2 readFile()

```
template<typename Container>
bool readFile (
            const std::string & filename,
            Container & Students)
```

Definition at line 50 of file fileHandling.cpp.

4.66 fileHandling.h

Go to the documentation of this file.

```
00001 #ifndef FILEHANDLING_H
00002 #define FILEHANDLING_H
00003
00004 #include <vector>
00005 #include <list>
00006 #include <deque>
00007 #include <string>
00008 #include "student.h"
00009
00010 template <typename Container>
00011 bool createFile(const Container& Students, int n);
00012 template <typename Container>
00013 bool readFile(const std::string& filename, Container& Students);
00014
00015 #endif
```

4.67 include/inputValid.h File Reference

```
#include <string>
```

Functions

- char inputLetter (const std::string &message)
- std::string inputWord (const std::string &message)
- int inputInt (const std::string &message, int min=INT_MIN, int max=INT_MAX)
- double inputR (const std::string &message, double min=INT_MIN, double max=INT_MAX)

4.67.1 Function Documentation

4.67.1.1 inputInt()

```
int inputInt (
            const std::string & message,
            int min = INT_MIN,
            int max = INT_MAX)
```

Definition at line 50 of file inputValid.cpp.

4.67.1.2 inputLetter()

```
char inputLetter (
            const std::string & message)
```

Definition at line 4 of file inputValid.cpp.

4.67.1.3 inputR()

```
double inputR (
            const std::string & message,
            double min = INT_MIN,
            double max = INT_MAX)
```

Definition at line 79 of file inputValid.cpp.

4.67.1.4 inputWord()

```
std::string inputWord (
            const std::string & message)
```

Definition at line 26 of file inputValid.cpp.

4.68 inputValid.h

Go to the documentation of this file.

```
#ifndef INPUTVALID_H
#define INPUTVALID_H

#include <string>

char inputLetter(const std::string& message);
std::string inputWord(const std::string& message);
int inputInt(const std::string& message, int min = INT_MIN, int max = INT_MAX);
double inputR(const std::string& message, double min = INT_MIN, double max = INT_MAX);

#endif
```

4.69 include/meniu.h File Reference

```
#include <vector>
#include <list>
#include <deque>
#include "student.h"
```

Functions

- int showMeniu ()
- template<typename Container>
  void manualInput (Container &Students)
- template<typename Container>
  void generateGrades (Container &Students)
- template<typename Container>
  void generateNamesGrades (Container &Students)
- template<typename Container>
  void scanFile (Container &Students)
- bool generateFile ()
- template<typename Container>
  void sortingStudentsTemplate ()
- template<typename Container>
  void sortingStudents (Container &Students)

4.69.1 Function Documentation

4.69.1.1 generateFile()

```
bool generateFile ()
```

Definition at line 129 of file meniu.cpp.

4.69.1.2 generateGrades()

```
template<typename Container>
void generateGrades (
            Container & Students)
```

Definition at line 57 of file meniu.cpp.

4.69.1.3 generateNamesGrades()

```
template<typename Container>
void generateNamesGrades (
            Container & Students)
```

Definition at line 83 of file meniu.cpp.

4.69.1.4 manualInput()

```
template<typename Container>
void manualInput (
            Container & Students)
```

Definition at line 30 of file meniu.cpp.

4.69.1.5 scanFile()

```
template<typename Container>
void scanFile (
            Container & Students)
```

Definition at line 112 of file meniu.cpp.

4.69.1.6 showMeniu()

```
int showMeniu ()
```

Definition at line 16 of file meniu.cpp.

4.69.1.7 sortingStudents()

```
template<typename Container>
void sortingStudents (
            Container & Students)
```

Definition at line 183 of file meniu.cpp.

4.69.1.8 sortingStudentsTemplate()

```
template<typename Container>
void sortingStudentsTemplate ()
```

4.70 meniu.h

Go to the documentation of this file.

```
00001 #ifndef MENIU_H
00002 #define MENIU_H
00003
00004 #include <vector>
00005 #include <list>
00006 #include <deque>
00007 #include "student.h"
00008
00009 int showMeniu();
00010 template <typename Container>
00011 void manualInput(Container& Students);
00012 template <typename Container>
00013 void generateGrades(Container& Students);
00014 template <typename Container>
00015 void generateNamesGrades(Container& Students);
00016 template <typename Container>
00017 void scanFile(Container& Students);
00018 bool generateFile();
00019
00020 template <typename Container>
00021 void sortingStudentsTemplate();
00022
00023 template <typename Container>
00024 void sortingStudents(Container& Students);
00025
00026 #endif
```

4.71 include/output.h File Reference

```
#include <vector>
#include <list>
#include <deque>
#include <string>
#include "student.h"
```

Functions

- template<typename Container>
  void printResults (const Container &Students, bool toFile=false, const std::string &filename="rezultatai.txt")
- template<typename Container>
  void outputToFileSorting (Container &Students)
- template<typename Container>
  void handleOutput (Container &Students)

4.71.1 Function Documentation

4.71.1.1 handleOutput()

```
template<typename Container>
void handleOutput (
            Container & Students)
```

Definition at line 71 of file output.cpp.

4.71.1.2 outputToFileSorting()

```
template<typename Container>
void outputToFileSorting (
            Container & Students)
```

Definition at line 53 of file output.cpp.

4.71.1.3 printResults()

```
template<typename Container>
void printResults (
            const Container & Students,
            bool toFile = false,
            const std::string & filename = "rezultatai.txt")
```

Definition at line 23 of file output.cpp.

4.72 output.h

Go to the documentation of this file.

```
00001 #ifndef OUTPUT_H
00002 #define OUTPUT_H
00003
00004 #include <vector>
00005 #include <list>
00006 #include <deque>
00007 #include <string>
00008 #include "student.h"
00009
00010 template <typename Container>
00011 void printResults(const Container& Students, bool toFile = false, const std::string& filename =
      "rezultatai.txt");
00012 template <typename Container>
00013 void outputToFileSorting(Container& Students);
00014 template <typename Container>
00015 void handleOutput(Container& Students);
00016
00017 #endif
```

4.73 include/person.h File Reference

```
#include <string>
```

Classes

- class Person

4.74 person.h

Go to the documentation of this file.

```
#ifndef PERSON_H
#define PERSON_H

#include <string>

class Person {
protected:
    std::string name_;
    std::string surname_;

public:
    Person(const std::string& name = "", const std::string& surname = "")
        : name_(name), surname_(surname) {}

    virtual ~Person() = default;

    Person(const Person& other);
    Person& operator=(const Person& other);

    Person(Person&& other) noexcept;
    Person& operator=(Person&& other) noexcept;

    virtual void print() const = 0;

    const std::string& getName() const { return name_; }
    const std::string& getSurname() const { return surname_; }
};

#endif
```

4.75 include/student.h File Reference

```
#include "person.h"
#include <vector>
#include <iostream>
```

Classes

- class Student

4.76 student.h

Go to the documentation of this file.

```
#ifndef STUDENT_H
#define STUDENT_H

#include "person.h"
#include <vector>
#include <iostream>

class Student : public Person {
private:
    std::vector<int> nd_;
    int egz_;

    double finalAvg_;
    double finalMed_;

    void calculate();

public:
    Student(const std::string name = "Julius", const std::string surname = "Petraitis", const
std::vector<int> nd = {1, 2}, int egz = 5);
```

```
    ~Student();

    Student(const Student& other);
    Student& operator=(const Student& other);

    Student(Student&& other) noexcept;
    Student& operator=(Student&& other) noexcept;

    int getEgz() const;
    const std::vector<int>& getNd() const;

    double getFinalAvg() const;
    double getFinalMed() const;

    void setNd(const std::vector<int>& nd);
    void setEgz(int egz);
    void addNd(int mark);

    void print() const override;

    friend std::ostream& operator«(std::ostream& os, const Student& s);
    friend std::istream& operator»(std::istream& is, Student& s);
};

#endif
```

4.77 obj/calculations.d File Reference

4.78 calculations.d

Go to the documentation of this file.

```
obj/calculations.o: src/calculations.cpp include/calculations.h \
 include/student.h
```

4.79 obj/fileHandling.d File Reference

4.80 fileHandling.d

Go to the documentation of this file.

```
obj/fileHandling.o: src/fileHandling.cpp include/fileHandling.h \
 include/student.h include/calculations.h include/inputValid.h
```

4.81 obj/inputValid.d File Reference

4.82 inputValid.d

Go to the documentation of this file.

```
obj/inputValid.o: src/inputValid.cpp include/inputValid.h
```

4.83 obj/main.d File Reference

4.84 main.d

Go to the documentation of this file.

```
00001 obj/main.o: src/main.cpp include/meniu.h include/student.h \
00002  include/output.h include/inputValid.h
```

4.85 obj/meniu.d File Reference

4.86 meniu.d

Go to the documentation of this file.

```
00001 obj/meniu.o: src/meniu.cpp include/meniu.h include/student.h \
00002  include/inputValid.h include/calculations.h include/output.h \
00003  include/fileHandling.h
```

4.87 obj/output.d File Reference

4.88 output.d

Go to the documentation of this file.

```
00001 obj/output.o: src/output.cpp include/output.h include/student.h \
00002  include/inputValid.h include/calculations.h
```

4.89 README.md File Reference

4.90 src/calculations.cpp File Reference

```
#include "calculations.h"
#include <algorithm>
#include <random>
```

Functions

- std::string GenerateName (const std::string &vardas, const std::vector< std::string > &saknys)
- bool compareByName (const Student &a, const Student &b)
- bool compareBySurname (const Student &a, const Student &b)
- bool compareByMed (const Student &a, const Student &b)
- bool compareByAvg (const Student &a, const Student &b)

4.90.1 Function Documentation

4.90.1.1 compareByAvg()

```
bool compareByAvg (
            const Student & a,
            const Student & b)
```

Definition at line 40 of file calculations.cpp.

4.90.1.2 compareByMed()

```
bool compareByMed (
            const Student & a,
            const Student & b)
```

Definition at line 36 of file calculations.cpp.

4.90.1.3 compareByName()

```
bool compareByName (
            const Student & a,
            const Student & b)
```

Definition at line 22 of file calculations.cpp.

4.90.1.4 compareBySurname()

```
bool compareBySurname (
            const Student & a,
            const Student & b)
```

Definition at line 29 of file calculations.cpp.

4.90.1.5 GenerateName()

```
std::string GenerateName (
            const std::string & vardas,
            const std::vector< std::string > & saknys)
```

Definition at line 6 of file calculations.cpp.

4.91 calculations.cpp

Go to the documentation of this file.

```
#include "calculations.h"
#include <algorithm>
#include <random>


std::string GenerateName(const std::string &vardas, const std::vector<std::string> &saknys) {
    std::random_device rd;
    std::mt19937 gen(rd());
    std::uniform_int_distribution<> dist(0, saknys.size() - 1);

    std::string saknis = saknys.at(dist(gen));

    if (vardas.substr(vardas.size() - 2) == "as")
        return saknis + "enis";

    if (vardas.back() == 'a')
        return saknis + "aite";

    return saknis + "is";
}

bool compareByName(const Student &a, const Student &b) {
    if (a.getName().length() != b.getName().length()) {
        return a.getName().length() < b.getName().length();
    }
    return a.getName() < b.getName();
}

bool compareBySurname(const Student &a, const Student &b) {
    if (a.getSurname().length() != b.getSurname().length()) {
        return a.getSurname().length() < b.getSurname().length();
    }
    return a.getSurname() < b.getSurname();
}

bool compareByMed(const Student &a, const Student &b) {
    return a.getFinalMed() < b.getFinalMed();
}

bool compareByAvg(const Student &a, const Student &b) {
    return a.getFinalAvg() < b.getFinalAvg();
}
```

4.92 src/fileHandling.cpp File Reference

```
#include "fileHandling.h"
#include "calculations.h"
#include "inputValid.h"
#include <fstream>
#include <sstream>
#include <iostream>
#include <iomanip>
#include <vector>
#include <list>
#include <deque>
```

Functions

- template<typename Container>
  bool createFile (const Container &Students, int n)
- template<typename Container>
  bool readFile (const std::string &filename, Container &Students)
- template bool createFile< std::vector< Student > > (const std::vector< Student > &, int)

- template bool createFile< std::list< Student > > (const std::list< Student > &, int)
- template bool createFile< std::deque< Student > > (const std::deque< Student > &, int)
- template bool readFile< std::vector< Student > > (const std::string &, std::vector< Student > &)
- template bool readFile< std::list< Student > > (const std::string &, std::list< Student > &)
- template bool readFile< std::deque< Student > > (const std::string &, std::deque< Student > &)

4.92.1 Function Documentation

4.92.1.1 createFile()

```
template<typename Container>
bool createFile (
            const Container & Students,
            int n)
```

Definition at line 13 of file fileHandling.cpp.

4.92.1.2 createFile< std::deque< Student > >()

```
template bool createFile< std::deque< Student > > (
            const std::deque< Student > & ,
            int )
```

4.92.1.3 createFile< std::list< Student > >()

```
template bool createFile< std::list< Student > > (
            const std::list< Student > & ,
            int )
```

4.92.1.4 createFile< std::vector< Student > >()

```
template bool createFile< std::vector< Student > > (
            const std::vector< Student > & ,
            int )
```

4.92.1.5 readFile()

```
template<typename Container>
bool readFile (
            const std::string & filename,
            Container & Students)
```

Definition at line 50 of file fileHandling.cpp.

4.92.1.6 readFile< std::deque< Student > >()

```
template bool readFile< std::deque< Student > > (
            const std::string & ,
            std::deque< Student > & )
```

4.92.1.7 readFile< std::list< Student > >()

```
template bool readFile< std::list< Student > > (
            const std::string & ,
            std::list< Student > & )
```

4.92.1.8 readFile< std::vector< Student > >()

```
template bool readFile< std::vector< Student > > (
            const std::string & ,
            std::vector< Student > & )
```

4.93 fileHandling.cpp

Go to the documentation of this file.

```
#include "fileHandling.h"
#include "calculations.h"
#include "inputValid.h"
#include <fstream>
#include <sstream>
#include <iostream>
#include <iomanip>
#include <vector>
#include <list>
#include <deque>

template <typename Container>
bool createFile (const Container &Students, int n) {
    try {
        std::ofstream kursiokai("kursiokai.txt");

        if (!kursiokai.is_open()) {
            throw std::runtime_error("Nepavyko sukurti failo kursiokai.txt");
        }
        kursiokai « std::left « std::setw(15) « "Vardas" « std::setw(15) « "Pavarde";
        for (int i = 0; i < n; i++) {
            kursiokai « std::setw(6) « ("ND" + std::to_string(i+1));
        }
        kursiokai « std::setw(9) « "Egzaminas\n";

        for (const auto &stud : Students) {
            kursiokai « std::left « std::setw(15) « stud.getName() « std::setw(15) « stud.getSurname();
            for (const auto &grade : stud.getNd()) {
                kursiokai « std::setw(6) « grade;
            }
            kursiokai « stud.getEgz() « "\n";

            if (!kursiokai) {
                throw std::runtime_error("Klaida rasant i kursiokai.txt");
            }
        }
        return true;
    }
    catch (const std::exception &e) {
        std::cerr « "Klaida: " « e.what() « std::endl;
        return false;
    }
    catch (...) {
        std::cerr « "Nezinoma klaida." « std::endl;
        return false;
```

```
    }
}

template <typename Container>
bool readFile(const std::string &filename, Container &Students) {
    try {
        std::ifstream stud_file(filename);
        if (!stud_file.is_open()) {
            throw std::runtime_error("Nepavyko atidaryti failo: " + filename);
        }
        std::cout « "Failas '" « filename « "' sekmingai atidarytas.\n";

        std::string line;
        std::getline(stud_file, line);

    while (std::getline(stud_file, line)) {
        if (line.empty()) continue;

        std::istringstream iss(line);

        std::string name, surname;
        iss » name » surname;

        std::vector<int> nd;
        int mark;

        while (iss » mark) {
            nd.push_back(mark);
        }

        int egz = nd.back();
        nd.pop_back();

        Student s(name, surname, nd, egz);

        Students.push_back(s);
    }

        stud_file.close();
        return true;

    } catch (const std::exception& e) {
        std::cerr « "Failo atidarymo klaida: " « e.what() « std::endl;
        return false;
    }
    catch (...) {
        std::cerr « "Nezinoma klaida." « std::endl;
        return false;
    }
}

template bool createFile<std::vector<Student>>(const std::vector<Student>&, int);
template bool createFile<std::list<Student>>(const std::list<Student>&, int);
template bool createFile<std::deque<Student>>(const std::deque<Student>&, int);

template bool readFile<std::vector<Student>>(const std::string&, std::vector<Student>&);
template bool readFile<std::list<Student>>(const std::string&, std::list<Student>&);
template bool readFile<std::deque<Student>>(const std::string&, std::deque<Student>&);
```

4.94 src/inputValid.cpp File Reference

```
#include "inputValid.h"
#include <iostream>
```

Functions

- char inputLetter (const std::string &message)
- std::string inputWord (const std::string &message)
- int inputInt (const std::string &message, int min, int max)
- double inputR (const std::string &message, double min, double max)

4.94.1 Function Documentation

4.94.1.1 inputInt()

```
int inputInt (
            const std::string & message,
            int min,
            int max)
```

Definition at line 50 of file inputValid.cpp.

4.94.1.2 inputLetter()

```
char inputLetter (
            const std::string & message)
```

Definition at line 4 of file inputValid.cpp.

4.94.1.3 inputR()

```
double inputR (
            const std::string & message,
            double min,
            double max)
```

Definition at line 79 of file inputValid.cpp.

4.94.1.4 inputWord()

```
std::string inputWord (
            const std::string & message)
```

Definition at line 26 of file inputValid.cpp.

4.95 inputValid.cpp

Go to the documentation of this file.

```
#include "inputValid.h"
#include <iostream>

char inputLetter(const std::string& message) {
    std::string input;

    while (true) {
        try {
            std::cout « message;
            std::cin » input;

            if (input.length() != 1)
                throw std::invalid_argument("Turi buti ivesta viena raide.");

            if (!std::isalpha(input[0]))
                throw std::invalid_argument("Leidziamos tik raides.");

            return input[0];
```

```
        }
        catch (const std::exception& e) {
            std::cout « "Klaida: " « e.what() « std::endl;
        }
    }
}

std::string inputWord(const std::string& message) {
    std::string text;

    while (true) {
        try {
            std::cout « message;
            std::cin » text;

            if (text.empty())
                throw std::invalid_argument("Zodis negali buti tuscias.");

            for (char c: text) {
                if (!std::isalpha(c))
                    throw std::invalid_argument("Zodyje gali buti tik raides.");
            }

            return text;
        }
        catch (const std::exception& e) {
            std::cout « "Klaida: " « e.what() « std::endl;
        }
    }
}

int inputInt(const std::string& message, int min, int max) {
    std::string input;
    int number;

    while (true) {
        try {
            std::cout « message;
            std::cin » input;

            size_t pos;
            number = std::stoi(input, &pos);

            if (pos != input.length())
                throw std::invalid_argument("Ivestis nera sveikasis skaicius.");

            if (number < min || number > max)
                throw std::out_of_range("Skaicius nepatenka i leidziamas ribas.");

            return number;
        }
        catch (const std::invalid_argument&) {
            std::cout « "Klaida: reikia ivesti sveikaji skaiciu.\n";
        }
        catch (const std::out_of_range&) {
            std::cout « "Klaida: ivestas skaicius uz leistinu ribu.\n";
        }
    }
}

double inputR(const std::string& message, double min, double max) {
    std::string input;
    double number;

    while (true) {
        try {
            std::cout « message;
            std::cin » input;

            size_t pos;
            number = std::stod(input, &pos);

            if (pos != input.length())
                throw std::invalid_argument("Ivestis nera realusis skaicius.");

            if (number < min || number > max)
                throw std::out_of_range("Skaicius nepatenka i leidziamas ribas.");

            return number;
        }
        catch (const std::invalid_argument&) {
            std::cout « "Klaida: reikia ivesti realuji skaiciu.\n";
        }
        catch (const std::out_of_range&) {
            std::cout « "Klaida: ivestas skaicius uz leistinu ribu.\n";
        }
    }
```

```
}
```

4.96 src/main.cpp File Reference

```
#include "meniu.h"
#include "output.h"
#include "inputValid.h"
#include <vector>
#include <ctime>
#include <list>
#include <deque>
```

Functions

- int main ()

4.96.1 Function Documentation

4.96.1.1 main()

```
int main ()
```

Definition at line 9 of file main.cpp.

4.97 main.cpp

Go to the documentation of this file.

```
#include "meniu.h"
#include "output.h"
#include "inputValid.h"
#include <vector>
#include <ctime>
#include <list>
#include <deque>

int main() {
    int meniu = inputInt(
        "Koki konteineri naudoti:\n"
        "1 - vector\n"
        "2 - list\n"
        "3 - deque\n", 1, 3);

        if (meniu == 1) {
            std::vector<Student> Students;
            while (true) {
            int choice = showMeniu();

            if (choice == 1) manualInput(Students), handleOutput(Students);
            else if (choice == 2) generateGrades(Students), handleOutput(Students);
            else if (choice == 3) generateNamesGrades(Students), handleOutput(Students);
            else if (choice == 4) scanFile(Students);
            else if (choice == 5) generateFile();
            else if (choice == 6) sortingStudents(Students);
            else if (choice == 7) break;
            }
        } else if (meniu == 2) {
        std::list<Student> Students;
        while (true) {
```

```
            int choice = showMeniu();

            if (choice == 1) manualInput(Students), handleOutput(Students);
            else if (choice == 2) generateGrades(Students), handleOutput(Students);
            else if (choice == 3) generateNamesGrades(Students), handleOutput(Students);
            else if (choice == 4) scanFile(Students);
            else if (choice == 5) generateFile();
            else if (choice == 6) sortingStudents(Students);
            else if (choice == 7) break;
            }
        } else if (meniu == 3) {
        std::deque<Student> Students;
        while (true) {
            int choice = showMeniu();

            if (choice == 1) manualInput(Students), handleOutput(Students);
            else if (choice == 2) generateGrades(Students), handleOutput(Students);
            else if (choice == 3) generateNamesGrades(Students), handleOutput(Students);
            else if (choice == 4) scanFile(Students);
            else if (choice == 5) generateFile();
            else if (choice == 6) sortingStudents(Students);
            else if (choice == 7) break;
            }
        }
    return 0;
}
```

4.98 src/meniu.cpp File Reference

```
#include "meniu.h"
#include "inputValid.h"
#include "calculations.h"
#include "output.h"
#include "fileHandling.h"
#include <iostream>
#include <chrono>
#include <iomanip>
#include <random>
#include <fstream>
#include <algorithm>
#include <vector>
#include <list>
#include <deque>
```

Functions

- int showMeniu ()
- template<typename Container>
  void manualInput (Container &Students)
- template<typename Container>
  void generateGrades (Container &Students)
- template<typename Container>
  void generateNamesGrades (Container &Students)
- template<typename Container>
  void scanFile (Container &Students)
- bool generateFile ()
- template<typename Container>
  void sortingStudents (Container &students)
- template void manualInput< std::vector< Student > > (std::vector< Student > &)
- template void manualInput< std::list< Student > > (std::list< Student > &)
- template void manualInput< std::deque< Student > > (std::deque< Student > &)

- template void generateGrades< std::vector< Student > > (std::vector< Student > &)
- template void generateGrades< std::list< Student > > (std::list< Student > &)
- template void generateGrades< std::deque< Student > > (std::deque< Student > &)
- template void generateNamesGrades< std::vector< Student > > (std::vector< Student > &)
- template void generateNamesGrades< std::list< Student > > (std::list< Student > &)
- template void generateNamesGrades< std::deque< Student > > (std::deque< Student > &)
- template void scanFile< std::vector< Student > > (std::vector< Student > &)
- template void scanFile< std::list< Student > > (std::list< Student > &)
- template void scanFile< std::deque< Student > > (std::deque< Student > &)
- template void sortingStudents< std::vector< Student > > (std::vector< Student > &)
- template void sortingStudents< std::list< Student > > (std::list< Student > &)
- template void sortingStudents< std::deque< Student > > (std::deque< Student > &)

4.98.1 Function Documentation

4.98.1.1 generateFile()

```
bool generateFile ()
```

Definition at line 129 of file meniu.cpp.

4.98.1.2 generateGrades()

```
template<typename Container>
void generateGrades (
            Container & Students)
```

Definition at line 57 of file meniu.cpp.

4.98.1.3 generateGrades< std::deque< Student > >()

```
template void generateGrades< std::deque< Student > > (
            std::deque< Student > & )
```

4.98.1.4 generateGrades< std::list< Student > >()

```
template void generateGrades< std::list< Student > > (
            std::list< Student > & )
```

4.98.1.5 generateGrades< std::vector< Student > >()

```
template void generateGrades< std::vector< Student > > (
            std::vector< Student > & )
```

4.98.1.6 generateNamesGrades()

```
template<typename Container>
void generateNamesGrades (
            Container & Students)
```

Definition at line 83 of file meniu.cpp.

4.98.1.7 generateNamesGrades< std::deque< Student > >()

```
template void generateNamesGrades< std::deque< Student > > (
            std::deque< Student > & )
```

4.98.1.8 generateNamesGrades< std::list< Student > >()

```
template void generateNamesGrades< std::list< Student > > (
            std::list< Student > & )
```

4.98.1.9 generateNamesGrades< std::vector< Student > >()

```
template void generateNamesGrades< std::vector< Student > > (
            std::vector< Student > & )
```

4.98.1.10 manualInput()

```
template<typename Container>
void manualInput (
            Container & Students)
```

Definition at line 30 of file meniu.cpp.

4.98.1.11 manualInput< std::deque< Student > >()

```
template void manualInput< std::deque< Student > > (
            std::deque< Student > & )
```

4.98.1.12 manualInput< std::list< Student > >()

```
template void manualInput< std::list< Student > > (
            std::list< Student > & )
```

4.98.1.13 manualInput< std::vector< Student > >()

```
template void manualInput< std::vector< Student > > (
            std::vector< Student > & )
```

4.98.1.14 scanFile()

```
template<typename Container>
void scanFile (
            Container & Students)
```

Definition at line 112 of file meniu.cpp.

4.98.1.15 scanFile< std::deque< Student > >()

```
template void scanFile< std::deque< Student > > (
            std::deque< Student > & )
```

4.98.1.16 scanFile< std::list< Student > >()

```
template void scanFile< std::list< Student > > (
            std::list< Student > & )
```

4.98.1.17 scanFile< std::vector< Student > >()

```
template void scanFile< std::vector< Student > > (
            std::vector< Student > & )
```

4.98.1.18 showMeniu()

```
int showMeniu ()
```

Definition at line 16 of file meniu.cpp.

4.98.1.19 sortingStudents()

```
template<typename Container>
void sortingStudents (
            Container & students)
```

Definition at line 183 of file meniu.cpp.

4.98.1.20 sortingStudents< std::deque< Student > >()

```
template void sortingStudents< std::deque< Student > > (
            std::deque< Student > & )
```

4.98.1.21 sortingStudents< std::list< Student > >()

```
template void sortingStudents< std::list< Student > > (
            std::list< Student > & )
```

4.98.1.22 sortingStudents< std::vector< Student > >()

```
template void sortingStudents< std::vector< Student > > (
            std::vector< Student > & )
```

4.99 meniu.cpp

Go to the documentation of this file.

```
#include "meniu.h"
#include "inputValid.h"
#include "calculations.h"
#include "output.h"
#include "fileHandling.h"
#include <iostream>
#include <chrono>
#include <iomanip>
#include <random>
#include <fstream>
#include <algorithm>
#include <vector>
#include <list>
#include <deque>

int showMeniu() {
    return inputInt(
        "1 - ivesti viska ranka.\n"
        "2 - generuoti tik pazymius.\n"
        "3 - generuoti studentu vardus, pavardes ir pazymius.\n"
        "4 - nuskaityti is failo.\n"
        "5 - generuoti failus.\n"
        "6 - rusiuoti studentus i du atskirus failus.\n"
        "7 - baigti darba.\n"
        "Pasirinkite (1-7): ",
        1, 7);
}

template <typename Container>
void manualInput(Container& Students) {
    while (true) {
        std::string name = inputWord(
            "Iveskite studento varda arba zodi STOP, jei norite baigti: \n"
        );
        if (name == "STOP") break;

        std::string surname = inputWord("Iveskite studento pavarde: \n");

        std::vector<int> nd;
        std::cout « "Iveskite namu darbu pazymius (0 - baigti): \n";

        while (true) {
            int mark = inputInt("Namu darbo pazymys: \n", 0, 10);
            if (mark == 0) break;
            nd.push_back(mark);
        }

        int egz = inputInt("Iveskite egzamino ivertinima: \n", 1, 10);

        Student s(name, surname, nd, egz);

        Students.push_back(s);
    }
}

template <typename Container>
void generateGrades(Container& Students) {
    int n = inputInt("Kiek namu darbu generuoti: \n", 1);

    std::random_device rd;
    std::mt19937 gen(rd());
    std::uniform_int_distribution<> dist(1, 10);

    while (true) {
        std::string name = inputWord("Iveskite varda arba STOP:\n");
        if (name == "STOP" || name == "stop") break;

        std::string surname = inputWord("Iveskite pavarde:\n");

        std::vector<int> nd;
```

```
        for (int i = 0; i < n; i++) {
            nd.push_back(dist(gen));
        }

        int egz = dist(gen);

        Student s(name, surname, nd, egz);
        Students.push_back(s);
    }
}

template <typename Container>
void generateNamesGrades(Container& Students) {
    int numb = inputInt("Kiek studentu generuoti: \n", 1);
    int n = inputInt("Kiek ND generuoti: \n", 1);

    std::vector<std::string> Names = {"Jonas","Domantas","Martynas","Edvinas","Evelina","Karolina","Gabija","Livija"};
    std::vector<std::string> Surnames = {"1Pavard","2Pavard","3Pavard","4Pavard","5Pavard"};

    std::random_device rd;
    std::mt19937 gen(rd());
    std::uniform_int_distribution<> gradeDist(1, 10);
    std::uniform_int_distribution<> nameDist(0, Names.size() - 1);

    for (int i = 0; i < numb; i++) {
        std::string name = Names.at(nameDist(gen));
        std::string surname = GenerateName(name, Surnames);

        std::vector<int> nd;
        for (int j = 0; j < n; j++) {
            nd.push_back(gradeDist(gen));
        }

        int egz = gradeDist(gen);

        Student s(name, surname, nd, egz);
        Students.push_back(s);
    }
}

template <typename Container>
void scanFile(Container& Students) {
    std::string filename;
    std::cout « "Iveskite failo pavadinima: ";
    system("powershell ls *.txt");
    std::cin » filename;
    Students.clear();

    try {
        if (readFile(filename, Students)) {
            std::cout « "Failas sekmingai nuskaitytas!\n";
            handleOutput(Students);
        }
    } catch (const std::runtime_error& e) {
        std::cerr « "Klaida atidarant faila: " « e.what() « std::endl;
    }
}

bool generateFile() {
    try {
    int ndCount = 5;
    int fnumb;
    long long records;

    fnumb = inputInt("Kiek failu generuoti: \n", 1);

    std::random_device rd;
    std::mt19937 gen(rd());
    std::uniform_int_distribution<> grade(1, 10);

        for (int f = 0; f < fnumb; f++) {

            records = inputInt("Kokio dydzio faila generuoti? Iveskite irasu skaiciu:\n", 1);
            std::string filename = "data/studentai_" + std::to_string(records) + ".txt";

            std::ofstream file(filename);

            if (!file.is_open()) {
                throw std::runtime_error("Nepavyko sukurti failo.");
            }

            file « std::left « std::setw(15) « "Vardas" « std::setw(15) « "Pavarde";

            for (int i = 0; i < ndCount; i++) {
                file « std::setw(6) « ("ND" + std::to_string(i + 1));
            }
```

```

            file « std::setw(9) « "Egzaminas.\n";

            for (long long i = 1; i <= records; ++i) {
                file « std::left « std::setw(15) « ("Vardas" + std::to_string(i)) « std::setw(15) « ("Pavarde" + std::to_string(i));

                for (int j = 0; j < ndCount; j++) {
                    file « std::setw(6) « grade(gen);
                }

                file « std::setw(10) « grade(gen) « "\n";
            }
        }
    return true;
}
    catch (const std::exception &e) {
        std::cerr « "Klaida: " « e.what() « std::endl;
        return false;
    }
    catch (...) {
        std::cerr « "Nezinoma klaida." « std::endl;
        return false;
    }
}

template <typename Container>
void sortingStudents(Container& students) {
    std::string filename;
    std::cout « "Koki faila rusiuoti i vargsiukus ir kietakius: \n";
    system("powershell ls data/*.txt");
    std::cin » filename;

    std::ifstream in("data/" + filename);
    if (!in.is_open()) return;

    students.clear();
    std::string line;
    std::getline(in, line);

    auto readStart = std::chrono::high_resolution_clock::now();

    int score;
    std::string name, surname;
    while (in » name » surname) {
        std::vector<int> nd;
        for (int i = 0; i < 5; i++) {
            in » score;
            nd.push_back(score);
        }
        int egz;
        in » egz;

        Student temp(name, surname, nd, egz);
        students.push_back(temp);
    }
    in.close();

    auto readEnd = std::chrono::high_resolution_clock::now();
    double readTime = std::chrono::duration<double>(readEnd - readStart).count();
    std::cout « "Failo skaitymas uztruko: " « readTime « " s\n";

    int choiceOutput = inputInt("Rusiuoti pagal:\n"
        "1 - Varda\n"
        "2 - Pavarde\n"
        "3 - Galutini bala\n", 1, 3);

    auto sortStart = std::chrono::high_resolution_clock::now();

    if constexpr (std::is_same_v<Container, std::list<Student») {
        if (choiceOutput == 1) students.sort(compareByName);
        else if (choiceOutput == 2) students.sort(compareBySurname);
        else students.sort(compareByAvg);
    } else {
        if (choiceOutput == 1) std::sort(students.begin(), students.end(), compareByName);
        else if (choiceOutput == 2) std::sort(students.begin(), students.end(), compareBySurname);
        else std::sort(students.begin(), students.end(), compareByAvg);
    }

    auto sortEnd = std::chrono::high_resolution_clock::now();
    double sortTime = std::chrono::duration<double>(sortEnd - sortStart).count();
    std::cout « "Bendras studentu rusiavimas uztruko: " « sortTime « " s\n";

    int strategy = inputInt("Pasirinkite studentu skirstymo strategija:\n"
        "1 - du nauji konteineriai\n"
        "2 - vienas naujas konteineris ir trynimas is bendro konteinerio\n"
        "3 - efektyvus metodai\n", 1, 3);
```

```

    Container vargsiukai, kietiakai;

    auto splitStart = std::chrono::high_resolution_clock::now();

    if (strategy == 1) {
        for (const auto& s : students) {
            if (s.getFinalAvg() < 5.0) vargsiukai.push_back(s);
            else kietiakai.push_back(s);
        }
    } else if (strategy == 2) {
    for (auto it = students.begin(); it != students.end(); ) {
        if (it->getFinalAvg() < 5.0) {
            vargsiukai.push_back(*it);
            it = students.erase(it);
        } else {
            ++it;
        }
    }
    kietiakai = students;
    } else if (strategy == 3) {
        auto it = std::partition(
            students.begin(), students.end(), [](const Student& s) {
                return s.getFinalAvg() < 5.0;
            }
        );

        vargsiukai.assign(students.begin(), it);
        kietiakai.assign(it, students.end());
    }

    auto splitEnd = std::chrono::high_resolution_clock::now();
    double splitTime = std::chrono::duration<double>(splitEnd - splitStart).count();
    std::cout « "Studentu skirstymas i dvi grupes uztruko: " « splitTime « " s\n";

    auto saveToFile = [](const auto& students, const std::string& filename) {
        std::ofstream out(filename);
        if (!out.is_open()) {
            std::cout « "Nepavyko atidaryti failo.\n";
            return;
        }

        out « std::left « std::setw(15) « "Vardas" « std::setw(15) « "Pavarde" « std::setw(17) « "Galutinis (Vid.)\n";
        out « "-------------------------------------------------------\n";
        for (const auto& s : students) {
            out « std::left « std::setw(15) « s.getName() « std::setw(15) « s.getSurname() « std::setw(17) « std::fixed « std::setprecision(2) « s.getFinalAvg() « "\n";
        }
    };

    saveToFile(vargsiukai, "data/vargsiukai.txt");
    saveToFile(kietiakai, "data/kietiakai.txt");

    std::cout « "Bendras vykdymo laikas: " « readTime + sortTime + splitTime « " s\n";
}

template void manualInput<std::vector<Student>>(std::vector<Student>&);
template void manualInput<std::list<Student>>(std::list<Student>&);
template void manualInput<std::deque<Student>>(std::deque<Student>&);

template void generateGrades<std::vector<Student>>(std::vector<Student>&);
template void generateGrades<std::list<Student>>(std::list<Student>&);
template void generateGrades<std::deque<Student>>(std::deque<Student>&);

template void generateNamesGrades<std::vector<Student>>(std::vector<Student>&);
template void generateNamesGrades<std::list<Student>>(std::list<Student>&);
template void generateNamesGrades<std::deque<Student>>(std::deque<Student>&);

template void scanFile<std::vector<Student>>(std::vector<Student>&);
template void scanFile<std::list<Student>>(std::list<Student>&);
template void scanFile<std::deque<Student>>(std::deque<Student>&);

template void sortingStudents<std::vector<Student>>(std::vector<Student>&);
template void sortingStudents<std::list<Student>>(std::list<Student>&);
template void sortingStudents<std::deque<Student>>(std::deque<Student>&);
```

4.100 src/output.cpp File Reference

```
#include "output.h"
#include "inputValid.h"
```

```
#include "calculations.h"
#include <iostream>
#include <iomanip>
#include <algorithm>
#include <fstream>
#include <vector>
#include <list>
#include <deque>
```

Functions

- template<typename Container, typename Compare>
  void sortContainer (Container &c, Compare comp)
- template<typename Compare>
  void sortContainer (std::list< Student > &c, Compare comp)
- template<typename Container>
  void printResults (const Container &Students, bool toFile, const std::string &filename)
- template<typename Container>
  void outputToFileSorting (Container &Students)
- template<typename Container>
  void handleOutput (Container &Students)
- template void printResults< std::vector< Student > > (const std::vector< Student > &, bool, const std::string &)
- template void printResults< std::list< Student > > (const std::list< Student > &, bool, const std::string &)
- template void printResults< std::deque< Student > > (const std::deque< Student > &, bool, const std::string &)
- template void outputToFileSorting< std::vector< Student > > (std::vector< Student > &)
- template void outputToFileSorting< std::list< Student > > (std::list< Student > &)
- template void outputToFileSorting< std::deque< Student > > (std::deque< Student > &)
- template void handleOutput< std::vector< Student > > (std::vector< Student > &)
- template void handleOutput< std::list< Student > > (std::list< Student > &)
- template void handleOutput< std::deque< Student > > (std::deque< Student > &)

4.100.1 Function Documentation

4.100.1.1 handleOutput()

```
template<typename Container>
void handleOutput (
            Container & Students)
```

Definition at line 71 of file output.cpp.

4.100.1.2 handleOutput< std::deque< Student > >()

```
template void handleOutput< std::deque< Student > > (
            std::deque< Student > & )
```

4.100.1.3 handleOutput< std::list< Student > >()

```
template void handleOutput< std::list< Student > > (
            std::list< Student > & )
```

4.100.1.4 handleOutput< std::vector< Student > >()

```
template void handleOutput< std::vector< Student > > (
            std::vector< Student > & )
```

4.100.1.5 outputToFileSorting()

```
template<typename Container>
void outputToFileSorting (
            Container & Students)
```

Definition at line 53 of file output.cpp.

4.100.1.6 outputToFileSorting< std::deque< Student > >()

```
template void outputToFileSorting< std::deque< Student > > (
            std::deque< Student > & )
```

4.100.1.7 outputToFileSorting< std::list< Student > >()

```
template void outputToFileSorting< std::list< Student > > (
            std::list< Student > & )
```

4.100.1.8 outputToFileSorting< std::vector< Student > >()

```
template void outputToFileSorting< std::vector< Student > > (
            std::vector< Student > & )
```

4.100.1.9 printResults()

```
template<typename Container>
void printResults (
            const Container & Students,
            bool toFile,
            const std::string & filename)
```

Definition at line 23 of file output.cpp.

4.100.1.10 printResults< std::deque< Student > >()

```
template void printResults< std::deque< Student > > (
            const std::deque< Student > & ,
            bool ,
            const std::string & )
```

4.100.1.11 printResults< std::list< Student > >()

```
template void printResults< std::list< Student > > (
            const std::list< Student > & ,
            bool ,
            const std::string & )
```

4.100.1.12 printResults< std::vector< Student > >()

```
template void printResults< std::vector< Student > > (
            const std::vector< Student > & ,
            bool ,
            const std::string & )
```

4.100.1.13 sortContainer() [1/2]

```
template<typename Container, typename Compare>
void sortContainer (
            Container & c,
            Compare comp)
```

Definition at line 13 of file output.cpp.

4.100.1.14 sortContainer() [2/2]

```
template<typename Compare>
void sortContainer (
            std::list< Student > & c,
            Compare comp)
```

Definition at line 18 of file output.cpp.

4.101 output.cpp

Go to the documentation of this file.

```
#include "output.h"
#include "inputValid.h"
#include "calculations.h"
#include <iostream>
#include <iomanip>
#include <algorithm>
#include <fstream>
#include <vector>
#include <list>
#include <deque>

template <typename Container, typename Compare>
void sortContainer(Container& c, Compare comp) {
    std::sort(c.begin(), c.end(), comp);
}

template <typename Compare>
void sortContainer(std::list<Student>& c, Compare comp) {
    c.sort(comp);
}

template <typename Container>
void printResults(const Container &Students, bool toFile, const std::string &filename) {
    if (Students.empty()) {
        throw std::runtime_error("Studentu sarasas tuscias.");
    }
    std::ofstream file;
    if (toFile) {
        file.open(filename);
        if (!file.is_open()) {
            throw std::runtime_error("Nepavyko sukurti failo " + filename);
        }
    }

    std::ostream &out = toFile ? file : std::cout;

    out « std::left « std::setw(15) « "Vardas" « std::setw(15) « "Pavarde" « std::setw(20) « "Galutinis (Vid.)" « std::setw(17) « "Galutinis (Med.)\n";
    out « "-----------------------------------------------------------------------\n";
    for (const auto& s : Students) {
        out « std::left « std::setw(15) « s.getName() « std::setw(15) « s.getSurname() « std::setw(20) « std::fixed « std::setprecision(2) « s.getFinalAvg() « std::setw(17) « std::fixed « std::setprecision(2) « s.getFinalMed() « "\n";
    }
    if (!out) {
        throw std::runtime_error("Klaida rasant rezultatus.");
    }

    if (toFile) {
        file.close();
    }

}

template <typename Container>
void outputToFileSorting(Container& Students) {
    int sorting = inputInt(
        "Pasirinkite pagal ka rusiuoti duomenis:\n"
        "1 - rusiuoti pagal varda.\n"
        "2 - rusiuoti pagal pavarde.\n"
        "3 - rusiuoti pagal galutini ivertinima (Vid.).\n"
        "4 - rusiuoti pagal galutini ivertinima (Med.).\n",
        1, 4);

    switch (sorting) {
        case 1: sortContainer(Students, compareByName); break;
        case 2: sortContainer(Students, compareBySurname); break;
        case 3: sortContainer(Students, compareByAvg); break;
        case 4: sortContainer(Students, compareByMed); break;
    }
}

template <typename Container>
void handleOutput(Container& Students) {
    outputToFileSorting(Students);
    int outputChoice = inputInt(
    "Kur isvesti rezultatus:\n"
    "1 - ekrane\n"
    "2 - i faila\n",
    1, 2);
    printResults(Students, outputChoice == 2, "rezultatai.txt");
}
```

```

template void printResults<std::vector<Student>>(const std::vector<Student>&, bool, const std::string&);
template void printResults<std::list<Student>>(const std::list<Student>&, bool, const std::string&);
template void printResults<std::deque<Student>>(const std::deque<Student>&, bool, const std::string&);

template void outputToFileSorting<std::vector<Student>>(std::vector<Student>&);
template void outputToFileSorting<std::list<Student>>(std::list<Student>&);
template void outputToFileSorting<std::deque<Student>>(std::deque<Student>&);

template void handleOutput<std::vector<Student>>(std::vector<Student>&);
template void handleOutput<std::list<Student>>(std::list<Student>&);
template void handleOutput<std::deque<Student>>(std::deque<Student>&);
```

4.102 src/person.cpp File Reference

```
#include "../include/person.h"
```

4.103 person.cpp

Go to the documentation of this file.

```
#include "../include/person.h"

Person::Person(const Person& other) : name_(other.name_), surname_(other.surname_) {}

Person& Person::operator=(const Person& other) {
    if (this != &other) {
        name_ = other.name_;
        surname_ = other.surname_;
    }
    return *this;
}

Person::Person(Person&& other) noexcept : name_(std::move(other.name_)), surname_(std::move(other.surname_)) {

    other.name_.clear();
    other.surname_.clear();
}

Person& Person::operator=(Person&& other) noexcept {
    if (this != &other) {
        name_ = std::move(other.name_);
        surname_ = std::move(other.surname_);

        other.name_.clear();
        other.surname_.clear();
    }
    return *this;
}
```

4.104 src/student.cpp File Reference

```
#include "student.h"
#include <algorithm>
#include <iostream>
```

Functions

- std::ostream & operator<< (std::ostream &os, const Student &s)
- std::istream & operator>> (std::istream &is, Student &s)

4.104.1 Function Documentation

4.104.1.1 operator<<()

```
std::ostream & operator<< (
            std::ostream & os,
            const Student & s)
```

Definition at line 109 of file student.cpp.

4.104.1.2 operator>>()

```
std::istream & operator>> (
            std::istream & is,
            Student & s)
```

Definition at line 118 of file student.cpp.

4.105 student.cpp

Go to the documentation of this file.

```
#include "student.h"
#include <algorithm>
#include <iostream>

Student::Student(const std::string name, const std::string surname, const std::vector<int> nd, int
    egz) : Person(name, surname), nd_(nd), egz_(egz) {
    calculate();
}

Student::~Student() {
    name_.clear();
    surname_.clear();
    nd_.clear();
    egz_ = 0;
    finalAvg_ = 0.0;
    finalMed_ = 0.0;
}

Student::Student(const Student& other)
    : Person(other),
      nd_(other.nd_),
      egz_(other.egz_),
      finalAvg_(other.finalAvg_),
      finalMed_(other.finalMed_) {}

Student& Student::operator=(const Student& other) {
    if (this != &other) {
        Person::operator=(other);
        nd_ = other.nd_;
        egz_ = other.egz_;
        finalAvg_ = other.finalAvg_;
        finalMed_ = other.finalMed_;
    }
    return *this;
}

Student::Student(Student&& other) noexcept
    : Person(std::move(other)),
      nd_(std::move(other.nd_)),
      egz_(other.egz_),
      finalAvg_(other.finalAvg_),
      finalMed_(other.finalMed_)
{
    other.nd_.clear();
    other.egz_ = 0;
    other.finalAvg_ = 0.0;
    other.finalMed_ = 0.0;
```

```
}

Student& Student::operator=(Student&& other) noexcept {
    if (this != &other) {
        Person::operator=(std::move(other));

        nd_ = std::move(other.nd_);
        egz_ = other.egz_;
        finalAvg_ = other.finalAvg_;
        finalMed_ = other.finalMed_;

        other.nd_.clear();
        other.egz_ = 0;
        other.finalAvg_ = 0.0;
        other.finalMed_ = 0.0;
    }
    return *this;
}

const std::vector<int>& Student::getNd() const { return nd_; }
int Student::getEgz() const { return egz_; }

double Student::getFinalAvg() const { return finalAvg_; }
double Student::getFinalMed() const { return finalMed_; }

void Student::setNd(const std::vector<int>& nd) {
    nd_ = nd;
    calculate();
}

void Student::setEgz(int egz) {
    egz_ = egz;
    calculate();
}

void Student::addNd(int mark) {
    nd_.push_back(mark);
    calculate();
}

void Student::calculate() {
    if (nd_.empty()) {
        finalAvg_ = finalMed_ = 0.6 * egz_;
        return;
    }

    double sum = 0;
    for (int x : nd_) sum += x;
    finalAvg_ = 0.4 * (sum / nd_.size()) + 0.6 * egz_;

    std::vector<int> temp = nd_;
    std::sort(temp.begin(), temp.end());

    double median;
    if (temp.size() % 2 == 0)
        median = (temp.at(temp.size()/2 - 1) + temp.at(temp.size()/2)) / 2.0;
    else
        median = temp.at(temp.size()/2);

    finalMed_ = 0.4 * median + 0.6 * egz_;
}

std::ostream& operator«(std::ostream& os, const Student& s) {
    os « s.name_ « " " « s.surname_ « " | ND: ";
    for (int x : s.nd_) os « x « " ";
    os « "| Egz: " « s.egz_;
    os « " | Galutinis (Vid): " « s.finalAvg_;
    os « " | Galutinis (Med): " « s.finalMed_;
    return os;
}

std::istream& operator»(std::istream& is, Student& s) {
    s.nd_.clear();

    int n;
    is » s.name_ » s.surname_ » n;

    for (int i = 0; i < n; i++) {
        int x;
        is » x;
        s.nd_.push_back(x);
    }

    is » s.egz_;
    s.calculate();

    return is;
```

```
}

void Student::print() const {
    std::cout « name_ « " " « surname_ « " | Galutinis (Vid): " « finalAvg_ « " | Galutinis (Med): " « finalMed_ « std::endl;
}
```

4.106 tests/student_test.cpp File Reference

```
#include <gtest/gtest.h>
#include "../include/student.h"
#include <iostream>
#include <cassert>
#include <sstream>
#include <fstream>
```

Functions

- TEST (StudentTest, Constructors)
- TEST (StudentTest, CopyConstructor)
- TEST (StudentTest, AssignmentOperator)
- TEST (StudentTest, MoveConstructor)
- TEST (StudentTest, MoveAssignment)
- TEST (StudentTest, DestructorDoesNotCrash)
- TEST (StudentTest, GettersSetters)
- TEST (StudentTest, InputOutput)
- TEST (StudentTest, Calculations)
- TEST (StudentTest, FileInput)
- TEST (StudentTest, FileOutput)
- TEST (StudentTest, Polymorphism)
- TEST (numberTest, equality)

4.106.1 Function Documentation

4.106.1.1 TEST() [1/13]

```
TEST (
            numberTest ,
            equality )
```

Definition at line 185 of file student_test.cpp.

4.106.1.2 TEST() [2/13]

```
TEST (
            StudentTest ,
            AssignmentOperator )
```

Definition at line 50 of file student_test.cpp.

4.106.1.3 TEST() [3/13]

```
TEST (
            StudentTest ,
            Calculations )
```

Definition at line 137 of file student_test.cpp.

4.106.1.4 TEST() [4/13]

```
TEST (
            StudentTest ,
            Constructors )
```

Definition at line 8 of file student_test.cpp.

4.106.1.5 TEST() [5/13]

```
TEST (
            StudentTest ,
            CopyConstructor )
```

Definition at line 30 of file student_test.cpp.

4.106.1.6 TEST() [6/13]

```
TEST (
            StudentTest ,
            DestructorDoesNotCrash )
```

Definition at line 102 of file student_test.cpp.

4.106.1.7 TEST() [7/13]

```
TEST (
            StudentTest ,
            FileInput )
```

Definition at line 146 of file student_test.cpp.

4.106.1.8 TEST() [8/13]

```
TEST (
            StudentTest ,
            FileOutput )
```

Definition at line 160 of file student_test.cpp.

4.106.1.9 TEST() [9/13]

```
TEST (
            StudentTest ,
            GettersSetters )
```

Definition at line 108 of file student_test.cpp.

4.106.1.10 TEST() [10/13]

```
TEST (
            StudentTest ,
            InputOutput )
```

Definition at line 124 of file student_test.cpp.

4.106.1.11 TEST() [11/13]

```
TEST (
            StudentTest ,
            MoveAssignment )
```

Definition at line 86 of file student_test.cpp.

4.106.1.12 TEST() [12/13]

```
TEST (
            StudentTest ,
            MoveConstructor )
```

Definition at line 68 of file student_test.cpp.

4.106.1.13 TEST() [13/13]

```
TEST (
            StudentTest ,
            Polymorphism )
```

Definition at line 178 of file student_test.cpp.

4.107 student_test.cpp

Go to the documentation of this file.

```
#include <gtest/gtest.h>
#include "../include/student.h"
#include <iostream>
#include <cassert>
#include <sstream>
#include <fstream>

TEST(StudentTest, Constructors) {
    Student s1;

    EXPECT_EQ(s1.getName(), "Julius");
    EXPECT_EQ(s1.getSurname(), "Petraitis");
    EXPECT_EQ(s1.getNd().size(), 2);
    EXPECT_EQ(s1.getNd()[0], 1);
    EXPECT_EQ(s1.getNd()[1], 2);
    EXPECT_EQ(s1.getEgz(), 5);

    std::vector<int> marks = {8, 9, 7};
    Student s2("Jonas", "Kazlauskas", marks, 8);

    EXPECT_EQ(s2.getName(), "Jonas");
    EXPECT_EQ(s2.getSurname(), "Kazlauskas");
    EXPECT_EQ(s2.getNd().size(), 3);
    EXPECT_EQ(s2.getEgz(), 8);

    double expectedAvg = 0.4 * ((8+9+7)/3.0) + 0.6 * 8;
    EXPECT_NEAR(s2.getFinalAvg(), expectedAvg, 0.0001);
}

TEST(StudentTest, CopyConstructor) {
    std::vector<int> marks = {7, 8, 9};
    Student original("Marius", "Jonaitis", marks, 7);

    Student copy(original);

    EXPECT_EQ(copy.getName(), original.getName());
    EXPECT_EQ(copy.getSurname(), original.getSurname());
    EXPECT_EQ(copy.getNd().size(), original.getNd().size());
    EXPECT_EQ(copy.getEgz(), original.getEgz());

    EXPECT_NEAR(copy.getFinalAvg(), original.getFinalAvg(), 0.0001);
    EXPECT_NEAR(copy.getFinalMed(), original.getFinalMed(), 0.0001);

    original.setNd({1,1,1});
    original.setEgz(1);

    EXPECT_EQ(copy.getEgz(), 7);
}

TEST(StudentTest, AssignmentOperator) {
    Student s1("Petras", "Petraitis", {8,9}, 9);
    Student s2("Linas", "Linauskas", {5,6,7}, 6);

    s2 = s1;

    EXPECT_EQ(s2.getName(), "Petras");
    EXPECT_EQ(s2.getSurname(), "Petraitis");
    EXPECT_EQ(s2.getNd().size(), 2);
    EXPECT_EQ(s2.getEgz(), 9);

    s1.setNd({10,10});
    EXPECT_EQ(s2.getNd()[0], 8);

    s2 = s2;
    EXPECT_EQ(s2.getName(), "Petras");
}

TEST(StudentTest, MoveConstructor) {
    std::vector<int> marks = {9,8,7};
    Student original("Tomas", "Tomauskas", marks, 9);

    std::string name = original.getName();
    double avg = original.getFinalAvg();

    Student moved(std::move(original));

    EXPECT_EQ(moved.getName(), name);
    EXPECT_NEAR(moved.getFinalAvg(), avg, 0.0001);

    EXPECT_TRUE(original.getName().empty());
    EXPECT_TRUE(original.getSurname().empty());
    EXPECT_TRUE(original.getNd().empty());
```

```
    EXPECT_EQ(original.getEgz(), 0);
}

TEST(StudentTest, MoveAssignment) {
    Student s1("Andrius", "Andriauskas", {8,9,10}, 8);
    Student s2("Giedrius", "Giedrauskas", {5,5}, 5);

    std::string name = s1.getName();
    double avg = s1.getFinalAvg();

    s2 = std::move(s1);

    EXPECT_EQ(s2.getName(), name);
    EXPECT_NEAR(s2.getFinalAvg(), avg, 0.0001);

    EXPECT_TRUE(s1.getName().empty());
    EXPECT_TRUE(s1.getNd().empty());
}

TEST(StudentTest, DestructorDoesNotCrash) {
    EXPECT_NO_THROW({
        Student s("Test", "Testauskas", {1, 2, 3}, 10);
    });
}

TEST(StudentTest, GettersSetters) {
    Student s("Laura", "Lauraite", {8,9}, 7);

    EXPECT_EQ(s.getName(), "Laura");
    EXPECT_EQ(s.getNd().size(), 2);

    s.setNd({10,9,8});
    EXPECT_EQ(s.getNd()[0], 10);

    s.setEgz(9);
    EXPECT_EQ(s.getEgz(), 9);

    s.addNd(7);
    EXPECT_EQ(s.getNd().size(), 4);
}

TEST(StudentTest, InputOutput) {
    std::stringstream ss;
    ss « "Rokas Rokauskas 3 9 8 7 8";

    Student s;
    ss » s;

    EXPECT_EQ(s.getName(), "Rokas");
    EXPECT_EQ(s.getSurname(), "Rokauskas");
    EXPECT_EQ(s.getNd().size(), 3);
    EXPECT_EQ(s.getEgz(), 8);
}

TEST(StudentTest, Calculations) {
    Student s1("Test1", "Test", {}, 10);
    EXPECT_NEAR(s1.getFinalAvg(), 6.0, 0.0001);

    Student s2("Test2", "Test", {8}, 7);
    double avg = 0.4 * 8 + 0.6 * 7;
    EXPECT_NEAR(s2.getFinalAvg(), avg, 0.0001);
}

TEST(StudentTest, FileInput) {
    std::ofstream fout("data/test_in.txt");
    fout « "Jonas Jonaitis 3 8 9 7 8";
    fout.close();

    std::ifstream fin("data/test_in.txt");

    Student s;
    fin » s;

    EXPECT_EQ(s.getName(), "Jonas");
    EXPECT_EQ(s.getSurname(), "Jonaitis");
}

TEST(StudentTest, FileOutput) {
    Student s("Petras", "Petraitis", {10, 9}, 8);

    std::ofstream fout("data/test_output.txt");
    ASSERT_TRUE(fout.is_open());

    fout « s;
    fout.close();

    std::ifstream fin("data/test_output.txt");
```

```
    ASSERT_TRUE(fin.is_open());

    std::string content;
    std::getline(fin, content);

    EXPECT_FALSE(content.empty());
}

TEST(StudentTest, Polymorphism) {
    Student s("Jonas", "Jonaitis", {8,9}, 10);
    Person* p = &s;

    EXPECT_NO_THROW(p->print());
}

TEST(numberTest, equality) {
    int a=5;
    EXPECT_EQ(a, 5);
}
```

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